

Linear Statistical Models

Lecture notes and Python companion
based on the IIT Madras BS Degree course by Prof. Siva Athreya

Compiled lecture notes

Part I: Weeks 1–5 (Lectures 1–18)

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Notation

The same notation is used throughout the book. Where the instructor's own notation differs, the last column says how to translate.

Symbol	Meaning	Instructor's notes
$Y = X\beta + \varepsilon$	the linear model; $Y \in \mathbb{R}^n$ observed, $\beta \in \mathbb{R}^p$ unknown	same, with m parameters
n	number of observations	same
p	number of parameters (columns of X)	written m
X	$n \times p$ known design matrix, $X = (x_{ij})$	same
$r = \text{rank}(X)$	rank of the design matrix	–
ε	error vector, $\mathbb{E}(\varepsilon) = 0$, $\text{Cov}(\varepsilon) = \sigma^2 I_n$	same
$(Y, X\beta, \sigma^2 I_n)$	shorthand for the linear model above	same
$\lambda^\top \beta$	a linear parametric function, $\lambda \in \mathbb{R}^p$	written $p^\top \beta$, $p \in \mathbb{R}^m$
$\ell^\top Y$	a linear estimator, $\ell \in \mathbb{R}^n$	same
$\mathcal{C}(A), \mathcal{N}(A), \mathcal{R}(A)$	column space, null space, row space of A	$\mathcal{C}(A), \mathcal{N}(A), \mathcal{R}(A)$
P_X	orthogonal projection of $\mathbb{R}^n$ onto $\mathcal{C}(X)$	same
A^-	a generalized inverse of A : $AA^-A = A$	–
$\hat{\beta}$	any solution of the normal equations $X^\top X\beta = X^\top Y$	$\hat{\beta}$ or $\tilde{\beta}$
$\hat{Y} = X\hat{\beta} = P_X Y$	fitted values	same
$\hat{e} = Y - \hat{Y}$	residual vector	same
$\text{RSS} = \text{SSE} = \sum \ Y - \hat{Y}\ ^2$	residual (error) sum of squares	R_0^2
TSS	total sum of squares $\sum (Y_i - \bar{Y})^2$	same
SSW, SSB	within- and between-group sums of squares	same
$\bar{Y}_{i\cdot}, \bar{Y}_{\cdot\cdot}$	group mean and grand mean	same
$\mathbf{1}_n$	the vector of n ones	–
$\mathbb{R}, \mathbb{E}, \mathbb{P}$	reals, expectation, probability	–
Var, Cov, tr	variance, covariance, trace	–
$\mathcal{N}(\mu, \sigma^2)$	normal distribution	$\text{Normal}(\mu, \sigma^2)$
$\chi_k^2, t_k, F_{k_1, k_2}$	chi-square, Student- t and F distributions	same
$\perp\!\!\!\perp$	independence	–

How to read these notes. Unmarked text follows the instructor's published material for the relevant week. Boxes whose title begins with *Supplementary* contain material added by the compiler: extra proofs, intuition or examples that are not in that material. Each lecture header states exactly which source was used. The video transcripts were *not* accessed. Because of that, video timestamps are not given, and the order of

topics follows the published notes.

Chapter 1

Basic Statistics, R and Data Visualisation

This chapter covers Week 1 of the course (Lectures 1–3). The instructor’s schedule starts with the computing environment and exploratory graphics. The linear model itself appears in Week 2. Note that the official syllabus lists “Review of estimation, hypothesis testing” for Week 1, but the published Week-1 material covers R and descriptive statistics. Estimation and testing are reviewed when they are needed (Sections 3.1 and 5.2).

1.1 Basic Statistics and Introduction to R

Lecture: 1

Video: <https://youtu.be/z1RfnVy80ak>

Timestamps: not available (video transcript not accessed)

Source used: Week-1 slides Jan26notes.pdf (with the instructor’s handwritten annotations)

Quiding Question

By the end of this lecture you should be able to:

- say what R is, where it came from, and how to install R and RStudio (and what the Python equivalents are);
- classify data as categorical, discrete numeric or continuous numeric;
- describe numerical data by its *centre*, *spread* and *shape*, and compute the mean, median, mode, variance, standard deviation and inter-quartile range;
- load a built-in data frame, look at it with `head`, `tail` and indexing, and make the five-number summary, histogram and scatter plots.

1.1.1 Motivation

The course is about *linear statistical models*, but every model starts with data. Before we write down $Y = X\beta + \varepsilon$ we have to know how to read a data set, summarise it, and draw pictures of it. The first week

therefore reviews basic descriptive statistics and introduces the computing environment. The instructor's slides open with a light-hearted "statutory warning": he tends to speak quickly, and students should ask him to repeat anything that is unclear. The same advice applies to these notes: work through every example yourself, in the companion notebook if you like.

1.1.2 The R environment

The slides describe R as follows.

- R is an open-source programming language for statistical computing. It runs on Linux, Windows and Mac OS, and it is free. The project web page is <http://www.r-project.org>.
- R is modelled on the languages S and S-Plus. S was developed in the late 1980s at AT&T Bell Labs.
- The R project was started in 1995 by Robert Gentleman and Ross Ihaka of the Statistics Department of the University of Auckland (*Journal of Computational and Graphical Statistics* 5:3, 299–314, 1996). It is now maintained by the R core development team with many contributors.
- R is downloaded from <https://cloud.r-project.org>. RStudio is an **integrated development environment** (IDE) for R.

There is plenty of help available on the web, for example the R-bloggers site.

Note 1.1.1 (Supplementary: the Python counterpart used in this book)

Throughout these notes the R demonstrations are translated into Python. The ingredients are `numpy` (arrays and linear algebra), `pandas` (data frames), `matplotlib/seaborn` (graphics), `scipy.stats` (distributions) and `statsmodels` (linear models). The IDE is VS Code with the Jupyter extension. Setup instructions are in the `README.md` at the root of the project.

1.1.3 Kinds of data

Definition 1.1.2 (Types of data)

The slides distinguish three kinds of data.

1. **Categorical data**: each observation belongs to one of a set of labels, for example $\{0, 1\}$ or $\{A, B, \dots\}$.
2. **Discrete numeric data**: numbers that naturally take only discrete values, for example counts.
3. **Continuous numeric data**: numbers that can take any value in an interval, for example temperature or a stock price.

The slides refer to S. S. Stevens, "On the theory of scales of measurement", *Science* 103 (1946), 677–680, for a finer classification of measurement data. As typical examples of numeric data the slides mention daily temperatures and the earnings of a stock in a stock market. Such data are visualised with histograms and boxplots.

Note 1.1.3 (Supplementary: Stevens' scales)

Stevens' paper is best known for its four scales of measurement: *nominal* (labels only), *ordinal* (labels with an order), *interval* (differences are meaningful, zero is arbitrary, e.g. temperature in °F) and *ratio* (a true zero, e.g. length). The scale determines which summaries make sense. A mean of nominal codes is meaningless, and a ratio of two Fahrenheit temperatures has no physical meaning.

1.1.4 Centre, spread and shape

For numerical data the slides single out three key features. We make each one precise. Let $x_1, \dots, x_n$ be the data and $x_{(1)} \leq \dots \leq x_{(n)}$ the ordered data.

- Definition 1.1.4 (Measures of centre)**
- The **mean** (average) is $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$. If the distinct values $x_1, \dots, x_k$ occur with frequencies $f_1, \dots, f_k$ ($\sum f_i = n$), then $\bar{x} = \frac{1}{n} \sum_{i=1}^k f_i x_i$. This frequency form is written on the slide.
 - The **median** is the middle ordered value: $x_{((n+1)/2)}$ if n is odd, and $\frac{1}{2} (x_{(n/2)} + x_{(n/2+1)})$ if n is even.
 - The **mode** is the most frequently occurring value.

- Definition 1.1.5 (Measures of spread)**
- The (sample) **variance** is $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$. This is R's var. The **standard deviation** is $s = \sqrt{s^2}$ (R's sd).
 - The **inter-quartile range** is $IQR = Q_3 - Q_1$, where Q_1 and Q_3 are the first and third quartiles (the 25% and 75% quantiles).

The *shape* of the data describes, for example, whether it is symmetric or skewed about its centre, and which values are more likely than others. The instructor's sketch shows a symmetric histogram around 0 and an asymmetric one with a long tail. Next to the sketch he writes that most of the data lie in an "essential range" from mean−3SD to mean+3SD. The next result makes this precise for every data set.

Proposition 1.1.6 (Supplementary: Chebyshev's inequality for data)

Let $\sigma_n^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ and suppose $\sigma_n > 0$. For every $k > 0$, the proportion of data points with $|x_i - \bar{x}| \geq k\sigma_n$ is at most $1/k^2$. In particular at least $8/9 \approx 89\%$ of any data set lies within $3\sigma_n$ of the mean.

Proof. Let $A = \{i : |x_i - \bar{x}| \geq k\sigma_n\}$ and let $\#A$ be its size. Every term in the sum below is non-negative, and for $i \in A$ we have $(x_i - \bar{x})^2 \geq k^2\sigma_n^2$. Therefore

$$n\sigma_n^2 = \sum_{i=1}^n (x_i - \bar{x})^2 \geq \sum_{i \in A} (x_i - \bar{x})^2 \geq \#A \cdot k^2\sigma_n^2.$$

Dividing by $nk^2\sigma_n^2 > 0$ gives $\#A/n \leq 1/k^2$. Taking $k = 3$ gives the second statement. $\square$

For roughly bell-shaped data the proportion inside mean $\pm 3SD$ is much higher, about 99.7%. That is why the instructor calls it the essential range.

Proposition 1.1.7 (Supplementary: the mean and the median as best constants)

For data $x_1, \dots, x_n$,

1. $c \mapsto \sum_{i=1}^n (x_i - c)^2$ is minimised uniquely at $c = \bar{x}$;
2. $c \mapsto \sum_{i=1}^n |x_i - c|$ is minimised at any median.

Proof. (1) Write $x_i - c = (x_i - \bar{x}) + (\bar{x} - c)$ and expand:

$$\sum_i (x_i - c)^2 = \sum_i (x_i - \bar{x})^2 + 2(\bar{x} - c) \sum_i (x_i - \bar{x}) + n(\bar{x} - c)^2.$$

The middle sum is $\sum_i x_i - n\bar{x} = 0$. So $\sum (x_i - c)^2 = \sum (x_i - \bar{x})^2 + n(\bar{x} - c)^2$, which is smallest exactly when $c = \bar{x}$.

(2) Let $g(c) = \sum_i |x_i - c|$. The function g is convex and piecewise linear. At a point c that is not a data value its derivative is $\#\{i : x_i < c\} - \#\{i : x_i > c\}$. This is negative while fewer than half of the points lie below c and positive once more than half do. A convex function is minimised where its slope changes sign, which is at a median. $\square$

The first identity, “total squared deviation = deviation about the mean + $n \times (\text{distance of the mean})^2$ ”, is the simplest case of the Pythagorean decompositions behind all of least squares. We will meet it again in Sections 3.1 and 4.2.

Example 1.1.8 (Summaries by hand)

Consider the nine observations 3, 7, 8, 5, 12, 14, 21, 13, 18.

Solution. Ordered: 3, 5, 7, 8, 12, 13, 14, 18, 21.

- Mean: $\sum x_i = 101$, so $\bar{x} = 101/9 = 11.2222$.
- Median: $n = 9$ is odd, so the median is $x_{(5)} = 12$.
- Variance: $\sum x_i^2 = 9 + 25 + 49 + 64 + 144 + 169 + 196 + 324 + 441 = 1421$. Hence $\sum (x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2 = 1421 - 101^2/9 = 1421 - 1133.4444 = 287.5556$, and $s^2 = 287.5556/8 = 35.9444$, $s = 5.9954$.
- Quartiles: R’s default rule (`type = 7`, also NumPy’s default) puts the q -quantile at ordered position $1 + (n - 1)q$. For $q = 0.25$ this is position 3, so $Q_1 = x_{(3)} = 7$. For $q = 0.75$ it is position 7, so $Q_3 = x_{(7)} = 14$. Hence IQR = 7.
- The five-number summary is $(\min, Q_1, \text{median}, Q_3, \max) = (3, 7, 12, 14, 21)$ and the mean is 11.22. The mean is below the median here: the lower tail (3, 5) pulls it down more than the upper tail pushes it up.

These numbers are reproduced in the notebook `notebooks/week01/L01_basic_stats_intro.ipynb`.

1.1.5 Data sets and data frames

R ships with many data sets. The command `data()` lists the installed ones, and the instructor strongly encourages students to pick a data set and repeat the class experiments on it.

Definition 1.1.9 (Data frame)

A **data frame** is a rectangular collection of *variables* (the columns) and *observations* (the rows).

Example 1.1.10 (The `airquality` data)

`?airquality` describes daily air-quality measurements in New York (May–September 1973). It is a data frame with 153 rows and 6 columns: `Ozone`, `Solar.R`, `Wind`, `Temp`, `Month` and `Day`. The slides use it to show the following commands.

- `head(airquality)` and `tail(airquality)` show the first and the last six rows; with `n = 10`, `head` shows ten rows.
- `airquality[148,4]` returns 63, the entry in row 148 and column 4 (`Temp`). The same value is `airquality$Temp[148]`.
- `airquality[148,]` is the whole 148th row. `airquality[,c(1,4)]` gives the `Ozone` and `Temp` columns for all rows; `c()` builds the vector of column numbers.
- `summary(airquality$Temp)` prints the five-number summary together with the mean:

Min 56, Q_1 72, Median 79, Mean 77.88, Q_3 85, Max 97.

Note 1.1.11 (Indexing pitfall)

R counts from 1, Python from 0. In pandas, R's `airquality[148,4]` is `aq.iloc[147, 3]` or `aq.loc[147, "Temp"]`. The value is again 63.

1.1.6 First plots

The slides then make three kinds of plot.

- `hist(airquality$Temp)`: a histogram. Temperatures range from 56 to 97 with a peak in the high 70s and low 80s, and the histogram is mildly left-skewed.
- `plot(airquality$Temp)`: the values plotted against their index. Because the rows are in date order, this is a time plot. It shows temperatures rising into the summer and falling in September.
- `plot(airquality$Ozone, airquality$Temp)`: a scatter plot. Days with higher ozone tend to be hotter.
- `plot(airquality)`: a matrix of scatter plots of every pair of variables.

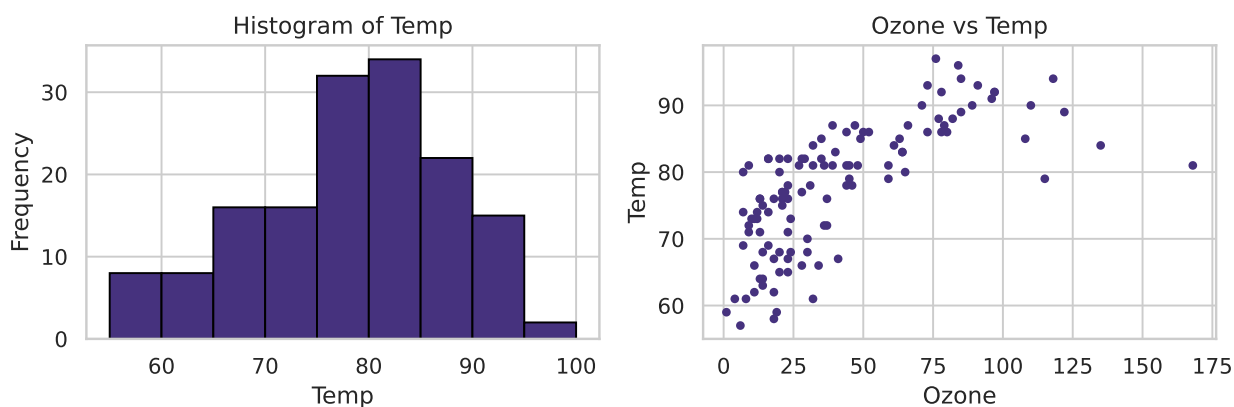


Figure 1.1: Histogram of Temp and scatter plot of Temp against Ozone (airquality).

1.1.7 External packages

R can be extended with packages. `install.packages("UsingR")` installs the package `UsingR`, which contains many data sets, and `library("UsingR")` adds it to the current workspace. Likewise `install.packages("tidyverse")` and `library("tidyverse")` load `ggplot2`, the graphics system used in Sections 1.2 and 1.3.

1.1.8 Computation

The R commands of this lecture translate to Python as follows.

R	Python (pandas / matplotlib)
<code>head(airquality, n=10)</code>	<code>aq.head(10)</code>
<code>airquality[148,4]</code>	<code>aq.iloc[147, 3]</code>
<code>airquality[,c(1,4)]</code>	<code>aq.iloc[:, [0, 3]]</code>
<code>summary(airquality\$Temp)</code>	<code>aq.Temp.describe()</code>
<code>hist(airquality\$Temp)</code>	<code>plt.hist(aq.Temp)</code>
<code>plot(airquality)</code>	<code>pd.plotting.scatter_matrix(aq)</code>

Everything, including Example 1.1.8, is in `notebooks/week01/L01_basic_stats_intro.ipynb`.

1.1.9 Summary

- R is a free language for statistics. The Python stack `pandas/statsmodels` does the same job in these notes.
- Data are categorical, discrete numeric or continuous numeric.
- Centre: $\bar{x}$, median, mode. Spread: $s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$, s , IQR. Shape: symmetry, skewness.
- $\sum (x_i - c)^2 = \sum (x_i - \bar{x})^2 + n(\bar{x} - c)^2$, so the mean is the least-squares constant.

- A data frame has variables as columns and observations as rows. Use `head`, `tail`, indexing and `summary` to look at it, and histograms and scatter plots to picture it.

1.1.10 Exercises

Problem 1.1.12

For the data 2, 4, 4, 5, 7, 9, 11 compute the mean, median, mode, sample variance and IQR (using R's default quantile rule).

Problem 1.1.13

Show that $\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2$.

Problem 1.1.14

Let $y_i = a + bx_i$ for constants a, b . Express $\bar{y}$, s_y^2 and IQR_y in terms of the corresponding quantities for x .

Problem 1.1.15

Using `airquality`, find the mean `Temp` for each month. Which month is hottest? (Use `groupby` in `pandas`.)

Problem 1.1.16

Give a data set of 10 numbers in which fewer than 90% of the observations lie within $2\sigma_n$ of the mean, and check that it still obeys Proposition 1.1.6 with $k = 2$.

1.1.11 Further reading

Christensen does not cover exploratory data analysis. For this lecture see instead the instructor's references: Verzani, *simpleR* (Sections 13–15), and Wickham & Grolemund, *R for Data Science*, Chapter 3. In Rao's *Linear Statistical Inference*, the sample moments appear in Chapter 2 (with the distribution of the sample mean and variance under normality in Chapter 3).

1.2 Data Visualization using ggplot2

Lecture: 2

Video: <https://youtu.be/7nmwfGpgDkM>

Timestamps: not available (video transcript not accessed)

Source used: Week-1 slides `Jan26notes.pdf`, `ggplot2` part (slides on aesthetics, scaling, `viridis`, facets)

Quiding Question

By the end of this lecture you should be able to:

- explain the idea of a *grammar of graphics* and the basic `ggplot` template;
- map a variable to an *aesthetic* (colour, shape, size) and explain what *scaling* does;
- use colour-blind-safe *viridis* palettes;
- split a plot into *facets* with `facet_wrap` and `facet_grid`;
- reproduce all of these plots in Python with `seaborn/matplotlib`.

1.2.1 Recap and motivation

In Section 1.1 we used R's base graphics (`hist`, `plot`). They are quick, but each kind of plot has its own function with its own arguments. The package `ggplot2` (loaded by `library(tidyverse)`) takes a different route: it is built on a *grammar of graphics*. The instructor's annotation sums up the benefit: "a system for describing and building graphs — learn one system and apply it at many places". In linear models we constantly plot a response against predictors, coloured or split by a factor. This is exactly what the grammar makes easy.

1.2.2 The mpg data

The running example is the tidyverse data set `mpg`. It contains observations collected by the US Environmental Protection Agency on 38 models of cars (234 rows, 11 columns). Two variables matter most here:

- `displ`: engine displacement in litres (the car's engine size);
- `hwy`: fuel efficiency on the highway, in miles per gallon.

Other variables used below are `class` (type of car: 2seater, compact, midsize, minivan, pickup, subcompact, suv), `drv` (f = front-wheel, r = rear-wheel, 4 = four-wheel drive), `cyl` (number of cylinders) and `trans` (type of transmission).

1.2.3 The basic template

The first plot on the slides is

```
ggplot(data = mpg) + geom_point(mapping = aes(x = displ, y = hwy))
```

It shows a *negative relationship between engine size and fuel efficiency*: cars with big engines use more fuel.

Definition 1.2.1 (The ggplot template)

A ggplot2 graph is built in layers:

1. `ggplot(data = <DATA>)` creates a coordinate system to which layers are added. On its own it draws an empty graph.
2. A **geom function** such as `geom_point()` adds a layer, here a layer of points.
3. Each geom takes a mapping argument, which is always paired with `aes()`. It says which variables go on which visual channels.

In short: `ggplot(data = <DATA>) + <GEOM_FUNCTION>(mapping = aes(<MAPPINGS>))`.

1.2.4 Aesthetic mappings and scaling

Definition 1.2.2 (Aesthetic, mapping, scaling)

An **aesthetic** is a visual property of the objects in the plot: position (x, y), colour, shape, size, and so on. To **map** a variable to an aesthetic we associate the aesthetic's name with the variable's name inside `aes()`. **Scaling** is the step in which ggplot2 assigns a unique level of the aesthetic (a colour, say) to each unique level of the variable. It also adds a legend that explains the correspondence.

Example 1.2.3 (Adding a third variable)

`ggplot(data=mpg) + geom_point(mapping=aes(x=displ, y=hwy, colour=class))` adds the variable `class` to the two-dimensional scatter plot by mapping it to the aesthetic colour. Here the aesthetic's name is `colour` and the variable is `class`. Scaling assigns 7 colours to the 7 classes, and the legend lists them. The plot shows that most of the cars with large engines but unusually *good* mileage are 2seaters. These are sports cars: big engines, but light bodies.

Other aesthetics include shape and size. The instructor notes on the slide that one can add shape in exactly the same way.

Note 1.2.4 (Supplementary: mapping versus setting)

Putting `colour = class` *inside* `aes()` maps a variable to colour. Putting `colour = "blue"` *outside* `aes()` just sets every point to blue. A common beginner's mistake is `aes(colour = "blue")`. This maps a constant *variable* whose single level is the string "blue", so the points get the first default colour (pinkish-red) and a pointless legend appears.

1.2.5 Viridis colour scales

The viridis scales give colour maps that are *perceptually uniform* in both colour and black-and-white. They are also designed to be read correctly by viewers with common forms of colour blindness (see <https://>

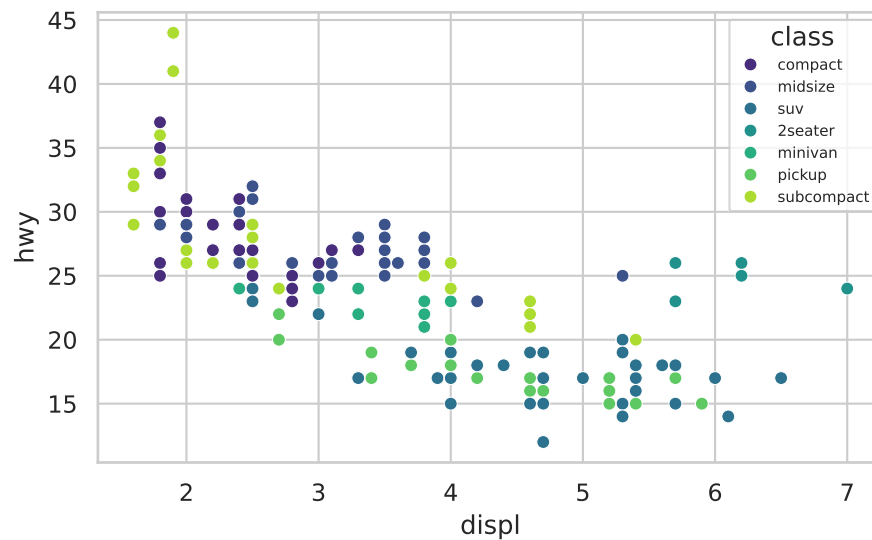


Figure 1.2: `hwy` against `displ`, coloured by `class` (viridis palette).

bids.github.io/colormap/). Adding `+ scale_colour_viridis_d()` to the previous plot switches to the default viridis palette for a discrete variable. `scale_colour_viridis_d(option = "plasma")` chooses the “plasma” variant.

1.2.6 Facets

Aesthetics are one way to add a variable. Another, especially useful for categorical variables, is to split the plot into **facets**.

Definition 1.2.5 (Facets)

`facet_wrap(~ class, nrow = 2)` splits a plot by a *single* variable into subplots, each showing one subset of the data. `facet_grid(drv ~ cyl)` splits by the *combination* of two variables: rows are levels of `drv`, columns are levels of `cyl`.

Example 1.2.6 (Facets on mpg)

With `facet_wrap(~class, nrow=2)` we get seven panels. The pickup and SUV panels show large engines and low mileage, while the compact and subcompact panels contain the high-mileage cars. With `facet_grid(drv~cyl)` the 3×4 grid shows, for instance, that there are *no* rear-wheel-drive cars with 4 or 5 cylinders in the data. Those cells are empty. Facets and colour can be combined, as on the last slide of this part: `geom_point(aes(colour=class)) + scale_colour_viridis_d(option="plasma") + facet_wrap(~class, nrow=2)`.

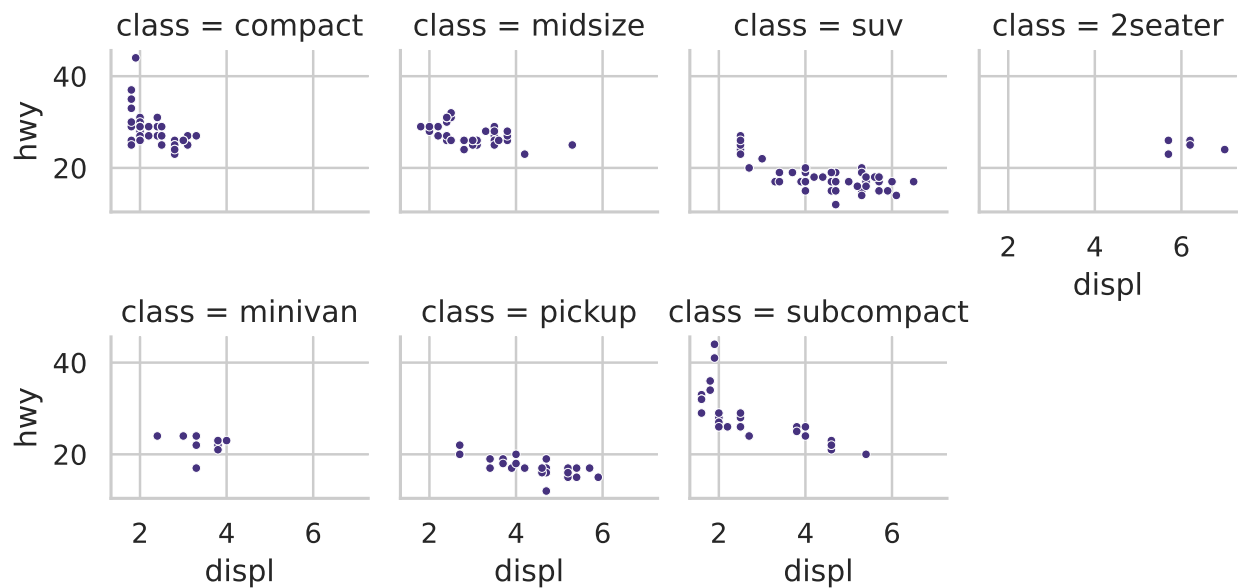


Figure 1.3: `facet_wrap(~ class)`: one panel per class.

Note 1.2.7 (Supplementary: why facets matter for linear models)

In a regression of `hwy` on `displ`, the relationship could differ between classes. For example, the slope within SUVs could differ from the slope within compacts. Faceting by a factor is the visual version of fitting a model with a *factor* $\times$ *covariate interaction*. That is the analysis of covariance (ANCOVA), which comes later in the course.

1.2.7 Computation

R (ggplot2)	Python (seaborn)
<code>ggplot(mpg)+geom_point(aes(displ,hwy))</code>	<code>sns.scatterplot(data=mpg, x="displ", y="hwy")</code>
<code>aes(..., colour=class)</code>	<code>hue="class"</code>
<code>scale_colour_viridis_d()</code>	<code>palette="viridis"</code>
<code>option="plasma"</code>	<code>palette="plasma"</code>
<code>facet_wrap(~class, nrow=2)</code>	<code>sns.relplot(..., col="class", col_wrap=4)</code>
<code>facet_grid(drv~cyl)</code>	<code>sns.relplot(..., row="drv", col="cyl")</code>

See notebooks/week01/L02_ggplot2.ipynb.

1.2.8 Summary

- Template: `ggplot(data) + geom(mapping = aes(...))`.
- Aesthetics (colour, shape, size) add variables. Scaling turns variable levels into aesthetic levels and makes a legend.

- Viridis palettes are perceptually uniform and colour-blind safe.
- `facet_wrap` splits by one variable; `facet_grid` by two.
- `mpg`: engine size and highway mileage are negatively related, and the 2-seaters are the exception.

1.2.9 Exercises

Problem 1.2.8

What goes wrong with `ggplot(mpg) + geom_point(aes(displ, hwy, colour = "blue"))`? How do you make all points blue?

Problem 1.2.9

Map the continuous variable `cty` to `colour`. How does the legend differ from the one for `class`? Do the same in `seaborn`.

Problem 1.2.10

Using `facet_grid(drv ~ cyl)` (or its `seaborn` equivalent), list the empty cells. What do they tell you about the cars in the data?

Problem 1.2.11

Count how many distinct values `class`, `drv` and `cyl` take in `mpg`. How many panels does `facet_grid(drv ~ cyl)` have?

1.2.10 Further reading

Wickham & Grolemund, *R for Data Science*, Chapter 3 (Data visualisation) — the source the instructor follows. Christensen and Rao do not cover graphics. Christensen's regression chapters (Chapter 6 onwards) use residual plots of the kind built with these tools.

1.3 Data Visualization using `ggplot()`-geoms

Lecture: 3

Video: <https://youtu.be/EP8uu2k5WfA>

Timestamps: not available (video transcript not accessed)

Source used: Week-1 slides `Jan26notes.pdf`, part “`ggplot()`-geoms” to “layered grammar of graphics”; recap slides at the start of `Feb2.pdf`

Guiding Question

By the end of this lecture you should be able to:

- explain what a *geom* is and switch between `geom_point` and `geom_smooth`;
- give mappings globally or per layer, and use a different data subset for one layer;
- explain the *statistical transformation* behind a bar chart (`stat_count`) and use `stat_summary`;
- use position adjustments (`dodge`) and coordinate systems (`coord_flip`, `coord_polar`);
- write down the full layered grammar-of-graphics template.

1.3.1 Recap

In Section 1.2 we built scatter plots with `ggplot(data) + geom_point(aes(...))`, added a third variable through an aesthetic, and split plots into facets. This lecture completes the grammar. We still use the `mpg` data.

1.3.2 Geoms

Definition 1.3.1 (Geom)

A **geom** is the geometrical object that a plot uses to represent data: points, lines, bars, boxes, and so on.

The same data can be drawn with different geoms.

- `ggplot(mpg) + geom_point(aes(x = displ, y = hwy))` draws the scatter plot.
- `ggplot(mpg) + geom_smooth(aes(x = displ, y = hwy))` draws a *smooth curve* fitted to the same points, with a grey confidence band.

The smooth curve decreases from about 29 mpg at 2 litres to about 17–18 mpg at 5 litres. The default R smoother (local *quadratic* fits) then turns up slightly for the largest engines. The upturn is caused by the 2-seaters we saw in Example 1.2.3. The local-*linear* LOWESS used in the Python notebook flattens out instead, and it gives about 15.5 mpg at 6 litres.

Note 1.3.2 (Supplementary: what `geom_smooth` fits)

For fewer than 1000 observations the default smoother is **LOESS** (locally estimated scatterplot smoothing). At each point x_0 it fits a low-degree polynomial to the nearby data by *weighted least squares*. Points closer to x_0 get larger weights. The value of that local fit at x_0 is the curve's height. So even the “model-free” smooth curve is built from many small least-squares fits: the subject of this course, applied locally. With `method = "lm"` one gets the single global least-squares line instead. The instructor uses this option in Week 5 (Section 5.1).

1.3.3 Global and local mappings

Mappings can be given once, in `ggplot()`, and are then inherited by every layer:

```
ggplot(mpg, aes(x = displ, y = hwy)) + geom_point(aes(color = drv)) + geom_smooth() +  
scale_colour_viridis_d()
```

Here `x` and `y` are *global*, while `color = drv` is *local* to the point layer. The points are coloured by drive type (4, f, r), and a single smooth curve is fitted to all the data.

A layer can even use different data. The instructor's second version replaces the smooth by

```
geom_smooth(data = filter(mpg, drv == "4"))
```

Here `filter` keeps only the four-wheel-drive cars, so the smooth curve describes only those cars while the points still show every car.

1.3.4 Bar charts and statistical transformations

Example 1.3.3 (Counts by class)

`ggplot(mpg) + stat_count(aes(x = class))` and `ggplot(mpg) + geom_bar(aes(x = class))` produce the *same* bar chart. The counts agree with `table(mpg$class)`:

2seater	compact	midsize	minivan	pickup	subcompact	suv
5	47	41	11	33	35	62

The seven counts add up to 234, the number of rows.

Definition 1.3.4 (Stat)

A **stat** is a *statistical transformation* applied to the data before display. For a bar chart the stat is count: for each level c of the x -variable it computes $n_c = \#\{i : x_i = c\}$ and passes the pairs (c, n_c) to the bar geom. Every geom has a default stat, and every stat has a default geom. `geom_bar` uses `stat_count`, and `stat_count` uses `geom_bar`. This is why the two commands above are interchangeable. `ggplot2` provides over 20 stats (see `?stat_bin`).

Example 1.3.5 (stat_summary)

```
ggplot(mpg) + stat_summary(aes(x = class, y = hwy), fun.min = min, fun.max = max,  
fun = median)
```

For each class this draws a point at the median highway mileage and a vertical line from the minimum to the maximum. The values plotted are:

	2seater	compact	midsize	minivan	pickup	subcompact	suv
min	23	23	23	17	12	20	12
median	25	27	27	23	17	26	17.5
max	26	44	32	24	22	44	27

Pickups and SUVs have the lowest median mileage. Compacts and subcompacts have the largest spread: their maxima of 44 belong to a few very efficient cars.

1.3.5 Position adjustments and coordinate systems

- `geom_bar(aes(x = class, fill = trans))` stacks the bars, splitting each class bar by transmission type. The `fill` aesthetic colours the inside of the bars.
- `position = "dodge"` puts the pieces *side by side*. This makes the counts within a class easier to compare.
- `coord_flip()` swaps the axes, giving horizontal bars. This is handy for long category names.
- `coord_polar()` uses polar coordinates. The bar chart becomes a “Coxcomb” chart.

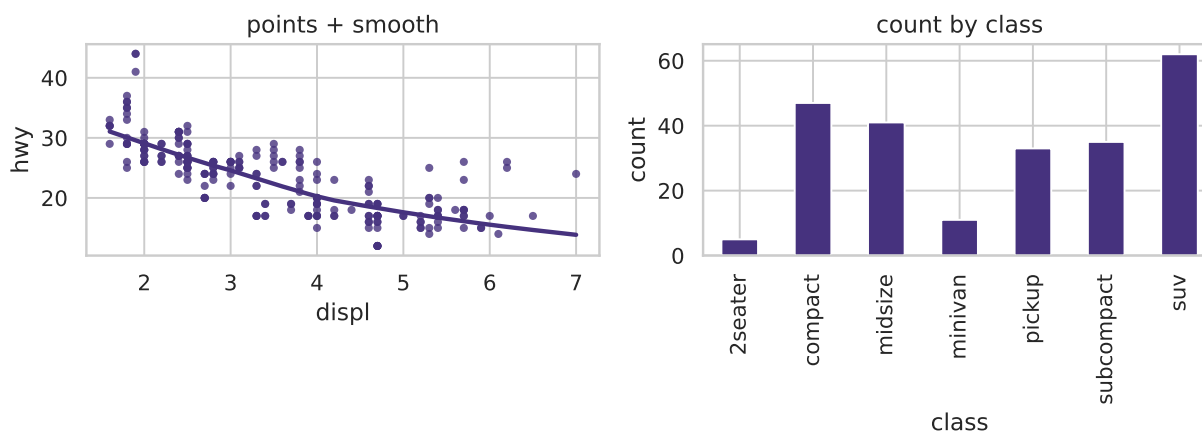


Figure 1.4: Left: points with a LOESS smooth. Right: the count stat behind a bar chart.

1.3.6 The layered grammar of graphics

Definition 1.3.6 (Layered grammar of graphics)

Every `ggplot2` graph fits the template

```
ggplot(data = <DATA>) +  
  <GEOM_FUNCTION>(  
    mapping = aes(<MAPPINGS>),  
    stat = <STAT>,  
    position = <POSITION>  
  ) +  
  <COORDINATE_FUNCTION> +  
  <FACET_FUNCTION>
```

The seven parameters (data, geom, mapping, stat, position, coordinates, facets) are enough to describe almost any plot.

The instructor's summary annotation lists what the week covered: scatter plots, viridis colours, bar charts,

facets, geoms and coordinate systems. The recap at the start of Week 2 calls the grammar the “key tool in data visualisation”.

1.3.7 Computation

R	Python
<code>geom_smooth()</code> (LOESS)	<code>sns.regplot(..., lowess=True)</code> or <code>statsmodels.nonparametric.lowess</code>
<code>filter(mpg, drv=="4")</code>	<code>mpg[mpg.drv == "4"]</code>
<code>geom_bar(aes(class))</code>	<code>sns.countplot(data=mpg, x="class")</code>
<code>stat_summary(fun=median, ...)</code>	<code>mpg.groupby("class").hwy.agg(["min", "median", "max"])</code>
<code>fill=trans, position="dodge"</code>	<code>sns.countplot(..., hue="trans")</code>
<code>coord_flip()</code>	<code>sns.countplot(..., y="class")</code>
<code>coord_polar()</code>	<code>plt.subplot(projection="polar")</code> with <code>ax.bar</code>

See notebooks/week01/L03_ggplot_geoms.ipynb.

1.3.8 Summary

- A geom is the object that represents data. A stat is the transformation applied first.
- Mappings in `ggplot()` are global; mappings in a geom are local. Each layer may use its own data.
- `geom_bar = stat_count + bars`. `stat_summary` draws min/median/max summaries.
- `position="dodge"`, `coord_flip()` and `coord_polar()` change the layout.
- The template has seven parameters: data, geom, mapping, stat, position, coordinates, facets.

1.3.9 Exercises

Problem 1.3.7

Which stat does `geom_histogram` use, and what does it compute? How is it related to `stat_count`?

Problem 1.3.8

Reproduce the `stat_summary` table of Example 1.3.5 for `cty` instead of `hwy`. Which class has the largest range?

Problem 1.3.9

Explain why `geom_smooth(method = "lm")` gives a straight line while the default gives a curve. What quantity does the `lm` line minimise?

Problem 1.3.10

Make a dodged bar chart of `drv` within each `class` in Python. Which class contains all three drive types?

1.3.10 Further reading

Wickham & Grolemund, *R for Data Science*, Chapter 3 (Sections on geometric objects, statistical transformations, position adjustments, coordinate systems and the layered grammar).

Solutions to the Exercises of Chapter 1

Problem 1.1.12. $n = 7$ and $\sum x_i = 42$, so $\bar{x} = 6$. The median is $x_{(4)} = 5$ and the mode is 4. $\sum x_i^2 = 312$, so $\sum (x_i - \bar{x})^2 = 312 - 7 \cdot 36 = 60$ and $s^2 = 60/6 = 10$. Quartiles (position $1 + 6q$): Q_1 is at position 2.5, i.e. $(4 + 4)/2 = 4$; Q_3 is at position 5.5, i.e. $(7 + 9)/2 = 8$. So $\text{IQR} = 4$.

Problem 1.1.13. $\sum (x_i - \bar{x})^2 = \sum x_i^2 - 2\bar{x} \sum x_i + n\bar{x}^2 = \sum x_i^2 - 2n\bar{x}^2 + n\bar{x}^2 = \sum x_i^2 - n\bar{x}^2$.

Problem 1.1.14. $\bar{y} = \frac{1}{n} \sum (a + bx_i) = a + b\bar{x}$. Then $y_i - \bar{y} = b(x_i - \bar{x})$, so $s_y^2 = b^2 s_x^2$ and $s_y = |b| s_x$. The map $x \mapsto a + bx$ is monotone. For $b > 0$ it keeps the order, so every quantile transforms as $q_y = a + bq_x$. For $b < 0$ it reverses the order, so $Q_1^{(y)} = a + bQ_3^{(x)}$ and $Q_3^{(y)} = a + bQ_1^{(x)}$. The positions $1 + (n-1)q$ and $1 + (n-1)(1-q)$ are symmetric, so this holds exactly for R's type-7 rule. In both cases $\text{IQR}_y = |b| \text{IQR}_x$. The shift a affects only the centre.

Problem 1.1.15. `aq.groupby("Month").Temp.mean()` gives 65.55 (May), 79.10 (June), 83.90 (July), 83.97 (August) and 76.90 (September). August is marginally the hottest month.

Problem 1.1.16. Take eight 0's and two 10's. Then $\bar{x} = 2$, $\sigma_n^2 = (8 \cdot 4 + 2 \cdot 64)/10 = 16$ and $\sigma_n = 4$. The two 10's are at distance $8 = 2\sigma_n$, so only 80% of the data lie strictly within $2\sigma_n$. The proportion at distance $\geq 2\sigma_n$ is $0.2 \leq 1/2^2 = 0.25$, as Chebyshev requires.

Problem 1.2.8. Inside `aes()`, "blue" is treated as a (constant) variable. `ggplot` maps its one level to the first default colour and adds a legend entry "blue". To colour every point blue, set the colour outside `aes()`: `geom_point(aes(displ, hwy), colour = "blue")`.

Problem 1.2.9. A continuous variable gets a *continuous* scale. The legend is a colour bar (a gradient), not a list of discrete keys. In `seaborn`, `hue="cty"` with a numeric column likewise uses a sequential palette and shows a few representative values.

Problem 1.2.10. The cross-tabulation of `drv` against `cyl` has zeros in three cells: four-wheel drive with 5 cylinders, and rear-wheel drive with 4 or 5 cylinders. All 5-cylinder cars are front-wheel drive, and all rear-wheel-drive cars have 6 or 8 cylinders.

Problem 1.2.11. `class` has 7 levels, `drv` has 3 and `cyl` has 4 (4, 5, 6, 8). The grid therefore has $3 \times 4 = 12$ panels, three of them empty (Problem 1.2.10).

Problem 1.3.7. `geom_histogram` uses `stat_bin`. It divides the range of the continuous x variable into intervals (bins) and counts the observations in each bin. `stat_count` counts observations at each *distinct value* of a discrete variable. `stat_bin` is its analogue for continuous data, where the “values” are the bins.

Problem 1.3.8. For `cty` (min/median/max): 2seater 15/15/16, compact 15/20/33, midsize 15/18/23, minivan 11/16/18, pickup 9/13/17, subcompact 14/19/35, suv 9/13/20. The largest range is for subcompacts: $35 - 14 = 21$.

Problem 1.3.9. `method="lm"` fits a single global linear model $y = a + bx$ by least squares. That is, it minimises $\sum_i (y_i - a - bx_i)^2$ over (a, b) (Section 2.2). The result is necessarily a straight line. The default LOESS fits a separate weighted least-squares polynomial near each x_0 , so the curve can bend.

Problem 1.3.10. `sns.countplot(data=mpg, x="class", hue="drv")`. From the cross-tabulation, only subcompact contains all three drive types (4: 4 cars, f: 22, r: 9).

Chapter 2

Linear Models: First Look

This chapter covers Week 2 (Lectures 4–7). The instructor introduces linear models *exploratorily*, with real and simulated data and R’s `lm`. The formal theory of estimation starts in Chapter 3. Week 2 of the official syllabus is “working with R”. Here the Python equivalents of every R command are given alongside.

2.1 Introduction to Linear Models

Lecture: 4

Video: <https://youtu.be/TPJmhTmm4h0>

Timestamps: not available (video transcript not accessed)

Source used: Week-2 slides Feb2.pdf (recap and “Model: Aims and Methods”, heights data), with the instructor’s handwritten annotations

Guiding Question

By the end of this lecture you should be able to:

- state the aim of modelling: to find a low-dimensional summary of the relationship between a response y and an explanatory variable x ;
- explain “all models are wrong but some are useful” with an example;
- describe the heights data and use a jittered scatter plot;
- explain the training / validation / test split used to judge models.

2.1.1 Recap

Week 1 introduced the layered grammar of graphics: scatter plots, smooth lines, bar charts, polar coordinates and faceting. The instructor calls it the key tool in data visualisation. Week 2 turns from *looking* at data to *modelling* it. The slides warn that there is no statistical analysis yet: this week introduces the linear model exploratorily, to see *how it works*. The formal theory starts in Week 3 (Section 3.1).

Note 2.1.1 (Practical instruction from the slides)

Download the data sets and the Week-2 R code from the course web page and keep them in the same directory, set as R's working directory. The Python notebooks follow the same idea. All data sets are in the project's `data/` folder, and the notebooks find it by a relative path.

2.1.2 Aims of modelling

Definition 2.1.2 (Response and explanatory variable)

We observe pairs $\{(x_i, y_i) : 1 \leq i \leq n\}$. The variable x is the **independent** (explanatory) variable and y the **dependent** (response) variable. A **model** is a low-dimensional summary of the data set that captures the patterns in the data, and in particular the way y depends on x . It can then be used for *prediction*: given a new x , what y should we expect?

Note 2.1.3 (Caution: all models are wrong, but some are useful)

The slides quote G. E. P. Box's aphorism. The instructor's example is the ideal gas law $PV = RT$. No real gas satisfies it exactly, yet it is a good approximation, and it helps us *understand the system*. A linear model is judged the same way: not by being true, but by being a good enough approximation to be useful.

In practice, the slides list the questions one asks:

- What hypotheses do we want to test with the data?
- How do we judge which model works better? One common method is to split the data into three parts: about 60% for **training** (fitting the models), 20% for **query** or validation (choosing between models), and 20% for **testing** (evaluating the chosen model's performance on data it has never seen).

2.1.3 The heights data

Example 2.1.4 (Pearson's mothers and daughters)

Between 1893 and 1898, Karl Pearson (the slides write E. S. Pearson) and his colleagues collected the heights of mothers and their adult daughters in Britain. The version used in the course (`heights.txt`; `Heights` in the R package `alr4`) has $n = 1375$ pairs. The annotations mention "about 1500 odd" and the restriction to daughters aged at least 18 and mothers under 65.

- `Mheight`: height of the mother (inches), mean 62.45;
- `Dheight`: height of one adult daughter (inches), mean 63.75.

The *object of study* is the question: *how does the daughter's height depend on the height of the mother?*

Heights were rounded to the nearest inch in the original records, so many points coincide. A **jitter plot** moves each data point by a small uniformly distributed amount, here $U(-0.5, 0.5)$ in each coordinate, so that overlapping points become visible.

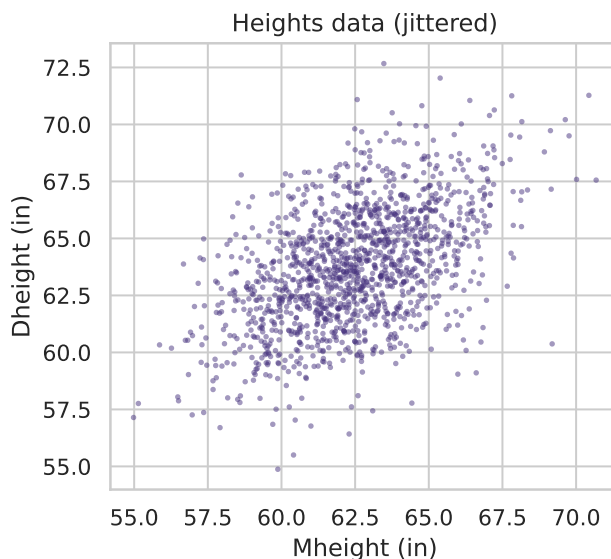


Figure 2.1: Jittered scatter plot of daughter's height against mother's height.

The instructor then restricts attention to the middle of the plot: mothers between about 58 and 68 inches. His annotations point out two features. There are *no unusually tall daughters of short mothers*, and *no short daughters of very tall mothers*. The scatter drifts upwards: taller mothers tend to have taller daughters. The plan is to develop a technique that fits a linear model to the data $\{(x_i, y_i)\}$, that is, to find a_1, a_2 such that

$$y = a_1 + a_2x$$

is a good approximation. The technique is least squares. It is built step by step in Section 2.2. It can also identify outliers: points far from the fitted line.

Note 2.1.5 (Supplementary: the fitted line and “regression to the mean”)

Anticipating Section 2.2, the least-squares line for these data is

$$\widehat{\text{Dheight}} = 29.917 + 0.542 \text{ Mheight}, \quad R^2 = 0.241.$$

The slope $0.54 < 1$ means that a mother who is 1 inch taller than average has a daughter who is on average only about half an inch taller than average. This is Galton's *regression to the mean*. The word “regression” comes from this phenomenon. R^2 is defined in Section 5.3. Only about a quarter of the variation in daughters' heights is explained by mothers' heights.

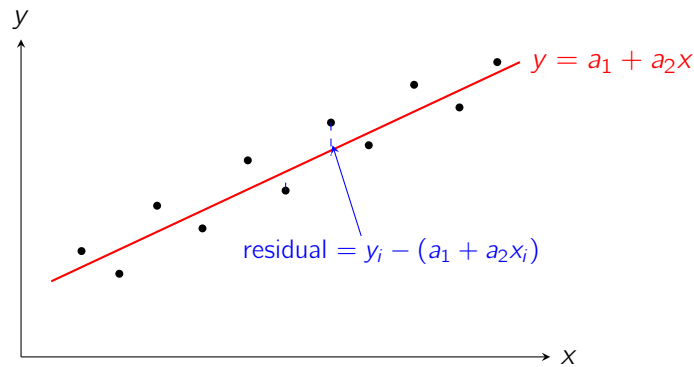


Figure 2.2: The goal: a line $y = a_1 + a_2 x$ that summarises the cloud of points. The vertical gaps are the residuals studied in Sections 2.2 and 2.3.

2.1.4 Computation

The R code reads `heights.txt` and plots `Dheight` against `Mheight`, with jitter. In Python:

```
h = pd.read_csv(DATA / "Heights.csv"); plt.scatter(h.mheight +
rng.uniform(-.5,.5,len(h)), h.dheight + rng.uniform(-.5,.5,len(h)))
```

The notebook `notebooks/week02/L04_intro_linear_models.ipynb` also demonstrates a 60/20/20 split.

2.1.5 Summary

- A model is a low-dimensional summary of how the response y depends on x . It is used for understanding and prediction.
- All models are wrong, but some are useful ($PV = RT$).
- Heights data: 1375 mother–daughter pairs. Jittering reveals overlapping points. Taller mothers tend to have taller daughters.
- Goal: find a_1, a_2 so that $y \approx a_1 + a_2 x$. Models are compared using training/validation/test splits.

2.1.6 Exercises

Problem 2.1.6

Why does jittering by $U(-0.5, 0.5)$ not change any conclusion drawn from the plot, while it would change a fitted line if we fitted to the jittered data? By how much can the mean of the jittered x 's differ from the true mean in expectation?

Problem 2.1.7

Split the heights data at random into 60/20/20 parts (with a fixed seed). Fit the line to the training part and compute the mean squared prediction error on the test part.

Problem 2.1.8

Give another physical law (like $PV = RT$) that is “wrong but useful”, and say in what regime it fails.

Problem 2.1.9

Suppose daughters’ heights were *exactly* equal to their mothers’. What would the scatter plot look like, and what would a_1, a_2 be?

2.1.7 Further reading

Weisberg, *Applied Linear Regression* (4th ed.), Chapter 1 (Scatterplots and Regression), which is the source of the heights, Forbes, bass and snowfall examples. Wickham & Grolemund, *R for Data Science*, “Model basics”. Christensen, Chapter 6 (Regression analysis) for the formal treatment.

2.2 Basics of Linear Models

Lecture: 5

Video: <https://youtu.be/Ec94oyf3bKw>

Timestamps: not available (video transcript not accessed)

Source used: Week-2 slides Feb2.pdf (“Linear Models” slides on the `sim1` data), with annotations;
Week-2 R code

Quiding Question

By the end of this lecture you should be able to:

- describe a family of linear models $y = a_1 + a_2x$ and the residuals of each member;
- explain why the *sum of squared* residuals is the criterion used to choose the best line;
- find the best line numerically (`optim`) and with `lm`;
- derive the least-squares estimates $\hat{a}_2 = S_{xy}/S_{xx}$, $\hat{a}_1 = \bar{y} - \hat{a}_2\bar{x}$.

2.2.1 Recap and motivation

In Section 2.1 we set ourselves the goal of finding a_1, a_2 such that $y \approx a_1 + a_2x$. To see how this works, the instructor uses a simulated data set in which we know there is a linear pattern: `sim1`, from the R package `modelr`.

2.2.2 A family of lines

Example 2.2.1 (The `sim1` data)

`sim1` has $n = 30$ points: three y -values at each of $x = 1, 2, \dots, 10$. The scatter plot shows a clear pattern: a linear relationship. To explore possible models, the instructor generates 250 lines $y = a_1 + a_2x$, with the intercept a_1 chosen uniformly from $[-20, 40]$ and the slope a_2 uniformly from $[-5, 5]$, and plots all 250 lines on top of the data (Fig. 2.3, left). Most of them are terrible. A few pass through the cloud of points.

To say which lines are good, we need a number that measures how far a line is from the data.

Definition 2.2.2 (Residual)

For the model $y = a_1 + a_2x$ and the data point (x_i, y_i) , the **residual** is

$$e_i(a_1, a_2) = y_i - (a_1 + a_2x_i),$$

the vertical distance (with sign) between the data point and the line. It is the part of y_i that the model does not explain.

The slides then choose one line from the 250 (line number 130), plot it, and draw the residuals as vertical segments from each point to the line. A good line should make the residuals small *overall*. The instructor's annotation lists candidate criteria:

1. minimise the sum of distances $\sum_i |e_i|$ ("hard", since $|\cdot|$ is not differentiable);
2. minimise the sum of squared distances $\sum_i e_i^2$: the **least squares** criterion (ticked on the slide).

Definition 2.2.3 (Least squares line)

For data $\{(x_i, y_i)\}_{i=1}^n$ let

$$f(a_1, a_2) = \sum_{i=1}^n (y_i - a_1 - a_2x_i)^2.$$

A pair $(\hat{a}_1, \hat{a}_2)$ minimising f gives the **least squares line** $y = \hat{a}_1 + \hat{a}_2x$.

2.2.3 Finding the best line numerically

The R code on the slides defines the model and a distance, then minimises the distance with a general-purpose optimiser:

```
modell1 <- function(a, data) { a[1] + data$x * a[2] }
measure_distance <- function(mod, data) {
  diff <- data$y - modell1(mod, data)
  sqrt(mean(diff^2)) }
best <- optim(c(0, 0), measure_distance, data = sim1)
best$par          # 4.222248 2.051204
measure_distance(best$par, sim1) # 2.128181
```

The distance used is the **root mean squared error** $\text{RMSE}(a_1, a_2) = \sqrt{\frac{1}{n} \sum_i e_i(a_1, a_2)^2} = \sqrt{f(a_1, a_2)/n}$. The map $t \mapsto \sqrt{t/n}$ is strictly increasing, so minimising RMSE is the same as minimising f . `optim` (Nelder–Mead) returns intercept 4.222248 and slope 2.051204, with RMSE 2.128181. The slide plots this line: intercept ≈ 4.22 , slope ≈ 2.05 . It seems to give a good approximation.

The slide then asks: *how do we judge the model, and is this the best possible solution?* The “to do” written on the slide is to minimise $f(a_1, a_2)$ exactly. R’s built-in function `lm(y ~ x, data = sim1)` does this and returns

$$\hat{a}_1 = 4.220822, \quad \hat{a}_2 = 2.051533.$$

The small difference from `optim` is numerical: Nelder–Mead stops once further progress falls below its tolerance. `lm` solves the problem exactly, as we now show.

2.2.4 The least-squares estimates

Theorem 2.2.4 (Least squares for a straight line)

Let $n \geq 2$ and suppose the x_i are not all equal. Write

$$\bar{x} = \frac{1}{n} \sum x_i, \quad \bar{y} = \frac{1}{n} \sum y_i, \quad S_{xx} = \sum_i (x_i - \bar{x})^2, \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

Then $f(a_1, a_2) = \sum (y_i - a_1 - a_2 x_i)^2$ has a unique minimiser

$$\hat{a}_2 = \frac{S_{xy}}{S_{xx}}, \quad \hat{a}_1 = \bar{y} - \hat{a}_2 \bar{x},$$

and the minimum value is $f(\hat{a}_1, \hat{a}_2) = S_{yy} - S_{xy}^2/S_{xx}$, where $S_{yy} = \sum (y_i - \bar{y})^2$.

Proof. We complete the square twice, so that no calculus is needed. For fixed a_2 write $z_i = y_i - a_2 x_i$. Then $f = \sum_i (z_i - a_1)^2$. By Proposition 1.1.7,

$$\sum_i (z_i - a_1)^2 = \sum_i (z_i - \bar{z})^2 + n(\bar{z} - a_1)^2, \quad \bar{z} = \bar{y} - a_2 \bar{x}.$$

Now $z_i - \bar{z} = (y_i - \bar{y}) - a_2(x_i - \bar{x})$, so

$$\sum_i (z_i - \bar{z})^2 = S_{yy} - 2a_2 S_{xy} + a_2^2 S_{xx} = S_{xx} \left(a_2 - \frac{S_{xy}}{S_{xx}} \right)^2 + S_{yy} - \frac{S_{xy}^2}{S_{xx}},$$

where we used $S_{xx} > 0$ (the x_i are not all equal). Altogether

$$f(a_1, a_2) = S_{yy} - \frac{S_{xy}^2}{S_{xx}} + S_{xx} \left(a_2 - \frac{S_{xy}}{S_{xx}} \right)^2 + n(\bar{y} - a_2 \bar{x} - a_1)^2.$$

The last two terms are non-negative. Both vanish if and only if $a_2 = S_{xy}/S_{xx}$ and $a_1 = \bar{y} - a_2 \bar{x}$. Hence this pair is the unique minimiser, and the minimum is $S_{yy} - S_{xy}^2/S_{xx}$. $\square$

Corollary 2.2.5

The least-squares line passes through $(\bar{x}, \bar{y})$. The residuals $\hat{e}_i = y_i - \hat{a}_1 - \hat{a}_2 x_i$ satisfy $\sum_i \hat{e}_i = 0$ and $\sum_i x_i \hat{e}_i = 0$.

Proof. At $x = \bar{x}$ the line gives $\hat{a}_1 + \hat{a}_2 \bar{x} = \bar{y}$. Next, $\hat{e}_i = (y_i - \bar{y}) - \hat{a}_2(x_i - \bar{x})$. Summing gives $0 - 0 = 0$. Also $\sum_i (x_i - \bar{x}) \hat{e}_i = S_{xy} - \hat{a}_2 S_{xx} = 0$. Since $\sum \hat{e}_i = 0$, it follows that $\sum x_i \hat{e}_i = \sum (x_i - \bar{x}) \hat{e}_i + \bar{x} \sum \hat{e}_i = 0$. $\square$

The two identities $\sum \hat{e}_i = 0$ and $\sum x_i \hat{e}_i = 0$ are exactly the *normal equations* of Section 4.2, written out for a straight line.

Example 2.2.6 (A least-squares line by hand)

Take $n = 4$ points with $x = (1, 2, 3, 4)$ and $y = (2, 3, 5, 6)$.

- $\bar{x} = 2.5$ and $\bar{y} = 16/4 = 4$.
- Deviations: $x_i - \bar{x} = (-1.5, -0.5, 0.5, 1.5)$ and $y_i - \bar{y} = (-2, -1, 1, 2)$.
- $S_{xx} = 2.25 + 0.25 + 0.25 + 2.25 = 5$ and $S_{xy} = 3 + 0.5 + 0.5 + 3 = 7$. Also $S_{yy} = 4 + 1 + 1 + 4 = 10$.
- $\hat{a}_2 = 7/5 = 1.4$ and $\hat{a}_1 = 4 - 1.4 \times 2.5 = 0.5$.
- Fitted values: $0.5 + 1.4x = (1.9, 3.3, 4.7, 6.1)$. Residuals: $(0.1, -0.3, 0.3, -0.1)$. They sum to 0, and $\sum x_i \hat{e}_i = 0.1 - 0.6 + 0.9 - 0.4 = 0$.
- Minimum: $f = 0.01 + 0.09 + 0.09 + 0.01 = 0.2$, which agrees with $S_{yy} - S_{xy}^2/S_{xx} = 10 - 49/5 = 0.2$. The RMSE is $\sqrt{0.2/4} = 0.2236$.

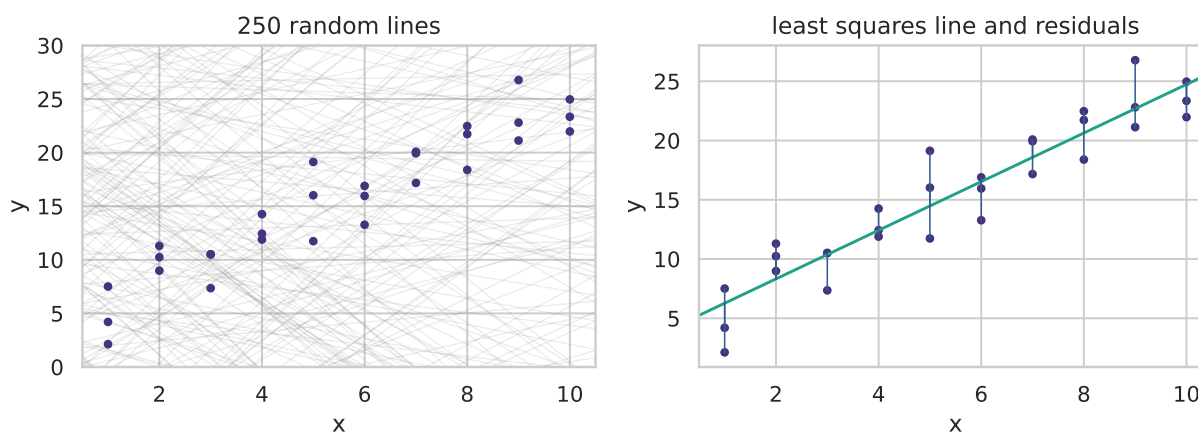


Figure 2.3: Left: 250 random lines over the `sim1` data. Right: the least-squares line and its residuals.

Note 2.2.7 (Supplementary: why squares?)

Squares make f a smooth, convex quadratic, so the minimiser has the closed form of Theorem 2.2.4. The absolute-value criterion has no closed form and may have many minimisers. Squares also lead to the Pythagorean geometry of Sections 4.2 and 4.4. Under normal errors, least squares coincides with maximum likelihood, and the Gauss–Markov theorem (Week 6) shows that it is optimal among linear unbiased estimators. The price is sensitivity to outliers: one large residual counts quadratically.

2.2.5 Computation

R	Python
<code>runif(250,-20,40), runif(250,-5,5)</code>	<code>rng.uniform(-20, 40, 250), rng.uniform(-5, 5, 250)</code>
<code>optim(c(0,0), measure_distance, data=sim1)</code>	<code>scipy.optimize.minimize(rmse, [0,0], method="Nelder-Mead")</code>
<code>lm(y ~ x, data=sim1)</code>	<code>smf.ols("y ~ x", data=sim1).fit().params</code>

The notebook `notebooks/week02/L05_basics_linear_models.ipynb` also computes $\hat{a}_1, \hat{a}_2$ from the formulas of Theorem 2.2.4 and checks them against `statsmodels`.

2.2.6 Summary

- Residual $e_i = y_i - (a_1 + a_2 x_i)$. Criterion: minimise $f(a_1, a_2) = \sum e_i^2$ (least squares). This is equivalent to minimising RMSE.
- $\hat{a}_2 = S_{xy}/S_{xx}$, $\hat{a}_1 = \bar{y} - \hat{a}_2 \bar{x}$, and $\min f = S_{yy} - S_{xy}^2/S_{xx}$.
- The fitted line passes through $(\bar{x}, \bar{y})$, and $\sum \hat{e}_i = \sum x_i \hat{e}_i = 0$.
- `sim1`: `lm` gives 4.220822 and 2.051533 (RMSE 2.128181). `optim` agrees to about three decimals.

2.2.7 Exercises

Problem 2.2.8

Show that $S_{xy} = \sum_i (x_i - \bar{x})y_i = \sum_i x_i y_i - n\bar{x}\bar{y}$.

Problem 2.2.9

Fit the least-squares line by hand to $x = (0, 1, 2)$, $y = (1, 1, 4)$. Give the residuals and the minimum of f .

Problem 2.2.10

Show that if we fit the line through the origin, $y = a_2 x$, the least-squares slope is $\sum x_i y_i / \sum x_i^2$.

Problem 2.2.11

What happens in Theorem 2.2.4 if all x_i are equal? Describe the set of minimisers.

Problem 2.2.12

Minimise $\sum |e_j|$ for `sim1` numerically (Nelder–Mead) and compare the line with the least-squares line.

2.2.8 Further reading

Wickham & Grolemund, *R for Data Science*, “Model basics” (the `sim1` example). Weisberg, *ALR*, Chapter 2 (Simple linear regression). Christensen, Chapter 6. Rao, §4a (least squares).

2.3 More on Linear Models: Residuals

Lecture: 6

Video: https://youtu.be/YlAdFp_3j1M

Timestamps: not available (video transcript not accessed)

Source used: Week-2 slides Feb2.pdf (“Linear Models: best line”, Forbes data, residual plots), with annotations; Week-2 R code

Quiding Question

By the end of this lecture you should be able to:

- fit a least-squares line with `lm` and read off its coefficients;
- draw and interpret a *residual plot*: residuals against the explanatory variable;
- recognise curvature and outliers from a residual plot;
- explain why a transformation (here log) can make a relationship linear.

2.3.1 Recap

In Section 2.2 we defined residuals $e_i = y_i - (a_1 + a_2 x_i)$ and chose the line minimising $\sum e_i^2$. For `sim1`, `lm` returned (4.220822, 2.051533). This lecture applies the method to real data. It also shows that the residuals left over after fitting are the main diagnostic tool: they tell us whether a straight line was the right model in the first place.

2.3.2 Forbes’ boiling-point data

Example 2.3.1 (Forbes, 1857)

In the 1840s and 1850s the Scottish physicist James D. Forbes collected data in the Alps and in Scotland. He wanted to estimate altitude from the *boiling point of water*, which could replace the fragile barometers of the time. The data set (`forbes.txt`; Forbes in `alr4`) has $n = 17$ pairs:

- `bp` = boiling point of water ($^{\circ}\text{F}$), from 194.3 to 212.2;
- `pres` = atmospheric pressure (inches of mercury), from 20.79 to 30.06.

The slides plot Pressure against Temperature and use `lm` to find the best line.

Fitting $\text{pres} = a_1 + a_2 \text{bp}$ by least squares gives

$$\widehat{\text{pres}} = -81.0637 + 0.5229 \text{bp}.$$

In the scatter plot the points lie very close to this line, and $R^2 = 0.9944$. Judged by the scatter plot alone, the fit looks almost perfect.

Definition 2.3.2 (Residual plot)

For a fitted line $\hat{y}_i = \hat{a}_1 + \hat{a}_2 x_i$, the **fitted residuals** are $\hat{e}_i = y_i - \hat{y}_i$. A **residual plot** plots $\hat{e}_i$ against x_i (or against $\hat{y}_i$). If the straight-line model is adequate, the residual plot should look like a structureless horizontal band around 0.

The residual plot for Forbes' data (Fig. 2.4, bottom left) tells a different story. The instructor's annotations mark:

- the *largest positive residual*: case 12 (`bp` = 204.6), with $\hat{e}_{12} = 0.650$, far above all the others;
- the *most negative residual*: case 11 (`bp` = 203.6), with $\hat{e}_{11} = -0.257$;
- a *curved* pattern in the rest. The residuals are positive at both ends (e.g. 0.151, 0.256 at the lowest temperatures and 0.143, 0.166 at the highest) and negative in the middle.

A systematic curve in the residuals means that the straight line is not the right shape. It underestimates at the ends and overestimates in the middle. The residual plot magnifies departures from the line that the scatter plot hides, because it removes the large linear trend.

2.3.3 A transformation

The slides then note that *physical theory* says that $\log(\text{Pressure})$ is linearly related to temperature. We therefore fit

$$\log(\text{pres}) = a_1 + a_2 \text{bp}.$$

With natural logarithms, least squares gives $\widehat{\log(\text{pres})} = -0.97087 + 0.020622 \text{bp}$. The `alr4` file also contains `lpres = 100 \log_{10}(\text{pres})`, which gives $-42.138 + 0.8955 \text{bp}$. The two fits are equivalent up to the constant factor $100/\ln 10$. The residual plot (bottom right) no longer shows curvature: apart from case 12, the

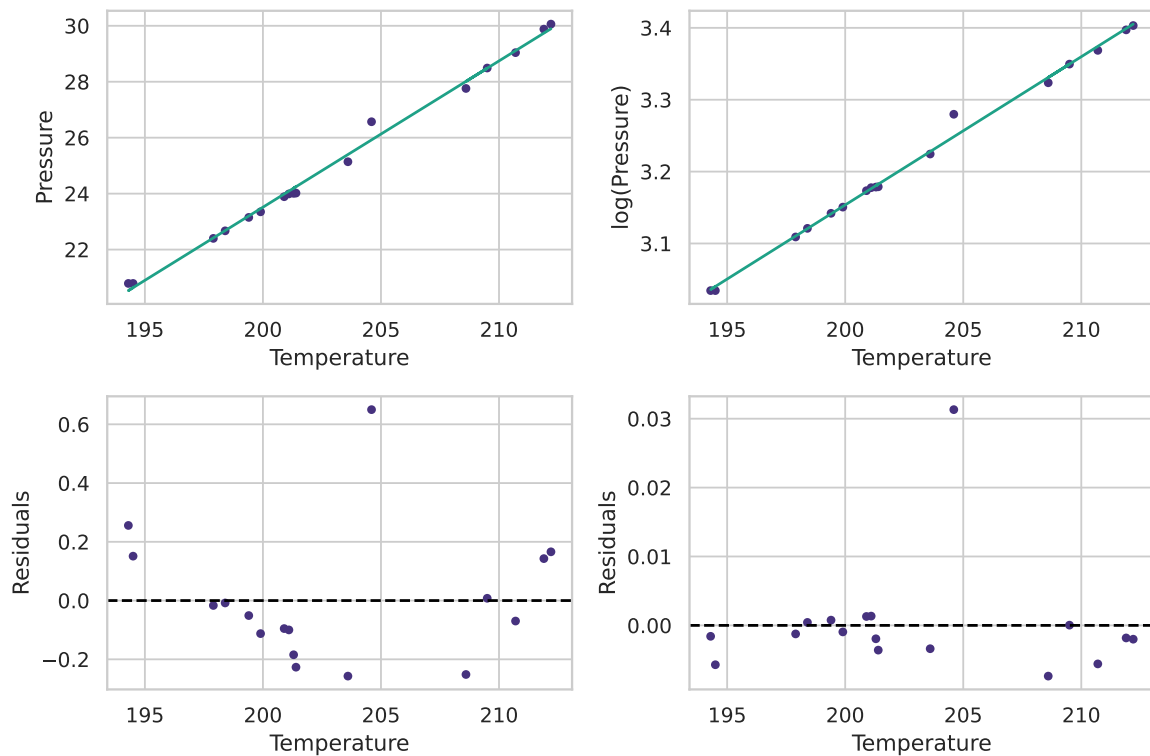


Figure 2.4: Forbes' data. Top: fitted lines for Pressure and for $\log(\text{Pressure})$. Bottom: the corresponding residual plots. Curvature is visible on the left; on the right only case 12 stands out.

residuals form a horizontal band. Case 12, with residual 0.0313 on the log scale, is now clearly an *outlier*. It could be a recording error.

Note 2.3.3 (Supplementary: the physics behind the log)

The Clausius–Clapeyron relation says that the vapour pressure P of a liquid at absolute temperature T satisfies, approximately, $\log P = c - L/(RT)$. Here L is the latent heat and R is the gas constant. Water boils when its vapour pressure equals the atmospheric pressure, so $\log(\text{pres})$ is linear in $1/T$. Over the narrow range 194–212 °F (about 363–373 K), $1/T$ is itself very nearly linear in T . This is why $\log(\text{pres})$ is linear in bp to high accuracy.

Note 2.3.4 (Supplementary: what one outlier does)

Refitting the log model without case 12 gives $\widehat{\log(\text{pres})} = -0.95177 + 0.020519 \text{ bp}$, and R^2 rises from 0.99496 to 0.99957. The slope hardly changes, so the outlier does not distort the line much. Its main effect is to inflate the residual variance. Whether to delete an outlier is a scientific decision, not a statistical one. It should be reported either way.

Proposition 2.3.5 (Supplementary: residuals carry no linear trend)

If a least-squares line is fitted to (x_i, y_i) , then the least-squares line fitted to the residual plot $(x_i, \hat{e}_i)$ is the horizontal line $y = 0$.

Proof. By Theorem 2.2.4, the slope for the data $(x_i, \hat{e}_i)$ is $\sum_i (x_i - \bar{x})(\hat{e}_i - \bar{\hat{e}})/S_{xx}$. By Corollary 2.2.5, $\bar{\hat{e}} = 0$ and $\sum_i (x_i - \bar{x})\hat{e}_i = 0$, so the slope is 0. The intercept is then $\bar{\hat{e}} - 0 \cdot \bar{x} = 0$. $\square$

So any *linear* tendency is removed by the fit. What remains visible in a residual plot is non-linear structure (curvature), unequal spread, or unusual points. These are exactly the features we look for.

2.3.4 Computation

R	Python
<code>m <- lm(pres ~ bp, data = forbes)</code>	<code>m = smf.ols("pres ~ bp", data=fb).fit()</code>
<code>residuals(m)</code>	<code>m.resid</code>
<code>plot(forbes\$bp, residuals(m))</code>	<code>plt.scatter(fb.bp, m.resid)</code>
<code>lm(log(pres) ~ bp, data = forbes)</code>	<code>smf.ols("np.log(pres) ~ bp", data=fb).fit()</code>

See notebooks/week02/L06_residuals.ipynb.

2.3.5 Summary

- Residual plot: $\hat{e}_i$ against x_i . It should be a structureless band around 0.
- Forbes: $\widehat{\text{pres}} = -81.06 + 0.523 \text{ bp}$. The residuals are curved, case 12 has the largest positive residual (0.650) and case 11 the most negative (−0.257).
- $\log(\text{pres})$ is linear in bp (physical theory). After transforming, only the outlier (case 12) remains.
- The fitted residuals have zero mean and zero correlation with x , so only non-linear structure can show up.

2.3.6 Exercises

Problem 2.3.6

Show that $\sum_i \hat{y}_i \hat{e}_i = 0$ for a least-squares line. Deduce that plotting residuals against fitted values also shows no linear trend.

Problem 2.3.7

Refit Forbes' log model without case 12 in Python and confirm the coefficients −0.95177 and 0.020519.

Problem 2.3.8

Show that if $\text{lpres} = 100 \log_{10}(\text{pres})$, the least-squares coefficients for lpres are exactly $100/\ln 10$ times those for $\ln(\text{pres})$.

Problem 2.3.9

Describe the residual plot you would expect if the spread of y around the true line grew with x .

2.3.7 Further reading

Weisberg, *ALR*, Chapter 1 (Forbes' data) and Chapter 9 (outliers and diagnostics). Christensen, Chapter 12 (Model diagnostics).

2.4 Playing with More Data Sets

Lecture: 7

Video: <https://youtu.be/V8SQrXbSA44>

Timestamps: not available (video transcript not accessed)

Source used: Week-2 slides Feb2.pdf (smallmouth bass and snowfall examples), with annotations;
Week-2 data sets `wblake.txt`, `snow.txt`

Quiding Question

By the end of this lecture you should be able to:

- explore a data set with a discrete explanatory variable (age) and compare the mean response at each level with a fitted line;
- explain why a regression of y on x cannot simply be inverted to predict x from y ;
- recognise from a scatter plot and a fitted line when two variables are essentially unrelated.

2.4.1 Recap

We can fit a least-squares line (Section 2.2) and check it with a residual plot (Section 2.3). The instructor now applies the same steps to further data sets from Weisberg's book. The point is to see how scatter plots guide the choice of a model.

2.4.2 Length at age for smallmouth bass

Example 2.4.1 (West Bearskin Lake bass)

The data (`wblake.txt`; `wblake` in `alr4`) record the `Length` (mm) and `Age` (years) of $n = 439$ small-mouth bass caught in West Bearskin Lake, Minnesota, USA, in 1991. Age is determined from annular rings on the fish's scales, like the rings of a tree. Ages run from 1 to 8, with counts

Age	1	2	3	4	5	6	7	8
number of fish	38	72	94	15	68	87	61	4

Because Age is discrete, the scatter plot consists of eight vertical strips. Within each strip there is a spread of lengths. Fish of the same age differ in length. We can summarise each strip by its mean length and compare the eight means with the least-squares line

$$\widehat{\text{Length}} = 65.53 + 30.32 \text{ Age}.$$

The means lie close to the line for young fish but bend below it for the oldest: growth slows with age. The instructor's annotation makes a second point. The graph shows how length depends on age, but *we cannot use the line to predict age from length*. Each strip is wide, and the strips overlap heavily. A fish of length 200 mm could plausibly be 3, 4 or 5 years old. Moreover, inverting the line $y = a_1 + a_2x$ is not the same as regressing x on y .

Proposition 2.4.2 (Supplementary: regression is not symmetric)

Let $\hat{a}_2^{(y|x)} = S_{xy}/S_{xx}$ be the least-squares slope of y on x and $\hat{b}_2^{(x|y)} = S_{xy}/S_{yy}$ the slope of x on y . Then

$$\hat{a}_2^{(y|x)} \cdot \hat{b}_2^{(x|y)} = r^2, \quad r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}.$$

The line of y on x , rewritten as a line for x in terms of y , has slope $1/\hat{a}_2^{(y|x)}$. This equals $\hat{b}_2^{(x|y)}$ only when $r^2 = 1$, i.e. when all points lie exactly on a line.

Proof. Both formulas come from Theorem 2.2.4, with the roles of x and y swapped for the second. Multiplying, $\hat{a}_2\hat{b}_2 = S_{xy}^2/(S_{xx}S_{yy}) = r^2$. The equality $1/\hat{a}_2 = \hat{b}_2$ is equivalent to $\hat{a}_2\hat{b}_2 = 1$, i.e. to $r^2 = 1$. By the Cauchy–Schwarz inequality $r^2 \leq 1$, with equality if and only if the vectors $(y_i - \bar{y})$ and $(x_i - \bar{x})$ are proportional. $\square$

2.4.3 Predicting the weather: snowfall

Example 2.4.3 (Fort Collins snowfall)

Can the snowfall of early winter predict the snowfall of late winter? The data (`snow.txt`; `ftcollinssnow` in `alr4`) give, for 93 years starting in 1900 at Fort Collins, Colorado, the snowfall (inches) from September to December (Early) and from January to June (Late).

The scatter plot of Late against Early is a shapeless cloud. The least-squares line is

$$\widehat{\text{Late}} = 28.64 + 0.2035 \text{ Early},$$

with $R^2 = 0.0258$ and sample correlation $r = 0.161$. The slope is small and, as we will be able to test in Section 5.2, not significantly different from 0 (p-value 0.124). The fitted line is hardly distinguishable from the horizontal line at the mean of Late. The instructor's conclusion, written on the slide: early and late snowfall are *not related*. A good-looking line can always be computed. Whether it means anything is a separate question.

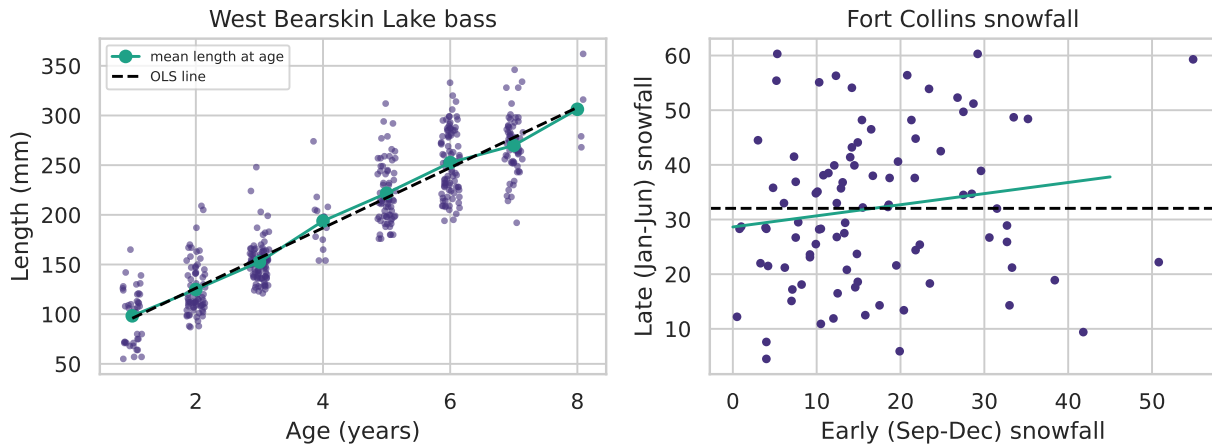


Figure 2.5: Left: bass length against age, with mean length at each age and the least-squares line. Right: late against early snowfall; the fitted line (solid) is close to the mean of Late (dashed).

2.4.4 Computation

The Week-2 R code reads each .txt file, plots it and calls `lm`. In Python:

```
w = pd.read_csv(DATA / "wblake.csv"); w.groupby("Age").Length.mean(); smf.ols("Length ~ Age", w).fit()
sn = pd.read_csv(DATA / "ftcollinssnow.csv"); smf.ols("Late ~ Early", sn).fit().summary()
See notebooks/week02/L07_more_datasets.ipynb.
```

2.4.5 Summary

- Bass: $\widehat{\text{Length}} = 65.53 + 30.32 \text{ Age}$. Mean length grows with age but levels off. Age cannot be predicted reliably from length.
- The regressions of y on x and of x on y differ unless $r^2 = 1$. Their slopes multiply to r^2 .
- Snowfall: slope 0.2035, $R^2 = 0.026$. Early and late snowfall are essentially unrelated.

2.4.6 Exercises

Problem 2.4.4

Compute the least-squares line of Age on Length for the bass data and compare its slope with $1/30.32$.

Problem 2.4.5

Show that $\hat{a}_2 = r s_y / s_x$, where s_x, s_y are the sample standard deviations.

Problem 2.4.6

For the snowfall data, compute $\sum_i (\text{Late}_i - \overline{\text{Late}})^2$ and the minimum of f from Theorem 2.2.4. What fraction of the total is removed by the line?

Problem 2.4.7

Fit a model in which each age has its own mean length (use $C(\text{Age})$ in the formula). Compare its residual sum of squares with that of the straight line. Which is smaller, and why must it be?

2.4.7 Further reading

Weisberg, *ALR*, Chapter 1 (bass and snowfall examples). Christensen, Chapter 6 (Regression) and Chapter 4 (one-way ANOVA, the “one mean per age” model).

Solutions to the Exercises of Chapter 2

Problem 2.1.6. Jitter only affects the picture, and each point moves by less than half an inch, so no visual conclusion changes. A line fitted to jittered data would change slightly, because it is a function of the data. The jitter $U_i \sim U(-0.5, 0.5)$ has mean 0, so the jittered mean $\bar{x} + \bar{U}$ equals the true mean in expectation. Its standard deviation is $\sqrt{1/12}/\sqrt{n} \approx 0.0078$ for $n = 1375$.

Problem 2.1.7. With seed 2024 (see the notebook), the training set has 825 pairs and the test set 275. The training fit is $\widehat{\text{Dheight}} = 32.281 + 0.5052 \text{Mheight}$, and the mean squared prediction error on the test set is 5.08 square inches. The exact numbers depend on the seed.

Problem 2.1.8. Newton's laws of motion: very accurate for everyday speeds, but they fail when speeds approach the speed of light (relativity) or at atomic scales (quantum mechanics). Another example is Hooke's law $F = -kx$, which fails for large extensions of a spring.

Problem 2.1.9. All points would lie on the line $y = x$, with no scatter, so $a_1 = 0$ and $a_2 = 1$. The real data have slope 0.54 and a large scatter, so heights are far from exactly inherited.

Problem 2.2.8. $\sum(x_i - \bar{x})(y_i - \bar{y}) = \sum(x_i - \bar{x})y_i - \bar{y} \sum(x_i - \bar{x}) = \sum(x_i - \bar{x})y_i$, since $\sum(x_i - \bar{x}) = 0$. Also $\sum(x_i - \bar{x})y_i = \sum x_i y_i - \bar{x} \sum y_i = \sum x_i y_i - n\bar{x}\bar{y}$.

Problem 2.2.9. $\bar{x} = 1$, $\bar{y} = 2$, $S_{xx} = 2$, $S_{xy} = (-1)(-1) + 0 \cdot (-1) + 1 \cdot 2 = 3$, $S_{yy} = 6$. So $\hat{a}_2 = 1.5$ and $\hat{a}_1 = 2 - 1.5 = 0.5$. Fitted values $(0.5, 2, 3.5)$, residuals $(0.5, -1, 0.5)$, and $\min f = 0.25 + 1 + 0.25 = 1.5 = S_{yy} - S_{xy}^2/S_{xx} = 6 - 4.5$.

Problem 2.2.10. $g(a) = \sum(y_i - ax_i)^2 = \sum y_i^2 - 2a \sum x_i y_i + a^2 \sum x_i^2$. If $\sum x_i^2 > 0$, complete the square: $g(a) = \sum x_i^2 (a - \sum x_i y_i / \sum x_i^2)^2 + \text{const}$. The minimum is at $a = \sum x_i y_i / \sum x_i^2$.

Problem 2.2.11. If all $x_i = c$, then $f(a_1, a_2) = \sum(y_i - (a_1 + a_2 c))^2$ depends only on $t = a_1 + a_2 c$. It is minimised when $t = \bar{y}$ (Proposition 1.1.7). The minimisers form the whole line $\{(a_1, a_2) : a_1 + a_2 c = \bar{y}\}$, so there is no unique slope. The slope is *not identifiable* from such data. This is a first example of the non-uniqueness studied in Sections 3.1 and 4.3.

Problem 2.2.12. See the notebook. The least-absolute-deviations line is close to the least-squares line for `sim1`, because the data have no outliers. Nelder–Mead may return slightly different answers from different starting points, because the L^1 objective is flat in places.

Problem 2.3.6. $\hat{y}_i = \hat{a}_1 + \hat{a}_2 x_i$, so $\sum \hat{y}_i \hat{e}_i = \hat{a}_1 \sum \hat{e}_i + \hat{a}_2 \sum x_i \hat{e}_i = 0$ by Corollary 2.2.5. Since also $\sum \hat{e}_i = 0$, the argument of Proposition 2.3.5 with x_i replaced by $\hat{y}_i$ shows that the least-squares line through $(\hat{y}_i, \hat{e}_i)$ is $y = 0$.

Problem 2.3.7. In Python, `smf.ols("np.log(pres) ~ bp", fb.drop(11)).fit().params` gives -0.951766 and 0.020519 . Case 12 is row 11 in zero-based indexing.

Problem 2.3.8. $100 \log_{10} p = c \ln p$ with $c = 100/\ln 10$. If $(\hat{a}_1, \hat{a}_2)$ minimises $\sum(\ln p_i - a_1 - a_2 x_i)^2$, then $\sum(c \ln p_i - b_1 - b_2 x_i)^2 = c^2 \sum(\ln p_i - b_1/c - (b_2/c)x_i)^2$ is minimised at $b_j = c\hat{a}_j$. Check: $0.020622 \times 43.4294 = 0.8956$.

Problem 2.3.9. A “fan” or “megaphone” shape: residuals scattered narrowly around 0 for small x and widely for large x . This indicates non-constant variance (heteroscedasticity), which violates the assumption $\text{Var}(\varepsilon_i) = \sigma^2$ introduced in Section 3.2.

Problem 2.4.4. Age on Length gives $\widehat{\text{Age}} = -0.993 + 0.02693 \text{ Length}$. The slope 0.02693 is smaller than $1/30.32 = 0.03298$, as Proposition 2.4.2 predicts: $30.32 \times 0.02693 = r^2 = 0.816$.

Problem 2.4.5. $\hat{a}_2 = S_{xy}/S_{xx} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \sqrt{\frac{S_{yy}}{S_{xx}}} = r \sqrt{\frac{S_{yy}/(n-1)}{S_{xx}/(n-1)}} = r s_y/s_x$.

Problem 2.4.6. $S_{yy} = 17572.41$ and $\min f = 17118.83$. The line removes $1 - 17118.83/17572.41 = 2.58\%$ of the total, which is exactly R^2 .

Problem 2.4.7. The residual sum of squares is 358595.5 for the line and 348491.7 for the separate-means model. The separate-means model is smaller, and it must be. Every straight line $a_1 + a_2 \text{Age}$ assigns one fitted value to each age, so it is a special case of “one mean per age”. Minimising over the larger family can only give a smaller (or equal) minimum. The difference is the lack of fit of the straight line. Testing whether it is significant uses the F -tests of Week 8.

Chapter 3

Linear Models and Estimability

This chapter covers Week 3 (Lectures 8–10): the one-way classification model, the general linear model $(Y, X\beta, \sigma^2 I_n)$, and the characterisation of estimable linear parametric functions. From here on, the source for each lecture is the instructor's handwritten notes. Every step the notes leave as an exercise is proved in full.

3.1 Fundamentals of Linear Models and the Estimation Problem

Lecture: 8

Video: <https://youtu.be/9RbRhz776-k>

Timestamps: not available (video transcript not accessed)

Source used: Week-3 handwritten notes Feb9.pdf, first part ("Model 1", within/between variation, design matrix, identifiability)

Quiding Question

By the end of this lecture you should be able to:

- write down the one-way classification model $Y_{ij} = \mu + \mu_i + \varepsilon_{ij}$ and state the three estimation questions for it;
- define SSW, SSB and TSS, prove $TSS = SSW + SSB$, and state and prove their distributions and independence;
- write the model as $Y = X\beta + \varepsilon$, find $\text{rank}(X)$, and explain why its parameters are *not identifiable*;
- show that contrasts such as $\mu_1 - \mu_2$ are identifiable, and that a constraint like $\sum \mu_i = 0$ makes all parameters identifiable.

3.1.1 Motivation

Week 2 fitted lines to data. From now on we develop the *theory*: when can parameters be estimated, what is the best estimate, and how are its errors distributed? The instructor begins with the model that underlies the analysis of variance.

3.1.2 The one-way classification model

Definition 3.1.1 (Model 1: one-way classification)

There are k populations (groups). From population i we observe n_i values $Y_{i1}, \dots, Y_{in_i}$:

$$Y_{ij} = \mu + \mu_i + \varepsilon_{ij}, \quad 1 \leq i \leq k, \quad 1 \leq j \leq n_i,$$

where $\mu, \mu_1, \dots, \mu_k \in \mathbb{R}$ are unknown and the errors $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ are independent. The total sample size is $n = n_1 + n_2 + \dots + n_k$. Here μ is a general level and μ_i the **effect** of group i . The mean of population i is $\mu + \mu_i$.

The data can be pictured as k columns:

Population 1	Population 2	...	Population k
Y_{11}	Y_{21}		Y_{k1}
$\vdots$	$\vdots$		$\vdots$
Y_{1n_1}	Y_{2n_2}		Y_{kn_k}

Question 3.1.2. The lecture poses three questions.

- (i) Can we estimate μ ? If yes, what is the estimate?
- (ii) Can we estimate $\mu + \mu_i$? If yes, what is the estimate?
- (iii) Can we estimate $\mu_i - \mu_j$ for $1 \leq i \neq j \leq k$? If yes, what is the estimate?

The answers are (i) no, (ii) yes, (iii) yes. This lecture shows why (i) fails and gives unbiased estimates for (iii). Sections 3.3 and 4.3 give the general theory.

3.1.3 Within-group and between-group variation

Write $\bar{Y}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} Y_{ij}$ for the mean of group i and $\bar{Y}_{..} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} Y_{ij}$ for the grand mean.

Definition 3.1.3 (Sums of squares)

• The variance of group i is estimated unbiasedly by the i th sample variance $S_i^2 = \frac{1}{n_i-1} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2$ (for $n_i \geq 2$).

- The **sum of squares within** groups is $SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_i)^2$.
- The **sum of squares between** groups is $SSB = \sum_{i=1}^k n_i (\bar{Y}_i - \bar{Y}_{..})^2$.
- The **total sum of squares** is $TSS = \sum_{i=1}^k \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{..})^2$.

Theorem 3.1.4 (Decomposition of the total sum of squares)

$$\text{TSS} = \text{SSW} + \text{SSB}.$$

Proof. Write $Y_{ij} - \bar{Y}_{..} = (Y_{ij} - \bar{Y}_{i.}) + (\bar{Y}_{i.} - \bar{Y}_{..})$ and square:

$$\text{TSS} = \sum_{i,j} (Y_{ij} - \bar{Y}_{i.})^2 + \sum_{i,j} (\bar{Y}_{i.} - \bar{Y}_{..})^2 + 2 \sum_{i=1}^k (\bar{Y}_{i.} - \bar{Y}_{..}) \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.}).$$

The inner sum in the cross term is $\sum_j Y_{ij} - n_i \bar{Y}_{i.} = 0$ for every i , so the cross term vanishes. The first sum is SSW. In the second the summand does not depend on j , so it equals $\sum_i n_i (\bar{Y}_{i.} - \bar{Y}_{..})^2 = \text{SSB}$. $\square$

The notes then list distributional facts, the first two posed as exercises. To prove them we need two tools about normal vectors.

Lemma 3.1.5 (Quadratic forms in projections)

Let $Z \sim \mathcal{N}(0, I_m)$ (i.e. $Z_1, \dots, Z_m$ i.i.d. $\mathcal{N}(0, 1)$) and let A be an $m \times m$ symmetric matrix with $A^2 = A$ and $\text{rank}(A) = r$. Then $Z^T A Z \sim \chi_r^2$.

Proof. By the spectral theorem, $A = Q \Lambda Q^T$ with Q orthogonal and Λ diagonal. From $A^2 = A$ we get $\Lambda^2 = \Lambda$, so every eigenvalue is 0 or 1. The number of 1's is $\text{rank}(\Lambda) = \text{rank}(A) = r$. Order them first. Put $W = Q^T Z$. W is a linear function of a normal vector, so it is normal, with mean 0 and covariance $Q^T I Q = I$. Hence $W_1, \dots, W_m$ are i.i.d. $\mathcal{N}(0, 1)$, and $Z^T A Z = W^T \Lambda W = W_1^2 + \dots + W_r^2 \sim \chi_r^2$. $\square$

Lemma 3.1.6 (Uncorrelated jointly normal vectors are independent)

If (U, V) is jointly normal and $\text{Cov}(U, V) = 0$ (every component of U is uncorrelated with every component of V), then U and V are independent. Hence so are $g(U)$ and $h(V)$ for any functions g, h .

Proof. Let m_U, m_V be the means and Σ_U, Σ_V the covariance matrices. The joint covariance matrix is block diagonal, $\text{diag}(\Sigma_U, \Sigma_V)$. The moment generating function of a normal vector W with mean m and covariance Σ is $\mathbb{E} e^{t^T W} = \exp(t^T m + \frac{1}{2} t^T \Sigma t)$. For $t = (s, u)$ the block structure gives

$$\mathbb{E} e^{s^T U + u^T V} = \exp(s^T m_U + \frac{1}{2} s^T \Sigma_U s) \exp(u^T m_V + \frac{1}{2} u^T \Sigma_V u) = \mathbb{E} e^{s^T U} \mathbb{E} e^{u^T V}.$$

A joint moment generating function that factorises determines a product distribution, so $U \perp\!\!\!\perp V$. $\square$

Theorem 3.1.7 (Distribution of the sums of squares)

In Model 1:

1. $\text{SSW}/\sigma^2 \sim \chi_{n-k}^2$: SSW has $n - k$ **degrees of freedom**;
2. SSW is independent of SSB;
3. if $\mu_1 = \mu_2 = \dots = \mu_k$, then $\text{SSB}/\sigma^2 \sim \chi_{k-1}^2$ ($k - 1$ degrees of freedom) and $\text{TSS}/\sigma^2 \sim \chi_{n-1}^2$ ($n - 1$ degrees of freedom).

Proof. (1) Fix i and let $\varepsilon_i = (\varepsilon_{i1}, \dots, \varepsilon_{in_i})^\top$. Since $Y_{ij} - \bar{Y}_{i\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}$,

$$\sum_j (Y_{ij} - \bar{Y}_{i\cdot})^2 = \varepsilon_i^\top \left(I_{n_i} - \frac{1}{n_i} \mathbf{1}\mathbf{1}^\top \right) \varepsilon_i.$$

The matrix $C_i = I - \frac{1}{n_i} \mathbf{1}\mathbf{1}^\top$ is symmetric and idempotent: $C_i^2 = I - \frac{2}{n_i} \mathbf{1}\mathbf{1}^\top + \frac{1}{n_i^2} \mathbf{1}(\mathbf{1}^\top \mathbf{1})\mathbf{1}^\top = C_i$. Its rank equals its trace, $n_i - 1$. The trace equals the rank for idempotent matrices, because the eigenvalues are 0 or 1. Applying Lemma 3.1.5 to $Z = \varepsilon_i/\sigma$ gives $\sum_j (Y_{ij} - \bar{Y}_{i\cdot})^2/\sigma^2 \sim \chi_{n_i-1}^2$. The k groups are independent. A sum of independent χ^2 variables is χ^2 with the degrees of freedom added: multiply the moment generating functions $(1 - 2t)^{-(n_i-1)/2}$. So $\text{SSW}/\sigma^2 \sim \chi_{\sum(n_i-1)}^2 = \chi_{n-k}^2$.

(2) SSW is a function of the vector $U = (\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})_{ij}$. SSB is a function of $V = (\bar{Y}_{1\cdot}, \dots, \bar{Y}_{k\cdot})$, because $\bar{Y}_{i\cdot} = \frac{1}{n} \sum n_i \bar{Y}_{i\cdot}$. Both are linear functions of the normal vector ε , so (U, V) is jointly normal. For $i' \neq i$, $\text{Cov}(\bar{Y}_{i'\cdot}, \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}) = 0$ because different groups are independent. For $i' = i$,

$$\text{Cov}(\bar{\varepsilon}_{i\cdot}, \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}) = \text{Cov}(\bar{\varepsilon}_{i\cdot}, \varepsilon_{ij}) - \text{Var}(\bar{\varepsilon}_{i\cdot}) = \frac{\sigma^2}{n_i} - \frac{\sigma^2}{n_i} = 0.$$

By Lemma 3.1.6, $U \perp V$, hence $\text{SSW} \perp \text{SSB}$.

(3) Suppose all μ_i equal a common value, so every group has mean $\vartheta = \mu + \mu_1$. Then $\bar{Y}_{i\cdot} \sim \mathcal{N}(\vartheta, \sigma^2/n_i)$ independently. Put $Z_i = \sqrt{n_i}(\bar{Y}_{i\cdot} - \vartheta)/\sigma$, so $Z = (Z_1, \dots, Z_k)^\top \sim \mathcal{N}(0, I_k)$. Let $u = (\sqrt{n_1}, \dots, \sqrt{n_k})^\top / \sqrt{n}$, a unit vector. By Proposition 1.1.7 (with weights n_i , same proof), $\sum_i n_i (\bar{Y}_{i\cdot} - \vartheta)^2 = \text{SSB} + n(\bar{Y}_{\cdot\cdot} - \vartheta)^2$. Dividing by σ^2 :

$$\|Z\|^2 = \frac{\text{SSB}}{\sigma^2} + (u^\top Z)^2, \quad \text{since } \sqrt{n}(\bar{Y}_{\cdot\cdot} - \vartheta)/\sigma = \sum_i \frac{\sqrt{n_i}}{\sqrt{n}} Z_i = u^\top Z.$$

So $\text{SSB}/\sigma^2 = Z^\top (I_k - uu^\top) Z$. The matrix $I_k - uu^\top$ is symmetric and idempotent with trace $k - 1$, and Lemma 3.1.5 gives χ_{k-1}^2 . Finally $\text{TSS} = \text{SSW} + \text{SSB}$ is a sum of independent χ_{n-k}^2 and χ_{k-1}^2 variables (times σ^2), hence $\text{TSS}/\sigma^2 \sim \chi_{n-1}^2$. $\square$

Note 3.1.8

Degrees of freedom: $n - k$ for SSW, $k - 1$ for SSB, $n - 1$ for TSS, and $(n - k) + (k - 1) = n - 1$. Parts of the handwritten notes are hard to read at this point. The degrees of freedom stated here are the standard ones, and they are the ones the proof gives.

Example 3.1.9 (Sums of squares by hand)

Three groups: group 1 = {5, 7}, group 2 = {8, 10, 12}, group 3 = {4, 6}. So $k = 3$, $n = 7$.

- Group means $\bar{Y}_1 = 6$, $\bar{Y}_2 = 10$, $\bar{Y}_3 = 5$. Grand mean $\bar{Y} = 52/7 = 7.4286$.
- $SSW = [(5-6)^2 + (7-6)^2] + [(8-10)^2 + 0 + (12-10)^2] + [(4-5)^2 + (6-5)^2] = 2 + 8 + 2 = 12$.
- $\bar{Y}_i - \bar{Y} = -10/7, 18/7, -17/7$, so $SSB = \frac{2 \cdot 100 + 3 \cdot 324 + 2 \cdot 289}{49} = \frac{1750}{49} = \frac{250}{7} = 35.714$.
- $TSS = SSW + SSB = 12 + 250/7 = 334/7 = 47.714$. Directly: $\sum Y_{ij}^2 - n\bar{Y}^2 = 434 - 2704/7 = 334/7$. ✓
- Degrees of freedom $n - k = 4$ and $k - 1 = 2$. The ratio $\frac{SSB/2}{SSW/4} = \frac{125/7}{3} = \frac{125}{21} = 5.952$ is the F statistic of Week 8. Under equal means it has the $F_{2,4}$ distribution. Here the p-value is 0.063.

3.1.4 The model in matrix form

Stack the data group by group: $Y = (Y_{11}, \dots, Y_{1n_1}, Y_{21}, \dots, Y_{2n_2}, \dots, Y_{k1}, \dots, Y_{kn_k})^T \in \mathbb{R}^n$, and similarly ε . The parameter vector is $\beta = (\mu, \mu_1, \dots, \mu_k)^T \in \mathbb{R}^{k+1}$ and the **design matrix** is the $n \times (k+1)$ matrix

$$X = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{1}_{n_1} & 0 & \cdots & 0 \\ \mathbf{1}_{n_2} & 0 & \mathbf{1}_{n_2} & \cdots & 0 \\ \vdots & & & \ddots & \\ \mathbf{1}_{n_k} & 0 & 0 & \cdots & \mathbf{1}_{n_k} \end{pmatrix}.$$

Then $Y_{ij} = \mu + \mu_i + \varepsilon_{ij}$ for all i, j is the same as $Y = X\beta + \varepsilon$, and $\mathbb{E}[Y] = X\beta$.

Proposition 3.1.10

If every $n_i \geq 1$, then $\text{rank}(X) = k$, while X has $k+1$ columns.

Proof. Let c_0 be the first column and $c_1, \dots, c_k$ the indicator columns of the groups. Every row has exactly one 1 among $c_1, \dots, c_k$, so $c_0 = c_1 + \dots + c_k$ and $\text{rank}(X) = \text{rank}[c_1, \dots, c_k]$. The columns $c_1, \dots, c_k$ are non-zero ($n_i \geq 1$) and have disjoint supports. If $\sum a_i c_i = 0$, then looking at a row of group i gives $a_i = 0$. So they are linearly independent, and $\text{rank}(X) = k$. $\square$

3.1.5 Identifiability

How many parameters are there? $k+1$ (plus σ^2). Can we estimate all of them? The notes' answer: *no!* Fix any $c \in \mathbb{R}$ and set

$$\tilde{\mu} = \mu + c, \quad \tilde{\mu}_i = \mu_i - c \quad (1 \leq i \leq k).$$

Then $\tilde{\mu} + \tilde{\mu}_i + \varepsilon_{ij} = \mu + \mu_i + \varepsilon_{ij} = Y_{ij}$. The two parameter vectors give *the same* distribution of the data: "the models are the same!" No procedure based on the data can tell μ from $\mu + c$.

Definition 3.1.11 (Identifiability)

A parameter, or a function $\psi(\beta)$ of the parameters, is **identifiable** if different values of it always produce different distributions of the data. In the linear model with $\mathbb{E}[Y] = X\beta$, this means: $X\beta_1 = X\beta_2 \Rightarrow \psi(\beta_1) = \psi(\beta_2)$.

Example 3.1.12 (What is identifiable in Model 1)

• μ is not identifiable: $\beta_1 = (\mu, \mu_1, \dots)$ and $\beta_2 = (\mu + c, \mu_1 - c, \dots)$ have $X\beta_1 = X\beta_2$.

- $\mu + \mu_i$ is identifiable. It is the mean of group i , which is an entry of $X\beta$.
- $\mu_1 - \mu_2$ is identifiable. It equals $(\mu + \mu_1) - (\mu + \mu_2)$, a difference of entries of $X\beta$. Moreover $Y_{11} - Y_{21}$ is an *unbiased* estimate: $\mathbb{E}[Y_{11} - Y_{21}] = (\mu + \mu_1) - (\mu + \mu_2) = \mu_1 - \mu_2$.

The notes ask: is $Y_{11} - Y_{21}$ *optimal*? What is an optimal estimate? This is answered in Week 5–6 (BLUEs, the Gauss–Markov theorem). The next note gives a first comparison.

Note 3.1.13 (Supplementary: a better unbiased estimate of $\mu_1 - \mu_2$)

$\bar{Y}_1 - \bar{Y}_2$ is also unbiased for $\mu_1 - \mu_2$, and $\text{Var}(\bar{Y}_1 - \bar{Y}_2) = \sigma^2(1/n_1 + 1/n_2) \leq 2\sigma^2 = \text{Var}(Y_{11} - Y_{21})$. The inequality is strict as soon as n_1 or n_2 exceeds 1. Using all the data in each group beats using one observation. The Gauss–Markov theorem shows that $\bar{Y}_1 - \bar{Y}_2$ is in fact the best linear unbiased estimate.

The second way out is a *constraint*. Suppose we assume $\sum_{i=1}^k \mu_i = 0$. Then all parameters become identifiable (“Yay!” in the notes).

Proposition 3.1.14

Under the constraint $\sum_{i=1}^k \mu_i = 0$, the map $\beta \mapsto X\beta$ is one-to-one on $\{\beta : \sum \mu_i = 0\}$. Explicitly, if $m_i = \mu + \mu_i$ are the group means, then $\mu = \frac{1}{k} \sum_i m_i$ and $\mu_i = m_i - \frac{1}{k} \sum_l m_l$.

Proof. $X\beta$ determines the group means $m_i = \mu + \mu_i$. Summing, $\sum_i m_i = k\mu + \sum \mu_i = k\mu$, so $\mu = \frac{1}{k} \sum m_i$, and then $\mu_i = m_i - \mu$. Thus β is a function of $X\beta$, and any two constrained β with the same $X\beta$ are equal. $\square$

Note 3.1.15

In either approach, only k parameters (or combinations of parameters) are identifiable: the k group means $\mu + \mu_i$ and everything computable from them. The notes ask *why* k ? Because $\text{rank}(X) = k$ (Proposition 3.1.10). $X\beta$ ranges over a k -dimensional space, so the data can pin down at most k independent linear combinations of β . Section 3.3 makes this precise.

3.1.6 Computation

The notebook `notebooks/week03/L08_oneway_sums_of_squares.ipynb`:

- recomputes Example 3.1.9 and checks it against `statsmodels' anova_lm`;
- builds X for Model 1 and checks $\text{rank}(X) = k$;
- shows non-identifiability numerically ($X\beta_1 = X\beta_2$ for $\beta_2 = \beta_1 + c(1, -1, \dots, -1)$);
- simulates Model 1 many times and compares the histograms of SSW/σ^2 and SSB/σ^2 with the χ^2_{n-k} and χ^2_{k-1} densities, and checks that their correlation is near 0.

3.1.7 Summary

- Model 1: $Y_{ij} = \mu + \mu_i + \varepsilon_{ij}$, ε_{ij} i.i.d. $\mathcal{N}(0, \sigma^2)$, $n = \sum n_i$.
- $\text{TSS} = \text{SSW} + \text{SSB}$, with degrees of freedom $n - 1 = (n - k) + (k - 1)$.
- $\text{SSW}/\sigma^2 \sim \chi^2_{n-k}$ and $\text{SSW} \perp \text{SSB}$. Under equal means, $\text{SSB}/\sigma^2 \sim \chi^2_{k-1}$.
- $Y = X\beta + \varepsilon$ with $\beta \in \mathbb{R}^{k+1}$ and $\text{rank}(X) = k$. The parameters are not identifiable ($\mu \rightarrow \mu + c$, $\mu_i \rightarrow \mu_i - c$).
- Group means $\mu + \mu_i$ and contrasts $\mu_i - \mu_j$ are identifiable. The constraint $\sum \mu_i = 0$ makes everything identifiable.

3.1.8 Exercises

Problem 3.1.16

Show that S_f^2 is an unbiased estimate of σ^2 , and that $\text{SSW}/(n - k)$ is unbiased for σ^2 (without assuming normality).

Problem 3.1.17

For the groups $\{1, 3\}$ and $\{2, 4, 6\}$ compute SSW , SSB and TSS and check Theorem 3.1.4.

Problem 3.1.18

Show that in Model 1 (normality not needed) $\mathbb{E}[\text{SSB}] = (k - 1)\sigma^2 + \sum_i n_i(\mu_i - \bar{\mu}_w)^2$, where $\bar{\mu}_w = \frac{1}{n} \sum_i n_i \mu_i$.

Problem 3.1.19

Prove that the trace of a symmetric idempotent matrix equals its rank.

Problem 3.1.20

Under the constraint $\sum_i n_i \mu_i = 0$ (instead of $\sum_i \mu_i = 0$), express μ and μ_i in terms of the group means.

3.1.9 Further reading

Christensen, §1.2–1.3 (multivariate normal, distribution of quadratic forms), §2.1 (identifiability and estimability) and Chapter 4 (one-way ANOVA). Rao, §3b (quadratic forms in normal variables) and §4a.

3.2 Theory of Linear Models

Lecture: 9

Video: <https://youtu.be/kAKqc2ftwaA>

Timestamps: not available (video transcript not accessed)

Source used: Week-3 handwritten notes Feb9.pdf, part “Theory of linear models” (definition, remark, examples, matrix notation)

Quiding Question

By the end of this lecture you should be able to:

- state the general definition of a linear model and explain why it is called *linear*;
- write the one-way, nested and two-way classification models in this form;
- express any linear model in matrix notation as $(Y, X\beta, \sigma^2 I_n)$ and compute $\mathbb{E}[Y]$ and $\text{Cov}(Y)$.

3.2.1 Recap

Section 3.1 studied one particular model, the one-way classification, and wrote it as $Y = X\beta + \varepsilon$. We now define the general class of models of this kind.

3.2.2 Definition

Definition 3.2.1 (Linear model)

Suppose we have n observations $\{y_1, \dots, y_n\}$ that are realisations of random variables $Y_1, \dots, Y_n$. Let $(\beta_1, \dots, \beta_p)$ be the parameters on which the distribution of the data depends. A **linear model** is

$$Y_i = x_{i1}\beta_1 + x_{i2}\beta_2 + \dots + x_{ip}\beta_p + \varepsilon_i, \quad 1 \leq i \leq n,$$

where $\varepsilon_1, \dots, \varepsilon_n$ are *uncorrelated* random variables with $\mathbb{E}(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$ for $1 \leq i \leq n$, and the x_{ij} are known constants.

(The instructor writes m for the number of parameters. We use p ; see the notation table.)

Remark 3.2.2. *The model can equivalently be described by*

$$\mathbb{E}[Y_i] = \sum_{j=1}^p x_{ij}\beta_j \quad (1 \leq i \leq n), \quad Y_1, \dots, Y_n \text{ uncorrelated with } \text{Var}(Y_i) = \sigma^2.$$

It is called linear because the expected value is linear in the parameters $\beta_1, \dots, \beta_p$. It need not be linear in the explanatory variables.

Note 3.2.3

Normality is *not* part of the definition. Only means, variances and uncorrelatedness are assumed. Everything about least squares and the Gauss–Markov theorem needs only these second-order assumptions. Normality is added later (Week 6, “Assumption of normality”) to obtain exact distributions for tests and confidence intervals, as we already did in Theorem 3.1.7.

Example 3.2.4 (Linear in β , not in x)

The quadratic regression $Y_i = \beta_1 + \beta_2 t_i + \beta_3 t_i^2 + \varepsilon_i$ is a linear model, with $x_{i1} = 1$, $x_{i2} = t_i$, $x_{i3} = t_i^2$. So is $Y_i = \beta_1 + \beta_2 \log t_i + \varepsilon_i$. The model $Y_i = \beta_1 e^{\beta_2 t_i} + \varepsilon_i$ is *not* linear, because β_2 enters non-linearly.

3.2.3 Examples: classification models

Example 3.2.5 (One-way classification)

$Y_{ij} = \mu + \mu_i + \varepsilon_{ij}$, $1 \leq i \leq k$, $1 \leq j \leq n_i$ (Definition 3.1.1). The parameters are $(\mu, \mu_1, \dots, \mu_k)$. For observation (i, j) the coefficient of μ is 1, the coefficient of μ_i is 1, and all others are 0.

Example 3.2.6 (Nested classification)

$$Y_{ijl} = \mu + \mu_i + \delta_{ij} + \varepsilon_{ijl}, \quad 1 \leq l \leq n_{ij}, \quad 1 \leq j \leq m_i, \quad 1 \leq i \leq k.$$

There are k groups. Group i contains m_i subgroups, and n_{ij} observations are taken in subgroup j of group i . For example, k schools, m_i classes in school i , and n_{ij} pupils in class j . δ_{ij} is the effect of subgroup j within group i . The subgroups are nested: class 1 of school 1 has nothing to do with class 1 of school 2. The parameter vector has $1 + k + \sum_i m_i$ entries.

Example 3.2.7 (Two-way classification with interaction)

$$Y_{ijl} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijl}, \quad 1 \leq l \leq n_{ij}, \quad 1 \leq j \leq J, \quad 1 \leq i \leq I.$$

Two factors cross each other: factor A with I levels (effects α_i) and factor B with J levels (effects β_j). γ_{ij} is the *interaction*, the extra effect of the combination (i, j) . The notes label this model “three way classification”. It is the model the Week-5 assignment calls the two-way classification. There are $1 + I + J + IJ$ parameters.

3.2.4 Matrix notation**Definition 3.2.8** (Matrix form)

Put

$$Y = (Y_1, \dots, Y_n)^T, \quad \beta = (\beta_1, \dots, \beta_p)^T, \quad X = (x_{ij})_{1 \leq i \leq n, 1 \leq j \leq p}, \quad \varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^T.$$

Then the linear model reads

$$Y_{n \times 1} = X_{n \times p} \beta_{p \times 1} + \varepsilon_{n \times 1}, \quad \mathbb{E}[\varepsilon] = 0, \quad \text{Cov}(\varepsilon) = \sigma^2 I_n.$$

We denote it by $(Y, X\beta, \sigma^2 I_n)$. The three entries are the observations, their expectation $\mathbb{E}[Y] = X\beta$, and the variance–covariance matrix of the data.

To work with Cov we recall its definition and the two rules used constantly from now on.

Definition 3.2.9 (Mean vector and covariance matrix)

For a random vector $W \in \mathbb{R}^n$, $\mathbb{E}[W] = (\mathbb{E}W_1, \dots, \mathbb{E}W_n)^T$ and $\text{Cov}(W) = \mathbb{E}[(W - \mathbb{E}W)(W - \mathbb{E}W)^T]$, the $n \times n$ matrix with entries $\text{Cov}(W_i, W_j)$.

Proposition 3.2.10

For a random vector $W \in \mathbb{R}^n$, a fixed $q \times n$ matrix A and a fixed $b \in \mathbb{R}^q$:

$$\mathbb{E}[AW + b] = A\mathbb{E}[W] + b, \quad \text{Cov}(AW + b) = A\text{Cov}(W)A^T.$$

In particular, in the linear model $\mathbb{E}[Y] = X\beta$ and $\text{Cov}(Y) = \sigma^2 I_n$. For a fixed vector $\ell \in \mathbb{R}^n$, $\mathbb{E}[\ell^T Y] = \ell^T X\beta$ and $\text{Var}(\ell^T Y) = \sigma^2 \|\ell\|^2$.

Proof. The r th entry of $AW + b$ is $\sum_l a_{rl}W_l + b_r$. Its expectation is $\sum_l a_{rl}\mathbb{E}W_l + b_r$ by linearity of expectation, which is the r th entry of $A\mathbb{E}W + b$. Next, $AW + b - \mathbb{E}[AW + b] = A(W - \mathbb{E}W)$, so

$$\text{Cov}(AW + b) = \mathbb{E}[A(W - \mathbb{E}W)(W - \mathbb{E}W)^T A^T] = A\mathbb{E}[(W - \mathbb{E}W)(W - \mathbb{E}W)^T]A^T = A\text{Cov}(W)A^T,$$

again by linearity (entry by entry). For the linear model take $W = \varepsilon$, $A = I$, $b = X\beta$. Then $Y = \varepsilon + X\beta$ gives $\mathbb{E}Y = X\beta$ and $\text{Cov}(Y) = \text{Cov}(\varepsilon) = \sigma^2 I$. Finally take $A = \ell^T$ ($q = 1$): $\text{Var}(\ell^T Y) = \ell^T (\sigma^2 I) \ell = \sigma^2 \|\ell\|^2$. $\square$

Example 3.2.11 (Matrix form of simple linear regression)

$Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$, $i = 1, \dots, 4$, with $x = (1, 2, 3, 4)$:

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{pmatrix}, \quad X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}, \quad \mathbb{E}[Y] = \begin{pmatrix} \beta_1 + \beta_2 \\ \beta_1 + 2\beta_2 \\ \beta_1 + 3\beta_2 \\ \beta_1 + 4\beta_2 \end{pmatrix}, \quad \text{Cov}(Y) = \sigma^2 I_4.$$

Here $n = 4$, $p = 2$ and $\text{rank}(X) = 2$. The two columns are not proportional.

Example 3.2.12 (Matrix form of a small nested model)

Take $k = 2$ groups, each with $m_i = 2$ subgroups and $n_{ij} = 1$ observation. With $\beta = (\mu, \mu_1, \mu_2, \delta_{11}, \delta_{12}, \delta_{21}, \delta_{22})^T$ and $Y = (Y_{111}, Y_{121}, Y_{211}, Y_{221})^T$:

$$X = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Here $n = 4$, $p = 7$ and $\text{rank}(X) = 4$: the last four columns form I_4 . There are many more parameters than identifiable combinations. Section 4.1 writes out larger examples.

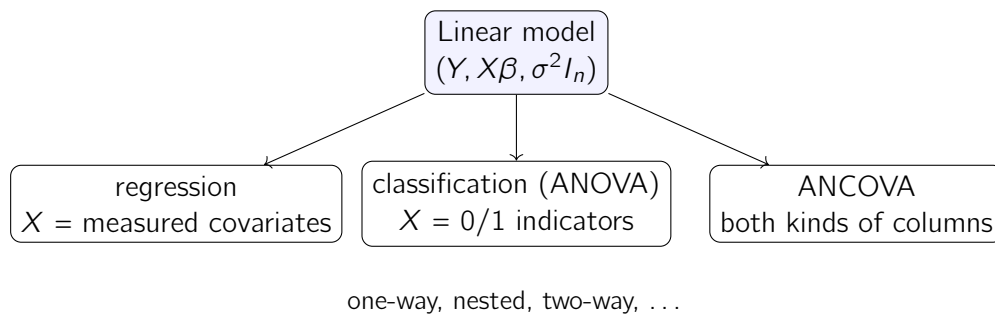


Figure 3.1: All models in this course are special cases of $(Y, X\beta, \sigma^2 I_n)$. They differ only in the design matrix.

3.2.5 Computation

The notebook `notebooks/week03/L09_theory_linear_models.ipynb` builds the design matrices of Examples 3.2.11 and 3.2.12 and of a two-way model with `numpy`, and also with `patsy/statsmodels` formulas. It computes their ranks and checks $\text{Cov}(AY) = A \text{Cov}(Y) A^T$ by simulation.

3.2.6 Summary

- Linear model: $Y_i = \sum_j x_{ij}\beta_j + \varepsilon_i$, with ε_i uncorrelated, $\mathbb{E}\varepsilon_i = 0$, $\text{Var } \varepsilon_i = \sigma^2$. It is linear in the *parameters*.
- Matrix form $(Y, X\beta, \sigma^2 I_n)$: $Y = X\beta + \varepsilon$, $\mathbb{E}Y = X\beta$, $\text{Cov } Y = \sigma^2 I_n$.
- $\mathbb{E}[AW + b] = A\mathbb{E}W + b$ and $\text{Cov}(AW + b) = A \text{Cov}(W)A^T$. In particular $\text{Var}(\ell^T Y) = \sigma^2 \|\ell\|^2$.
- Classification models (one-way, nested, two-way) are linear models with 0/1 design matrices.

3.2.7 Exercises

Problem 3.2.13

Write the design matrix of the two-way model with interaction for $I = J = 2$ and $n_{ij} = 1$. What is its rank?

Problem 3.2.14

Which of the following are linear models? (a) $Y_i = \beta_1 + \beta_2 \sin t_i + \varepsilon_i$; (b) $Y_i = \beta_1 t_i^{\beta_2} + \varepsilon_i$; (c) $Y_i = (\beta_1 + \beta_2)t_i + \varepsilon_i$.

Problem 3.2.15

In the linear model, find $\text{Cov}(\bar{Y}, Y_1 - \bar{Y})$ where $\bar{Y} = \frac{1}{n} \sum Y_i$.

Problem 3.2.16

Let W be a random vector with $\text{Cov}(W) = \Sigma$. Show that Σ is symmetric and positive semi-definite.

3.2.8 Further reading

Christensen, §1.1 (random vectors and matrices) and the introduction to Chapter 2. Rao, §1b (matrices) and §4a.1 (the Gauss–Markoff set-up). Weisberg, *ALR*, Chapter 3.

3.3 The Problem of Parameter Estimation

Lecture: 10

Video: https://youtu.be/TnJ_1LZXLBk

Timestamps: not available (video transcript not accessed)

Source used: Week-3 handwritten notes Feb9.pdf, part “The problem of estimation of the parameters” (linear parametric functions, estimability theorem); Week-5 assignment Q2, Q5, Q6

Quiding Question

By the end of this lecture you should be able to:

- define a *linear parametric function* $\lambda^T \beta$ and say when it is *estimable*;
- prove that $\lambda^T \beta$ is estimable if and only if $\lambda \in \mathcal{C}(X^T)$;
- decide estimability in concrete models, including the one-way classification;
- relate estimability to the null space of X and to identifiability.

3.3.1 Recap

In the linear model $(Y, X\beta, \sigma^2 I_n)$ of Section 3.2, the unknown parameters are $\beta \in \mathbb{R}^p$ and $\sigma^2 > 0$, i.e. $(\beta, \sigma^2) \in \mathbb{R}^p \times \mathbb{R}_+$. In Section 3.1 we saw that in the one-way model some linear combinations of β (such as $\mu_1 - \mu_2$) have unbiased estimates, while others (such as μ) cannot even be identified. This lecture characterises the good combinations.

3.3.2 Linear parametric functions

Definition 3.3.1 (Linear parametric function)

For $\lambda = (\lambda_1, \dots, \lambda_p)^T \in \mathbb{R}^p$, the linear combination

$$\lambda^T \beta = \lambda_1 \beta_1 + \lambda_2 \beta_2 + \dots + \lambda_p \beta_p$$

of the parameter vector is called a **linear parametric function**.

In a linear model we will be interested only in linear parametric functions. The instructor writes $\lambda^T \beta$ with $\lambda \in \mathbb{R}^p$; we write $\lambda^T \beta$ because p denotes the number of parameters in this book.

Definition 3.3.2 (Estimable)

Let $(Y, X\beta, \sigma^2 I_n)$ be a linear model and $\lambda \in \mathbb{R}^p$. The linear parametric function $\lambda^T \beta$ is **estimable** if there exists $\ell \in \mathbb{R}^n$ such that

$$\mathbb{E}[\ell^T Y] = \lambda^T \beta \quad \text{for every } \beta \in \mathbb{R}^p.$$

In other words, $\lambda^T \beta$ is estimable if it has a *linear unbiased estimate* $\ell^T Y$.

Unbiasedness must hold for every value of β . An estimate that is unbiased only for some β 's is useless, because we do not know β .

3.3.3 The estimability theorem

Theorem 3.3.3 (Characterisation of estimability)

Let $(Y, X\beta, \sigma^2 I_n)$ be a linear model and $\lambda \in \mathbb{R}^p$. Then

$$\lambda^\top \beta \text{ is estimable} \iff \lambda \in \mathcal{C}(X^\top),$$

where $\mathcal{C}(X^\top) = \{z \in \mathbb{R}^p : z = X^\top \ell \text{ for some } \ell \in \mathbb{R}^n\}$ is the column space of $X^\top$, which is the row space of X .

Proof. We follow the chain of equivalences in the notes and justify each step.

$$\begin{aligned} \lambda^\top \beta \text{ estimable} &\iff \exists \ell \in \mathbb{R}^n : \mathbb{E}[\ell^\top Y] = \lambda^\top \beta \quad \forall \beta \in \mathbb{R}^p && \text{(definition)} \\ &\iff \exists \ell \in \mathbb{R}^n : \ell^\top X\beta = \lambda^\top \beta \quad \forall \beta \in \mathbb{R}^p && \text{(linearity of expectation)} \\ &\iff \exists \ell \in \mathbb{R}^n : X^\top \ell = \lambda && \text{(linear algebra, below)} \\ &\iff \lambda \in \mathcal{C}(X^\top). && \text{(definition of column space)} \end{aligned}$$

Step 2: $\mathbb{E}[\ell^\top Y] = \ell^\top \mathbb{E}[Y] = \ell^\top X\beta$ by Proposition 3.2.10. Step 3: $\ell^\top X\beta = \lambda^\top \beta$ for all β means $(X^\top \ell - \lambda)^\top \beta = 0$ for all β . Taking $\beta = X^\top \ell - \lambda$ gives $\|X^\top \ell - \lambda\|^2 = 0$, i.e. $X^\top \ell = \lambda$. Conversely, if $X^\top \ell = \lambda$ then $\ell^\top X\beta = (X^\top \ell)^\top \beta = \lambda^\top \beta$ for all β . $\square$

Note 3.3.4

The proof shows more than the statement. The linear unbiased estimates of $\lambda^\top \beta$ are *exactly* the $\ell^\top Y$ with $X^\top \ell = \lambda$ (this is Week-5 assignment Q2). If some ℓ_0 works, then so does $\ell_0 + v$ for every $v \in \mathcal{N}(X^\top)$. So there are usually many linear unbiased estimates, and the question “which one is best?” (Weeks 5–6) makes sense.

Corollary 3.3.5

Every linear parametric function is estimable $\iff \mathcal{C}(X^\top) = \mathbb{R}^p \iff \text{rank}(X) = p$. In that case the parameters $\beta_1, \dots, \beta_p$ themselves are estimable.

Proof. All λ lie in $\mathcal{C}(X^\top)$ iff $\mathcal{C}(X^\top) = \mathbb{R}^p$, i.e. iff $\dim \mathcal{C}(X^\top) = \text{rank}(X) = p$. Then $\beta_j = e_j^\top \beta$ is estimable for each standard basis vector e_j . $\square$

Corollary 3.3.6 (Estimable functions form a subspace)

If $\lambda_1^\top \beta$ and $\lambda_2^\top \beta$ are estimable, so is $(a\lambda_1 + b\lambda_2)^\top \beta$ for all $a, b \in \mathbb{R}$. Each row $x_i^\top \beta = \mathbb{E}Y_i$ of $X\beta$ is estimable (take $\ell = e_i$). Every estimable function is a linear combination of these.

Proof. $\mathcal{C}(X^\top)$ is a subspace spanned by the rows $x_1, \dots, x_n$ of X . If $\lambda_1 = X^\top \ell_1$ and $\lambda_2 = X^\top \ell_2$ then $a\lambda_1 + b\lambda_2 = X^\top (a\ell_1 + b\ell_2)$. $\square$

Proposition 3.3.7 (Week-5 assignment Q5)

$\lambda^\top \beta$ is estimable $\iff \lambda$ is orthogonal to the null space $\mathcal{N}(X) = \{v : Xv = 0\}$.

Proof. By Theorem 3.3.3 we must show $\mathcal{C}(X^\top) = \mathcal{N}(X)^\perp$. ($\subseteq$) If $\lambda = X^\top \ell$ and $Xv = 0$, then $\lambda^\top v = \ell^\top Xv = 0$. ($\supseteq$) Both are subspaces of $\mathbb{R}^p$. $\dim \mathcal{C}(X^\top) = \text{rank}(X) = r$, and by rank-nullity $\dim \mathcal{N}(X) = p - r$, so $\dim \mathcal{N}(X)^\perp = r$. A subspace contained in another of the same finite dimension equals it. $\square$

Note 3.3.8 (Supplementary: estimable $\Rightarrow$ identifiable)

If $\lambda^\top \beta$ is estimable and $X\beta_1 = X\beta_2$, then $\beta_1 - \beta_2 \in \mathcal{N}(X)$, so $\lambda^\top \beta_1 = \lambda^\top \beta_2$ by Proposition 3.3.7. Estimable functions are therefore identifiable in the sense of Definition 3.1.11. Conversely, a linear function that is identifiable (constant on each set $\{\beta : X\beta = \text{const}\}$) must vanish on $\mathcal{N}(X)$, so it is estimable. For linear parametric functions the two notions coincide.

3.3.4 Examples**Example 3.3.9** (One-way model with $k = 2$, $n_1 = n_2 = 2$)

With $\beta = (\mu, \mu_1, \mu_2)^\top$ and $Y = (Y_{11}, Y_{12}, Y_{21}, Y_{22})^\top$,

$$X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathcal{C}(X^\top) = \text{span}\{(1, 1, 0)^\top, (1, 0, 1)^\top\}.$$

- $\mu_1 - \mu_2$: $\lambda = (0, 1, -1)^\top = (1, 1, 0)^\top - (1, 0, 1)^\top \in \mathcal{C}(X^\top)$, so it is *estimable*. Two linear unbiased estimates are $\ell = (1, 0, -1, 0)^\top$, i.e. $Y_{11} - Y_{21}$, and $\ell = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})^\top$, i.e. $\bar{Y}_1 - \bar{Y}_2$. Both satisfy $X^\top \ell = (0, 1, -1)^\top$. Their variances are $\sigma^2 \|\ell\|^2 = 2\sigma^2$ and σ^2 respectively.
- $\mu + \mu_1$: $\lambda = (1, 1, 0)^\top$ is a row of X , so it is *estimable* (e.g. by Y_{11} , or better by $\bar{Y}_1$).
- μ_1 : $\lambda = (0, 1, 0)^\top$. If $a(1, 1, 0) + b(1, 0, 1) = (0, 1, 0)$ then $a + b = 0$, $a = 1$, $b = 0$, a contradiction. So it is *not estimable*. Equivalently, $\mathcal{N}(X) = \text{span}\{(1, -1, -1)^\top\}$ and $\lambda^\top (1, -1, -1)^\top = -1 \neq 0$.
- μ : $\lambda = (1, 0, 0)^\top$ has $\lambda^\top (1, -1, -1)^\top = 1 \neq 0$, so it is *not estimable*.

Example 3.3.10 (General one-way model)

In Model 1 with $\beta = (\mu, \mu_1, \dots, \mu_k)^T$, the rows of X are $(1, e_i^T)$. By Proposition 3.1.10, $\mathcal{N}(X)$ is one-dimensional, spanned by $v = (1, -1, \dots, -1)^T$. Hence, by Proposition 3.3.7,

$$\lambda_0 \mu + \sum_{i=1}^k \lambda_i \mu_i \text{ is estimable} \iff \lambda_0 = \sum_{i=1}^k \lambda_i.$$

Consequences: any **contrast** $\sum \lambda_i \mu_i$ with $\sum \lambda_i = 0$ is estimable (e.g. $\mu_1 - \mu_2$, $\mu_1 + \mu_2 - 2\mu_3$), and so is $\mu + \mu_i$. The functions μ and μ_i alone are not. This answers Question 3.1.2.

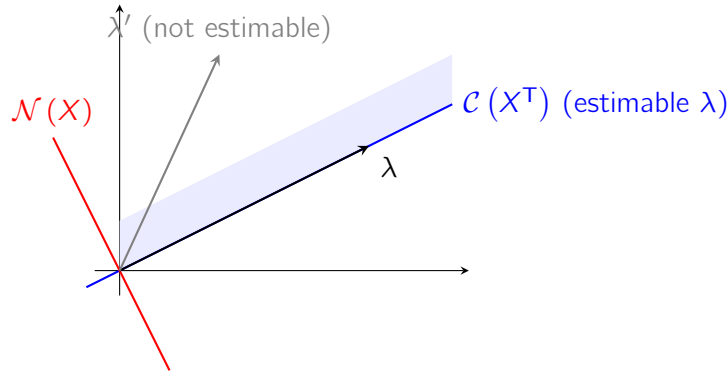


Figure 3.2: $\lambda^T \beta$ is estimable exactly when λ lies in the row space $\mathcal{C}(X^T) = \mathcal{N}(X)^\perp$ (schematic, $p = 2$, $\text{rank } X = 1$).

3.3.5 Computation

To test whether $\lambda \in \mathcal{C}(X^T)$ numerically, compare $\text{rank}(X^T)$ with $\text{rank}([X^T \ \lambda])$, or check whether λ is orthogonal to a basis of $\mathcal{N}(X)$ (`scipy.linalg.null_space`). A third test uses the Moore–Penrose inverse: $\lambda \in \mathcal{C}(X^T) \iff X^T(X^T)^+ \lambda = \lambda$. All three are implemented, and checked on Example 3.3.9, in `notebooks/week03/L10_estimability.ipynb`. The notebook also finds ℓ with $X^T \ell = \lambda$ and verifies unbiasedness by simulation.

3.3.6 Summary

- Linear parametric function: $\lambda^T \beta$. Estimable: $\exists \ell$ with $\mathbb{E}[\ell^T Y] = \lambda^T \beta$ for all β .
- **Theorem:** $\lambda^T \beta$ is estimable $\iff \lambda \in \mathcal{C}(X^T) \iff \lambda \perp \mathcal{N}(X)$. The linear unbiased estimates are $\ell^T Y$ with $X^T \ell = \lambda$.
- If $\text{rank } X = p$, everything is estimable. Estimable functions form the r -dimensional space spanned by the rows of X .
- One-way model: $\lambda_0 \mu + \sum \lambda_i \mu_i$ is estimable iff $\lambda_0 = \sum \lambda_i$.

3.3.7 Exercises

Problem 3.3.11

In the one-way model with $k = 3$, decide which are estimable: μ , $\mu_1 - \mu_2$, $\mu + \mu_1$, $\mu_1 + \mu_2 - 2\mu_3$, $2\mu + \mu_1 + \mu_2$.

Problem 3.3.12

(Week-5 assignment Q6.) Let $k, l \geq 2$ be integers and let X have rows $(1, i, ik)$ for $i = 1, \dots, k$ followed by rows $(1, i, il)$ for $i = 1, \dots, l$, with $\beta = (\beta_1, \beta_2, \beta_3)^T$. (a) Show that β_1 and $\beta_2 + \beta_3(l + k)/2$ are always estimable. (b) Find a necessary and sufficient condition on k, l for every linear function of β to be estimable.

Problem 3.3.13

Show that if $\lambda^T \beta$ is estimable, then any two linear unbiased estimates $\ell_1^T Y, \ell_2^T Y$ differ by $v^T Y$ with $v \in \mathcal{N}(X^T)$, and that $\mathbb{E}[v^T Y] = 0$.

Problem 3.3.14

In simple linear regression $Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$, when is β_2 estimable? Relate your answer to Problem 2.2.11.

Problem 3.3.15

In the nested model of Example 3.2.12, show that $\delta_{11} - \delta_{12}$ is estimable but $\mu_1 - \mu_2$ is not.

3.3.8 Further reading

Christensen, §2.1 (Identifiability and Estimability), especially the result that $\lambda^T \beta$ is estimable iff $\lambda^T = \rho^T X$ for some ρ . Rao, §4a.1 (estimable parametric functions).

Solutions to the Exercises of Chapter 3

Problem 3.1.16. Within group i , $\sum_j (Y_{ij} - \bar{Y}_{i.})^2 = \sum_j (\varepsilon_{ij} - \bar{\varepsilon}_{i.})^2 = \sum_j \varepsilon_{ij}^2 - n_i \bar{\varepsilon}_{i.}^2$. Taking expectations, $\mathbb{E} \sum_j \varepsilon_{ij}^2 = n_i \sigma^2$ and $\mathbb{E}[n_i \bar{\varepsilon}_{i.}^2] = n_i \text{Var}(\bar{\varepsilon}_{i.}) = \sigma^2$. So the expectation is $(n_i - 1)\sigma^2$, and $\mathbb{E} S_i^2 = \sigma^2$. Summing over groups, $\mathbb{E} \text{SSW} = \sum_i (n_i - 1)\sigma^2 = (n - k)\sigma^2$. Only $\mathbb{E} \varepsilon = 0$, $\text{Var} \varepsilon_{ij} = \sigma^2$ and uncorrelatedness were used.

Problem 3.1.17. Means: 2 and 4; grand mean $16/5 = 3.2$. $\text{SSW} = (1 + 1) + (4 + 0 + 4) = 10$. $\text{SSB} = 2(2 - 3.2)^2 + 3(4 - 3.2)^2 = 2.88 + 1.92 = 4.8$. $\text{TSS} = (1 - 3.2)^2 + (3 - 3.2)^2 + (2 - 3.2)^2 + (4 - 3.2)^2 + (6 - 3.2)^2 = 4.84 + 0.04 + 1.44 + 0.64 + 7.84 = 14.8 = 10 + 4.8$. ✓

Problem 3.1.18. Let $m_i = \mu + \mu_i$ and $\bar{m} = \frac{1}{n} \sum n_i m_i = \mu + \bar{\mu}_w$. Write $\bar{Y}_{i.} = m_i + \bar{\epsilon}_{i.}$ and $\bar{Y}_{..} = \bar{m} + \bar{\epsilon}_{..}$. By the weighted version of Proposition 1.1.7, $SSB = \sum n_i \bar{Y}_{i.}^2 - n \bar{Y}_{..}^2$. Now $\mathbb{E} \bar{Y}_{i.}^2 = m_i^2 + \sigma^2/n_i$ and $\mathbb{E} \bar{Y}_{..}^2 = \bar{m}^2 + \sigma^2/n$. Hence $\mathbb{E} SSB = \sum n_i m_i^2 + k\sigma^2 - n\bar{m}^2 - \sigma^2 = (k-1)\sigma^2 + \sum n_i (m_i - \bar{m})^2$, and $m_i - \bar{m} = \mu_i - \bar{\mu}_w$.

Problem 3.1.19. From the proof of Lemma 3.1.5, $A = Q\Lambda Q^T$ with Λ diagonal with 0/1 entries. Then $\text{tr } A = \text{tr}(\Lambda Q^T Q) = \text{tr } \Lambda =$ the number of 1's $= \text{rank } \Lambda = \text{rank } A$.

Problem 3.1.20. $\sum_i n_i m_i = n\mu + \sum n_i \mu_i = n\mu$. So $\mu = \frac{1}{n} \sum n_i m_i$, which is the weighted average of the group means, and $\mu_i = m_i - \mu$.

Problem 3.2.13. Order the observations as (1, 1), (1, 2), (2, 1), (2, 2) and the parameters as $\mu, \alpha_1, \alpha_2, \beta_1, \beta_2, \gamma_{11}, \gamma_{12}, \gamma_{21}, \gamma_{22}$:

$$X = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \text{rank } X = 4$$

(the γ columns form I_4), while $p = 9$.

Problem 3.2.14. (a) Linear, with $x_{i2} = \sin t_i$. (b) Not linear, because β_2 is in the exponent. (c) The mean $(\beta_1 + \beta_2)t_i$ is linear in (β_1, β_2) , so it is formally a linear model with $x_{i1} = x_{i2} = t_i$. The design matrix has two equal columns, and only $\beta_1 + \beta_2$ is estimable.

Problem 3.2.15. $\text{Cov}(\bar{Y}, Y_1) = \frac{1}{n} \sum_l \text{Cov}(Y_l, Y_1) = \sigma^2/n$ and $\text{Var}(\bar{Y}) = \sigma^2/n$, so $\text{Cov}(\bar{Y}, Y_1 - \bar{Y}) = 0$.

Problem 3.2.16. Symmetry: $\text{Cov}(W_i, W_j) = \text{Cov}(W_j, W_i)$. For any $a \in \mathbb{R}^n$, $a^T \Sigma a = \text{Var}(a^T W) \geq 0$ by Proposition 3.2.10.

Problem 3.3.11. Use Example 3.3.10: $\lambda_0 \mu + \sum \lambda_i \mu_i$ is estimable iff $\lambda_0 = \sum \lambda_i$. μ : $1 \neq 0$, not estimable. $\mu_1 - \mu_2$: $0 = 1 - 1$, estimable. $\mu + \mu_1$: $1 = 1$, estimable. $\mu_1 + \mu_2 - 2\mu_3$: $0 = 0$, estimable. $2\mu + \mu_1 + \mu_2$: $2 = 2$, estimable. It is $(\mu + \mu_1) + (\mu + \mu_2)$.

Problem 3.3.12. Each row $(1, i, ik) = (1, 0, 0) + i(0, 1, k)$. Since $k \geq 2$, the first block contains $i = 1, 2$, whose rows span $\{(1, 0, 0), (0, 1, k)\}$. Similarly the second block spans $\{(1, 0, 0), (0, 1, l)\}$. So $\mathcal{C}(X^T) = \text{span}\{(1, 0, 0), (0, 1, k), (0, 1, l)\}$. (a) $(1, 0, 0)^T$ is in it, so β_1 is estimable. Also $(0, 1, \frac{k+l}{2}) = \frac{1}{2}(0, 1, k) + \frac{1}{2}(0, 1, l)$ is in it, so $\beta_2 + \beta_3(k+l)/2$ is estimable, whatever k, l are. (b) $(0, 1, k)$ and $(0, 1, l)$ are linearly independent iff $k \neq l$. So $\text{rank } X = 3$, and every linear function is estimable (Corollary 3.3.5), iff $k \neq l$. If $k = l$, $\text{rank } X = 2$ and e.g. β_3 is not estimable.

Problem 3.3.13. $X^T \ell_1 = \lambda = X^T \ell_2$ gives $X^T(\ell_1 - \ell_2) = 0$, i.e. $v = \ell_1 - \ell_2 \in \mathcal{N}(X^T)$. Then $\mathbb{E}[v^T Y] = v^T X\beta = (X^T v)^T \beta = 0$.

Problem 3.3.14. $X = [\mathbf{1}, x]$. If the x_i are not all equal, $\text{rank } X = 2 = p$ and β_2 is estimable. If all $x_i = c$, the rows are all $(1, c)$, so $\mathcal{C}(X^T) = \text{span}\{(1, c)\}$. Then $(0, 1)$ is not in it, and β_2 is not estimable. This matches Problem 2.2.11, where the least-squares slope was not unique.

Problem 3.3.15. $\lambda = (0, 0, 0, 1, -1, 0, 0)^T$ equals row 1 $-$ row 2 of X , so $\delta_{11} - \delta_{12}$ is estimable (by $Y_{111} - Y_{121}$). For $\mu_1 - \mu_2$, $\lambda = (0, 1, -1, 0, 0, 0, 0)^T$. The vector $v = (0, 1, 0, -1, -1, 0, 0)^T$ satisfies $Xv = 0$: rows 1 and 2 give $1 - 1 = 0$, and rows 3 and 4 give 0. But $\lambda^T v = 1 \neq 0$, so by Proposition 3.3.7 $\mu_1 - \mu_2$ is not estimable. Only combinations such as $(\mu_1 + \frac{1}{2}(\delta_{11} + \delta_{12})) - (\mu_2 + \frac{1}{2}(\delta_{21} + \delta_{22}))$ are estimable.

Chapter 4

Least Squares and the Normal Equations

This chapter covers Week 4 (Lectures 11–14). The instructor’s handwritten notes for this week are divided into “Lecture 2”, “Lecture 3” and “Lecture 4”. We match them to the videos by title: the normal equations (video 12), the uniqueness theorem (video 13), and the projection and matrix review (video 14). The first part of the notes, which recalls the models and classifications, goes with video 11.

4.1 Different Classifications of Linear Models

Lecture: 11

Video: https://youtu.be/pNIF__GOLGo

Timestamps: not available (video transcript not accessed)

Source used: Week-4 handwritten notes Feb14.pdf, first part (“Recall: linear model”, examples 1–3, exercise “express 1, 2, 3 as $(Y, X\beta, \sigma^2 I)$ ”); Week-5 assignment Q3

Quiding Question

By the end of this lecture you should be able to:

- recall the linear model $(Y, X\beta, \sigma^2 I_n)$ and its ingredients;
- write down explicitly the design matrices of the one-way, nested and two-way classification models;
- compute the rank of these design matrices and say how many linear combinations of the parameters can be estimated.

4.1.1 Recall: the linear model

The data are n observations $Y_1, \dots, Y_n$, and the parameters are $\beta_1, \dots, \beta_p$ (the notes write m). The model is

$$Y_i = \sum_{j=1}^p x_{ij}\beta_j + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad \varepsilon_i \text{ uncorrelated},$$

or $Y = X\beta + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$ and $\text{Cov}(\varepsilon) = \sigma^2 I_n$. We write it as $(Y, X\beta, \sigma^2 I_n)$: the data, their expectation, and their variance–covariance matrix (Definition 3.2.8).

The three examples of Section 3.2 are recalled:

1. one-way classification: $Y_{ij} = \mu + \mu_i + \varepsilon_{ij}$, $1 \leq j \leq n_i$, $1 \leq i \leq k$;
2. nested classification: $Y_{ijl} = \mu + \mu_i + \delta_{ij} + \varepsilon_{ijl}$, $1 \leq l \leq n_{ij}$, $1 \leq j \leq m_i$, $1 \leq i \leq k$;
3. two-way classification: $Y_{ijl} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijl}$, $1 \leq l \leq n_{ij}$, $1 \leq j \leq J$, $1 \leq i \leq I$.

The exercise in the notes (also Week-5 assignment Q3) is to express each of them as $(Y, X\beta, \sigma^2 I_n)$. We do this in general and then for small cases.

4.1.2 A general principle

Definition 4.1.1 (Indicator columns)

For a classification with observations labelled by cells, the **indicator column** of a level (or cell) c is the 0/1 vector with a 1 exactly in the rows of observations belonging to c . In all three models, X is obtained by listing the observations in a fixed order (say lexicographic in (i, j, l)) and writing one column for each parameter: $\mathbf{1}_n$ for μ , and the indicator column of the corresponding level or cell for every other parameter.

Indeed, the coefficient of a parameter in the equation for Y_{ijl} is 1 if the parameter appears in that equation and 0 otherwise.

4.1.3 Model 1: one-way classification

Example 4.1.2 ($k = 3$, $n_1 = n_2 = n_3 = 2$)

$Y = (Y_{11}, Y_{12}, Y_{21}, Y_{22}, Y_{31}, Y_{32})^T$ and $\beta = (\mu, \mu_1, \mu_2, \mu_3)^T$:

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \quad n = 6, \quad p = 4, \quad \text{rank } X = 3.$$

By Proposition 3.1.10, $\text{rank } X = k$ in general.

4.1.4 Model 2: nested classification

Example 4.1.3 ($k = 2, m_1 = m_2 = 2, n_{ij} = 2$)

$\beta = (\mu, \mu_1, \mu_2, \delta_{11}, \delta_{12}, \delta_{21}, \delta_{22})^T$ and $Y = (Y_{111}, Y_{112}, Y_{121}, Y_{122}, Y_{211}, Y_{212}, Y_{221}, Y_{222})^T$:

$$X = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad n = 8, \quad p = 7, \quad \text{rank } X = 4.$$

Proposition 4.1.4

In the nested model with every $n_{ij} \geq 1$, $\text{rank } X = \sum_{i=1}^k m_i$, the number of subgroups.

Proof. Let d_{ij} be the indicator column of subgroup (i, j) . These columns are non-zero and have disjoint supports, so, as in the proof of Proposition 3.1.10, they are linearly independent. The remaining columns are sums of them: the column of μ_i is $\sum_{j=1}^{m_i} d_{ij}$ and the column of μ is $\sum_{i,j} d_{ij}$. Hence $\mathcal{C}(X) = \text{span}\{d_{ij}\}$, which has dimension $\sum_i m_i$. $\square$

4.1.5 Model 3: two-way classification**Example 4.1.5** ($I = J = 2, n_{ij} = 2$)

$\beta = (\mu, \alpha_1, \alpha_2, \beta_1, \beta_2, \gamma_{11}, \gamma_{12}, \gamma_{21}, \gamma_{22})^T$ and Y ordered as $Y_{111}, Y_{112}, Y_{121}, Y_{122}, Y_{211}, Y_{212}, Y_{221}, Y_{222}$:

$$X = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad n = 8, \quad p = 9, \quad \text{rank } X = 4.$$

Proposition 4.1.6

In the two-way model with interaction and every $n_{ij} \geq 1$, $\text{rank } X = IJ$, the number of cells. In the additive two-way model $Y_{ijl} = \mu + \alpha_i + \beta_j + \varepsilon_{ijl}$ (no γ_{ij}), with every $n_{ij} \geq 1$, $\text{rank } X = I + J - 1$.

Proof. With interaction: the IJ cell indicators g_{ij} (the columns of the γ_{ij}) are non-zero with disjoint supports, so they are independent. The other columns are sums of them: $\alpha_i \mapsto \sum_j g_{ij}$, $\beta_j \mapsto \sum_i g_{ij}$, $\mu \mapsto \sum_{i,j} g_{ij}$. So $\mathcal{C}(X) = \text{span}\{g_{ij}\}$ has dimension IJ .

Additive model: the columns are $\mathbf{1}$, $a_i = \sum_j g_{ij}$ and $b_j = \sum_i g_{ij}$. Since $\sum_i a_i = \sum_j b_j = \mathbf{1}$, the column space is spanned by $a_1, \dots, a_I, b_1, \dots, b_J$, and there is at least the one relation $\sum a_i - \sum b_j = 0$. So the rank is at most $I + J - 1$. To see that it is exactly $I + J - 1$, suppose $\sum_i s_i a_i + \sum_j t_j b_j = 0$. Every cell is non-empty, so reading the row of an observation in cell (i, j) gives $s_i + t_j = 0$ for all i, j . Hence $s_i = -t_1$ for all i and $t_j = -s_1$ for all j , so all s_i equal some s and all $t_j = -s$. The relations form a 1-dimensional space, so the $I + J$ columns a_i, b_j span a space of dimension $I + J - 1$. $\square$

Note 4.1.7 (How many parameters can we estimate?)

The rank tells us how many linearly independent linear parametric functions are estimable (Corollary 3.3.6): k of the $k + 1$ in Model 1, $\sum m_i$ of the $1 + k + \sum m_i$ in Model 2, and IJ of the $1 + I + J + IJ$ in Model 3. The estimable functions are exactly the combinations of cell means. In the next lectures we find *how* to estimate them: by least squares.

Note 4.1.8 (Supplementary: Kronecker-product form)

With equal replication the design matrices have a compact form. In Model 1 with $n_i \equiv m$: $X = [\mathbf{1}_k \otimes \mathbf{1}_m, I_k \otimes \mathbf{1}_m]$. In Model 3 with $n_{ij} \equiv m$: $X = [\mathbf{1}_I \otimes \mathbf{1}_J \otimes \mathbf{1}_m, I_I \otimes \mathbf{1}_J \otimes \mathbf{1}_m, \mathbf{1}_I \otimes I_J \otimes \mathbf{1}_m, I_I \otimes I_J \otimes \mathbf{1}_m]$. The notebook builds the matrices of this lecture both ways.

4.1.6 Computation

notebooks/week04/L11_classification_designs.ipynb writes a general function that builds the design matrix from factor labels, checks the ranks of Propositions 4.1.4 and 4.1.6, and compares with the full-rank codings that statsmodels formulas produce.

4.1.7 Summary

- Each parameter contributes one column: $\mathbf{1}$ for μ , an indicator column for each level or cell effect.
- Ranks: one-way k ; nested $\sum m_i$; two-way with interaction IJ ; additive two-way $I + J - 1$ (all cells non-empty).
- The rank is the number of linearly independent estimable functions.

4.1.8 Exercises

Problem 4.1.9

Write the design matrix of the additive two-way model with $I = 2$, $J = 3$, $n_{ij} = 1$, and verify that its rank is 4.

Problem 4.1.10

In the two-way model with interaction, $I = J = 2$, suppose cell $(2, 2)$ is empty. What is $\text{rank } X$ now?

Problem 4.1.11

In the nested model, show that $\mu + \mu_i + \delta_{ij}$ is estimable for each subgroup, and give a linear unbiased estimate.

Problem 4.1.12

Show that in the additive two-way model $\alpha_1 - \alpha_2$ is estimable when all cells are non-empty.

4.1.9 Further reading

Christensen, Chapter 4 (one-way ANOVA), Chapter 7 (multifactor ANOVA) and Chapter 8 (nested models); Rao, §4d (analysis of variance, two-way and nested classifications).

4.2 Normal Equations and Existence of Least Squares Estimates

Lecture: 12

Video: <https://youtu.be/Ek-3Ph1AFes>

Timestamps: not available (video transcript not accessed)

Source used: Week-4 handwritten notes Feb14.pdf, part “Least square estimation” and “Lecture 2” (normal equations, proof of the theorem)

Quiding Question

By the end of this lecture you should be able to:

- define a least squares estimate of β and of a linear parametric function;
- derive the *normal equations* $X^T X \beta = X^T Y$ and prove that their solutions are exactly the least squares estimates;
- prove that the normal equations always have a solution, although it need not be unique;
- picture least squares as dropping a perpendicular from Y onto the column space of X .

4.2.1 Recap and motivation

In Week 2 we found the best line by minimising $\sum_i (y_i - a_1 - a_2 x_i)^2$ (Theorem 2.2.4). We now do the same for a general linear model $(Y, X\beta, \sigma^2 I_n)$, where X may have any number of columns and need not

have full rank. Recall that for $z \in \mathbb{R}^n$, $\|z\|^2 = \sum_i z_i^2$ is the squared distance of z from the origin. So $\|Y - X\beta\|^2 = \sum_i (Y_i - \sum_j x_{ij}\beta_j)^2$ is the squared distance between the data and the point $X\beta$.

4.2.2 Least squares estimates

Definition 4.2.1 (Least squares estimate)

Let $(Y, X\beta, \sigma^2 I_n)$ be a linear model. A (random) vector $\hat{\beta} \in \mathbb{R}^p$ is a **least squares estimate** of β if

$$\|Y - X\hat{\beta}\|^2 = \min_{\beta \in \mathbb{R}^p} \|Y - X\beta\|^2.$$

For a linear parametric function $\lambda^T \beta$, the least squares estimate is $\lambda^T \hat{\beta}$. We write $R_0^2 = \min_{\beta} \|Y - X\beta\|^2$ for the minimum. It is called the **residual sum of squares** (RSS or SSE).

Remark 4.2.2. The definition requires that $X\hat{\beta}$ be the point closest to Y among all possible choices of $X\beta$, $\beta \in \mathbb{R}^p$, i.e. among all points of the column space $\mathcal{C}(X)$.

4.2.3 The normal equations

Theorem 4.2.3 (Normal equations)

Consider a linear model $(Y, X\beta, \sigma^2 I_n)$.

1. The **normal equations**

$$X^T X \beta = X^T Y \tag{4.1}$$

always have at least one solution.

2. $\hat{\beta}$ is a least squares estimate of β if and only if it solves (4.1).
3. If $\tilde{\beta}$ is any solution of (4.1), then for every $\beta \in \mathbb{R}^p$

$$\|Y - X\beta\|^2 = \|Y - X\tilde{\beta}\|^2 + \|X(\tilde{\beta} - \beta)\|^2, \tag{4.2}$$

and $R_0^2 = \|Y - X\tilde{\beta}\|^2$.

The remark in the notes: a solution *need not be unique*, but it always *exists*, and that is all we show here. Uniqueness is the subject of Section 4.3. The proof uses the following fact from linear algebra, which is proved in Section 4.4 (Theorem 4.4.6).

Lemma 4.2.4

For every matrix X , $\mathcal{C}(X^T X) = \mathcal{C}(X^T)$.

Proof of Theorem 4.2.3. Existence. First of all, $X^T Y \in \mathcal{C}(X^T) = \mathcal{C}(X^T X)$ by Lemma 4.2.4. By the definition of a column space, there exists $\tilde{\beta} \in \mathbb{R}^p$ with $(X^T X)\tilde{\beta} = X^T Y$.

The identity (4.2). Let $\tilde{\beta}$ solve the normal equations and let $\beta \in \mathbb{R}^p$ be arbitrary. Add and subtract $X\tilde{\beta}$:

$$\|Y - X\beta\|^2 = \|(Y - X\tilde{\beta}) + X(\tilde{\beta} - \beta)\|^2.$$

For $u, w \in \mathbb{R}^n$ we have $\|u + w\|^2 = \sum_i (u_i + w_i)^2 = \|u\|^2 + \|w\|^2 + 2\langle u, w \rangle$, where $\langle u, w \rangle = u^T w = \sum_i u_i w_i$. Hence

$$\|Y - X\beta\|^2 = \|Y - X\tilde{\beta}\|^2 + \|X(\tilde{\beta} - \beta)\|^2 + 2\langle Y - X\tilde{\beta}, X(\tilde{\beta} - \beta) \rangle.$$

We now understand the cross term. It is a scalar, so it equals its own transpose:

$$\langle Y - X\tilde{\beta}, X(\tilde{\beta} - \beta) \rangle = (Y - X\tilde{\beta})^T X(\tilde{\beta} - \beta) = (\tilde{\beta} - \beta)^T X^T (Y - X\tilde{\beta}) = (\tilde{\beta} - \beta)^T (X^T Y - X^T X\tilde{\beta}) = 0,$$

by the choice of $\tilde{\beta}$. This proves (4.2).

Solutions are least squares estimates. From (4.2), $\|Y - X\beta\|^2 \geq \|Y - X\tilde{\beta}\|^2$ for every β , so $\tilde{\beta}$ attains the minimum and $R_0^2 = \|Y - X\tilde{\beta}\|^2$.

Least squares estimates are solutions. Let $\hat{\beta}$ be a least squares estimate and fix a solution $\tilde{\beta}$. Then $\|Y - X\hat{\beta}\|^2 = R_0^2 = \|Y - X\tilde{\beta}\|^2$, and (4.2) with $\beta = \hat{\beta}$ gives $\|X(\tilde{\beta} - \hat{\beta})\|^2 = 0$, i.e. $X\hat{\beta} = X\tilde{\beta}$. Therefore $X^T X\hat{\beta} = X^T X\tilde{\beta} = X^T Y$. $\square$

Corollary 4.2.5

All least squares estimates give the same fitted vector $\hat{Y} = X\hat{\beta}$ and the same residual vector $\hat{e} = Y - X\hat{\beta}$. The residual vector is orthogonal to every column of X : $X^T \hat{e} = 0$. If $\text{rank } X = p$, then $X^T X$ is invertible and $\hat{\beta} = (X^T X)^{-1} X^T Y$ is the unique least squares estimate.

Proof. The first statement was shown at the end of the proof above. $X^T \hat{e} = X^T Y - X^T X\hat{\beta} = 0$ is the normal equations. If $\text{rank } X = p$ then $\text{rank } X^T X = \text{rank } X = p$ by Lemma 4.2.4. The $p \times p$ matrix $X^T X$ is then invertible, and (4.1) has exactly one solution. $\square$

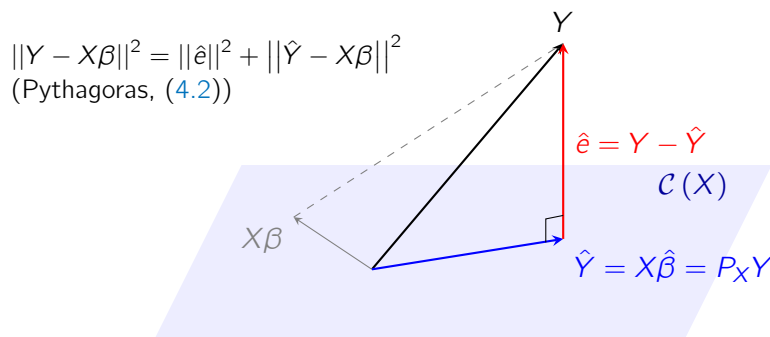


Figure 4.1: Least squares geometry. $X\hat{\beta}$ is the foot of the perpendicular from Y to the plane $\mathcal{C}(X)$. The residual $\hat{e}$ is orthogonal to $\mathcal{C}(X)$ (the normal equations), and every other $X\beta$ is farther from Y .

The name “normal” refers to this picture: the residual is *normal* (perpendicular) to the column space.

Example 4.2.6 (Simple linear regression in matrix form)

Take $x = (1, 2, 3, 4)$, $Y = (2, 3, 5, 6)^T$ and $X = [\mathbf{1}, x]$ as in Example 3.2.11. Then

$$X^T X = \begin{pmatrix} 4 & 10 \\ 10 & 30 \end{pmatrix}, \quad X^T Y = \begin{pmatrix} 16 \\ 47 \end{pmatrix}, \quad \det(X^T X) = 120 - 100 = 20.$$

So

$$\hat{\beta} = (X^T X)^{-1} X^T Y = \frac{1}{20} \begin{pmatrix} 30 & -10 \\ -10 & 4 \end{pmatrix} \begin{pmatrix} 16 \\ 47 \end{pmatrix} = \frac{1}{20} \begin{pmatrix} 480 - 470 \\ -160 + 188 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 1.4 \end{pmatrix},$$

exactly as in Example 2.2.6, with $R_0^2 = 0.2$. Written out, the two normal equations are $\sum_i \hat{e}_i = 0$ and $\sum_i x_i \hat{e}_i = 0$ (Corollary 2.2.5).

Example 4.2.7 (A rank-deficient model: many solutions)

One-way model, $k = 2$, $n_1 = n_2 = 2$, $\beta = (\mu, \mu_1, \mu_2)^T$ and data $Y = (3, 5, 8, 10)^T$. With X from Example 3.3.9:

$$X^T X = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}, \quad X^T Y = \begin{pmatrix} 26 \\ 8 \\ 18 \end{pmatrix}.$$

The normal equations are $4\mu + 2\mu_1 + 2\mu_2 = 26$, $2\mu + 2\mu_1 = 8$, $2\mu + 2\mu_2 = 18$. The first is the sum of the other two, so they reduce to $\mu + \mu_1 = 4$ and $\mu + \mu_2 = 9$. The solution set is the line

$$\hat{\beta}(t) = (t, 4 - t, 9 - t)^T, \quad t \in \mathbb{R}.$$

Every $\hat{\beta}(t)$ gives the same fitted vector $X\hat{\beta}(t) = (4, 4, 9, 9)^T$ (the group means), the same residuals $(-1, 1, -1, 1)^T$, and $R_0^2 = 4$. The solution with the smallest norm minimises $t^2 + (4 - t)^2 + (9 - t)^2$. Setting the derivative to zero gives $3t = 13$, so $t = 13/3$ and $\hat{\beta} = (13/3, -1/3, 14/3)^T$. This is the solution returned by the Moore–Penrose inverse (below).

Note 4.2.8 (Supplementary: solutions via a generalized inverse)

A **generalized inverse** of a matrix A is any matrix A^- with $AA^-A = A$. Every matrix has one; the Moore–Penrose inverse A^+ (`numpy.linalg.pinv`) is a particular choice. Then

$$\hat{\beta} = (X^T X)^- X^T Y$$

solves the normal equations. By Lemma 4.2.4, $X^T Y = X^T X b$ for some b , so $X^T X (X^T X)^- X^T Y = X^T X (X^T X)^- X^T X b = X^T X b = X^T Y$. With A^+ one gets the minimum-norm solution of Example 4.2.7.

4.2.4 Computation

`notebooks/week04/L12_normal_equations.ipynb` solves the normal equations for both examples (`numpy.linalg.solve` for full rank; `pinv` and `lstsq` for rank-deficient X). It checks the Pythagorean

identity (4.2) numerically, shows that different solutions give the same fitted values, and compares with statsmodels OLS.

4.2.5 Summary

- $\hat{\beta}$ is a least squares estimate iff it minimises $\|Y - X\beta\|^2$ iff $X^T X \hat{\beta} = X^T Y$.
- Solutions always exist, because $X^T Y \in \mathcal{C}(X^T) = \mathcal{C}(X^T X)$. They are unique iff $\text{rank } X = p$, in which case $\hat{\beta} = (X^T X)^{-1} X^T Y$.
- Pythagoras: $\|Y - X\beta\|^2 = \|Y - X\hat{\beta}\|^2 + \|X(\hat{\beta} - \beta)\|^2$.
- The fitted values $X\hat{\beta}$ and the residuals are the same for every solution, and $X^T \hat{e} = 0$.

4.2.6 Exercises

Problem 4.2.9

Write the normal equations for $Y_i = \beta_1 + \beta_2 x_i + \varepsilon_i$ in terms of $\sum x_i$, $\sum x_i^2$, $\sum Y_i$, $\sum x_i Y_i$, and solve them to recover Theorem 2.2.4.

Problem 4.2.10

For the one-way model with general k and n_i , show that the normal equations reduce to $\mu + \mu_i = \bar{Y}_{i\cdot}$, $i = 1, \dots, k$, and that $R_0^2 = \text{SSW}$.

Problem 4.2.11

Show that $\sum_i \hat{e}_i = 0$ whenever $\mathbf{1}_n \in \mathcal{C}(X)$.

Problem 4.2.12

Verify that $G = \text{diag}(0, \frac{1}{2}, \frac{1}{2})$ is a generalized inverse of $X^T X$ in Example 4.2.7, and find the corresponding solution $G X^T Y$.

4.2.7 Further reading

Christensen, §2.2 (least squares estimation) and Appendix B (generalized inverses); Rao, §4a.2 (normal equations) and §1b.5 (generalized inverse of a matrix).

4.3 A Theorem on Least Squares Estimates

Lecture: 13

Video: <https://youtu.be/XLT3vSQphCg>

Timestamps: not available (video transcript not accessed)

Source used: Week-4 handwritten notes Feb14.pdf, part “Lecture 3” (estimable $\iff$ least squares estimate unique)

Guiding Question

By the end of this lecture you should be able to:

- state and prove: $\lambda^T \beta$ is estimable if and only if its least squares estimate $\lambda^T \hat{\beta}$ is the same for every solution $\hat{\beta}$ of the normal equations;
- compute the least squares estimate of an estimable function in a rank-deficient model;
- show that the least squares estimate of an estimable function is a *linear unbiased* estimate.

4.3.1 Recall

For the linear model $(Y, X\beta, \sigma^2 I_n)$ and $\lambda \in \mathbb{R}^p$, the linear parametric function $\lambda^T \beta$ is estimable if there is $\ell \in \mathbb{R}^n$ with $\mathbb{E}[\ell^T Y] = \lambda^T \beta$ for all β , i.e. if $\ell^T Y$ is an unbiased estimate of $\lambda^T \beta$. By Theorem 3.3.3 this happens iff $\lambda \in \mathcal{C}(X^T)$. In Section 4.2 we saw that the normal equations may have many solutions (Example 4.2.7). So the “least squares estimate” $\lambda^T \hat{\beta}$ might depend on which solution we pick. The theorem of this lecture says exactly when it does not.

4.3.2 The theorem

Theorem 4.3.1

Let $(Y, X\beta, \sigma^2 I_n)$ be a linear model and $\lambda \in \mathbb{R}^p$. Then $\lambda^T \beta$ is estimable if and only if its least squares estimate is unique, i.e. $\lambda^T \hat{\beta}_1 = \lambda^T \hat{\beta}_2$ for any two solutions $\hat{\beta}_1, \hat{\beta}_2$ of the normal equations $X^T X \beta = X^T Y$.

Proof. ($\Rightarrow$) Suppose $\lambda^T \beta$ is estimable. Then $\lambda \in \mathcal{C}(X^T) = \mathcal{C}(X^T X)$ (recall the earlier theorem, Theorem 3.3.3, and Lemma 4.2.4). So there exists $b \in \mathbb{R}^p$ with $\lambda = X^T X b$. Let $\hat{\beta}$ be any solution of the normal equations. Then

$$\lambda^T \hat{\beta} = (X^T X b)^T \hat{\beta} = b^T X^T X \hat{\beta} = b^T X^T Y,$$

which does not depend on the choice of $\hat{\beta}$. Hence the least squares estimate of $\lambda^T \beta$ is unique.

($\Leftarrow$) Suppose $\lambda^T \hat{\beta}$ is the same for all solutions $\hat{\beta}$. Take one solution $\hat{\beta}_1$ and any $v \in \mathcal{N}(X^T X)$, i.e. $X^T X v = 0$. Then $\hat{\beta}_2 = \hat{\beta}_1 + v$ is also a solution: $X^T X \hat{\beta}_2 = X^T X \hat{\beta}_1 + 0 = X^T Y$. By assumption $\lambda^T \hat{\beta}_2 = \lambda^T \hat{\beta}_1$, i.e. $\lambda^T v = 0$. So λ is orthogonal to $\mathcal{N}(X^T X)$. For the symmetric matrix $A = X^T X$,

$$\mathcal{N}(A)^\perp = \mathcal{C}(A^T) = \mathcal{C}(A)$$

(the proof of Proposition 3.3.7 applied to A). Hence $\lambda \in \mathcal{C}(X^T X) = \mathcal{C}(X^T)$ (Lemma 4.2.4), and $\lambda^T \beta$ is estimable by Theorem 3.3.3. $\square$

Note 4.3.2

The set of all solutions of the normal equations is $\hat{\beta}_1 + \mathcal{N}(X^T X)$, and $\mathcal{N}(X^T X) = \mathcal{N}(X)$ (Theorem 4.4.6). The theorem therefore says: $\lambda^T \hat{\beta}$ is constant on this affine set iff $\lambda \perp \mathcal{N}(X)$. This is the condition of Proposition 3.3.7, now seen from the side of the estimates rather than the side of the expectations.

Proposition 4.3.3 (Least squares estimates of estimable functions are unbiased)

If $\lambda^T \beta$ is estimable, with $\lambda = X^T \ell$, then its least squares estimate satisfies

$$\lambda^T \hat{\beta} = \ell^T X \hat{\beta} = \ell^T \hat{Y},$$

which is a linear function of Y , and $\mathbb{E}[\lambda^T \hat{\beta}] = \lambda^T \beta$.

Proof. $\lambda^T \hat{\beta} = \ell^T X \hat{\beta}$ holds for every solution. By Corollary 4.2.5, $X \hat{\beta} = \hat{Y}$ is the same for all solutions. In the proof of Theorem 4.3.1 we wrote $\lambda^T \hat{\beta} = b^T X^T Y$, which is linear in Y . Then $\mathbb{E}[b^T X^T Y] = b^T X^T X \beta = (X^T X b)^T \beta = \lambda^T \beta$. $\square$

Thus, for an estimable $\lambda^T \beta$, $\lambda^T \hat{\beta}$ is one of the linear unbiased estimates of Section 3.3. Week 6 (the Gauss–Markov theorem) shows that it is the *best* one.

4.3.3 Examples

Example 4.3.4 (Continuing Example 4.2.7)

The solutions are $\hat{\beta}(t) = (t, 4 - t, 9 - t)^T$, $t \in \mathbb{R}$, and $\mathcal{N}(X) = \text{span}\{(1, -1, -1)^T\}$.

- $\mu_1 - \mu_2$, $\lambda = (0, 1, -1)^T$: $\lambda^T \hat{\beta}(t) = (4 - t) - (9 - t) = -5$ for every t . It is unique, as the theorem predicts for an estimable function. The estimate is $\bar{Y}_1 - \bar{Y}_2 = 4 - 9$.
- $\mu + \mu_2$, $\lambda = (1, 0, 1)^T$: $t + 9 - t = 9 = \bar{Y}_2$ for every t .
- μ_1 , $\lambda = (0, 1, 0)^T$: $\lambda^T \hat{\beta}(t) = 4 - t$ depends on t . It is not unique, so μ_1 is not estimable.
- μ : $\hat{\mu} = t$ can be any real number. With the constraint $\mu_1 + \mu_2 = 0$ of Proposition 3.1.14, we would pick t with $(4 - t) + (9 - t) = 0$, i.e. $t = 6.5$, so $\hat{\beta} = (6.5, -2.5, 2.5)^T$.

Example 4.3.5 (A regression design with a redundant column)

Let $Y_i = \beta_1 + \beta_2 x_i + \beta_3 z_i + \varepsilon_i$ with $z_i = 2x_i$ for all i (for example, the same length recorded in two units). Then $\mathcal{N}(X) = \text{span}\{(0, 2, -1)^T\}$. The function $\beta_2 + 2\beta_3$ (the total effect of x) is estimable, since $(0, 1, 2) \cdot (0, 2, -1) = 0$. The separate slopes β_2 and β_3 are not. Software fitting this model must either drop a column or report a warning: `statsmodels` uses the Moore–Penrose solution and flags a singular design. The notebook shows this with the data of Example 4.2.6.

4.3.4 Computation

`notebooks/week04/L13_uniqueness_of_lse.ipynb` generates many solutions $\hat{\beta} + v$, $v \in \mathcal{N}(X)$, of the normal equations and plots $\lambda^T \hat{\beta}$ for estimable and non-estimable λ (constant versus varying). It also verifies Proposition 4.3.3 by simulation.

4.3.5 Summary

- **Theorem:** $\lambda^T \beta$ is estimable $\iff \lambda^T \hat{\beta}$ is the same for all solutions of $X^T X \beta = X^T Y$.
- If $\lambda = X^T X b$, then $\lambda^T \hat{\beta} = b^T X^T Y$. If $\lambda = X^T \ell$, then $\lambda^T \hat{\beta} = \ell^T \hat{Y}$. It is linear in Y and unbiased.
- Non-estimable functions (like μ or μ_i in the one-way model) have least squares “estimates” that depend on an arbitrary choice.

4.3.6 Exercises

Problem 4.3.6

In the one-way model with general k , show that the least squares estimate of an estimable contrast $\sum_i c_i \mu_i$ ($\sum c_i = 0$) is $\sum_i c_i \bar{Y}_{i..}$.

Problem 4.3.7

Show that $\text{Var}(\lambda^T \hat{\beta}) = \sigma^2 \lambda^T (X^T X)^- \lambda$ for estimable $\lambda^T \beta$ and any generalized inverse, and that the value does not depend on the choice of $(X^T X)^-$.

Problem 4.3.8

In Example 4.3.5, find all solutions of the normal equations for $x = (1, 2, 3, 4)$, $z = 2x$, $Y = (2, 3, 5, 6)^T$, and the least squares estimate of $\beta_2 + 2\beta_3$.

Problem 4.3.9

Give an example where $\lambda^T \hat{\beta}$ is the same for two particular solutions but $\lambda^T \beta$ is not estimable. Why does this not contradict the theorem?

4.3.7 Further reading

Christensen, §2.2 (Theorem 2.2.3 and the discussion of estimable functions and least squares). Rao, §4a.2 (least squares estimators of estimable functions are unique).

4.4 Fundamentals of Matrices and the Projection P_X

Lecture: 14

Video: https://youtu.be/K_j3NHsQ4t8

Timestamps: not available (video transcript not accessed)

Source used: Week-4 handwritten notes Feb14.pdf, part “Lecture 4” (proposition on $X\beta$ and P_X ; linear algebra review of the four subspaces, $\mathcal{N}(A) = \mathcal{N}(A^T A)$, rank–nullity)

Guiding Question

By the end of this lecture you should be able to:

- show that $X\beta$ is estimable and that its least squares estimate is $X\hat{\beta} = P_X Y$, the orthogonal projection of Y onto $\mathcal{C}(X)$;
- work with the four fundamental subspaces of a matrix, and compute them for small examples;
- prove $\mathcal{N}(A) = \mathcal{N}(A^T A)$ and $\mathcal{C}(A^T A) = \mathcal{C}(A^T)$ using the rank–nullity theorem;
- construct P_X and list its properties.

4.4.1 Recap

Least squares estimates are the solutions of $X^T X \beta = X^T Y$ (Theorem 4.2.3). The estimate of $\lambda^T \beta$ is unique exactly when $\lambda^T \beta$ is estimable (Theorem 4.3.1). Both proofs used $\mathcal{C}(X^T X) = \mathcal{C}(X^T)$ (Lemma 4.2.4). This lecture proves it and identifies the fitted vector geometrically.

4.4.2 $X\beta$ and the orthogonal projection

Definition 4.4.1 (Orthogonal projection)

Let $V \subseteq \mathbb{R}^n$ be a subspace. A vector $w \in V$ is the **orthogonal projection** of $y \in \mathbb{R}^n$ onto V if $y - w \perp V$, i.e. $\langle y - w, v \rangle = 0$ for all $v \in V$. The matrix P_V with $P_V y =$ (projection of y) is the **orthogonal projection matrix** onto V . For $V = \mathcal{C}(X)$ we write P_X .

Lemma 4.4.2 (Projection theorem)

For every subspace V and every $y \in \mathbb{R}^n$ the orthogonal projection w exists and is unique. It is the unique closest point of V to y : $\|y - v\|^2 = \|y - w\|^2 + \|w - v\|^2$ for all $v \in V$. The map $y \mapsto w$ is linear, given by the matrix $P_V = \sum_{i=1}^r u_i u_i^T$ for any orthonormal basis $u_1, \dots, u_r$ of V . The matrix is symmetric and idempotent ($P_V^T = P_V = P_V^2$), with $\mathcal{C}(P_V) = V$ and $\text{rank } P_V = \text{tr } P_V = \dim V$.

Proof. Choose an orthonormal basis $u_1, \dots, u_r$ of V (Gram–Schmidt) and put $w = \sum_i (u_i^T y) u_i \in V$. For each j , $u_j^T (y - w) = u_j^T y - \sum_i (u_i^T y) u_j^T u_i = u_j^T y - u_j^T y = 0$. So $y - w$ is orthogonal to every basis vector, hence to V . *Uniqueness:* if $w, w' \in V$ both have $y - w, y - w' \perp V$, then $w - w' = (y - w') - (y - w) \in V \cap V^\perp$. So $\|w - w'\|^2 = \langle w - w', w - w' \rangle = 0$. *Closest point:* for $v \in V$, $y - v = (y - w) + (w - v)$ with $w - v \in V \perp y - w$, and expanding the square gives the identity. *Matrix:* $w = \sum_i u_i (u_i^T y) = (\sum_i u_i u_i^T) y$, so $P_V = \sum_i u_i u_i^T$. It is clearly symmetric, and $P_V^2 = \sum_{i,j} u_i (u_i^T u_j) u_j^T = \sum_i u_i u_i^T = P_V$. Its columns lie in V , and $P_V u_j = u_j$, so $\mathcal{C}(P_V) = V$. Finally, $\text{tr } P_V = \sum_i \text{tr}(u_i u_i^T) = \sum_i u_i^T u_i = r$. $\square$

Proposition 4.4.3

In the linear model $(Y, X\beta, \sigma^2 I_n)$, $X\beta$ is estimable (every component is), and its least squares estimate is

$$X\hat{\beta} = P_X Y.$$

Proof. $X\beta = (x_1^T \beta, \dots, x_n^T \beta)^T$, where x_i^T is the i th row of X . Since $x_i = X^T e_i \in \mathcal{C}(X^T)$, every component is estimable (Theorem 3.3.3). (The notes leave this as an exercise; $\mathbb{E}[Y_i] = x_i^T \beta$ shows it directly.) Let $\hat{\beta}$ solve the normal equations. By Theorem 4.2.3,

$$R_0^2 = \|Y - X\hat{\beta}\|^2 = \min_{\beta \in \mathbb{R}^p} \|Y - X\beta\|^2 = \min_{v \in \mathcal{C}(X)} \|Y - v\|^2.$$

Since $X\hat{\beta} \in \mathcal{C}(X)$, it is a closest point of $\mathcal{C}(X)$ to Y . By Lemma 4.4.2 the closest point is unique and equals $P_X Y$. Hence $X\hat{\beta} = P_X Y$ and $R_0^2 = \|Y - P_X Y\|^2$. (Alternatively: $X\hat{\beta} \in \mathcal{C}(X)$, and $Y - X\hat{\beta} \perp \mathcal{C}(X)$ because $X^T(Y - X\hat{\beta}) = 0$; this is the definition of the projection.) $\square$

4.4.3 Linear algebra review: the four fundamental subspaces

For an $m \times n$ matrix A (a map $\mathbb{R}^n \rightarrow \mathbb{R}^m$, $x \mapsto Ax$):

- **column space** $\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} = \{b \in \mathbb{R}^m : Ax = b \text{ has a solution}\} \subseteq \mathbb{R}^m$;
- **null space** $\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n$;
- **row space** $\mathcal{R}(A) = \mathcal{C}(A^T) \subseteq \mathbb{R}^n$;
- **left null space** $\mathcal{N}(A^T) \subseteq \mathbb{R}^m$.

The **rank** of A is $\dim \mathcal{C}(A) = \dim \mathcal{R}(A)$ (the number of linearly independent columns, equal to the number of independent rows). The **nullity** is $\dim \mathcal{N}(A)$. The row space and the null space are orthogonal complements in $\mathbb{R}^n$, and $\mathcal{C}(A)$ and $\mathcal{N}(A^T)$ are orthogonal complements in $\mathbb{R}^m$.

Theorem 4.4.4 (Rank–nullity)

For an $m \times n$ matrix A , $\text{rank}(A) + \dim \mathcal{N}(A) = n$.

Proof. Let $v_1, \dots, v_s$ be a basis of $\mathcal{N}(A)$ and extend it to a basis $v_1, \dots, v_n$ of $\mathbb{R}^n$. Every Ax is a combination of $Av_{s+1}, \dots, Av_n$, because $Av_1 = \dots = Av_s = 0$. These vectors are independent: if $A \sum_{i>s} c_i v_i = 0$, then $\sum_{i>s} c_i v_i \in \mathcal{N}(A) = \text{span}\{v_1, \dots, v_s\}$, which forces all $c_i = 0$ by the independence of $v_1, \dots, v_n$. So $\text{rank } A = n - s$. $\square$

Example 4.4.5 (The four subspaces of a rank-one matrix)

The example on the board is only partly legible in the published notes. It is a 2×3 matrix whose column space is $\{b : b_2 = 2b_1\}$. We take

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \end{pmatrix}.$$

- $\mathcal{C}(A)$: every column is a multiple of $(1, 2)^T$, so $\mathcal{C}(A) = \text{span}\{(1, 2)^T\}$. By row reduction, $Ax = b$ is solvable iff $R_2 - 2R_1$ gives $0 = b_2 - 2b_1$, i.e. iff $b_2 = 2b_1$ (e.g. $b = (1, 2)^T$). The rank is 1.
- $\mathcal{R}(A) = \mathcal{C}(A^T) = \text{span}\{(1, 2, 1)^T\}$, a line in $\mathbb{R}^3$.
- $\mathcal{N}(A) = \{x : x_1 + 2x_2 + x_3 = 0\} = \text{span}\{(-2, 1, 0)^T, (-1, 0, 1)^T\}$, a plane in $\mathbb{R}^3$ orthogonal to $(1, 2, 1)^T$.
- $\mathcal{N}(A^T) = \{y : y_1 + 2y_2 = 0\} = \text{span}\{(2, -1)^T\}$, orthogonal to $\mathcal{C}(A)$ in $\mathbb{R}^2$.
- Rank–nullity: $1 + 2 = 3 = n$.

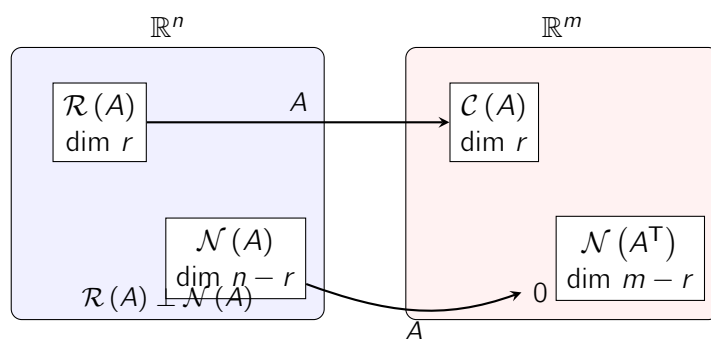


Figure 4.2: The four fundamental subspaces. A maps the row space one-to-one onto the column space and kills the null space.

Theorem 4.4.6

For every $m \times n$ matrix A :

1. $\mathcal{N}(A) = \mathcal{N}(A^T A)$;
2. $\text{rank}(A^T A) = \text{rank}(A)$;
3. $\mathcal{C}(A^T A) = \mathcal{C}(A^T)$.

In particular, Lemma 4.2.4 holds.

Proof. (1) If $x \in \mathcal{N}(A)$, then $Ax = 0$, so $A^T Ax = 0$ and $x \in \mathcal{N}(A^T A)$. Conversely, if $A^T Ax = 0$, then

$$0 = x^T A^T Ax = \langle Ax, Ax \rangle = \|Ax\|^2,$$

so $Ax = 0$ and $x \in \mathcal{N}(A)$. (2) A and $A^T A$ both have n columns. By rank-nullity and (1), $\text{rank}(A^T A) = n - \dim \mathcal{N}(A^T A) = n - \dim \mathcal{N}(A) = \text{rank}(A)$. (3) $\mathcal{C}(A^T A) \subseteq \mathcal{C}(A^T)$, because $A^T Ax = A^T(Ax)$. Both spaces have dimension $\text{rank}(A^T A) = \text{rank}(A) = \text{rank}(A^T)$ by (2). A subspace contained in another of the same finite dimension equals it. $\square$

Note 4.4.7

The notes add: “in the recording it was mentioned $\mathcal{C}(A)$ instead of $\mathcal{C}(A^T)$ ”. The correct statement is $\mathcal{C}(A^T A) = \mathcal{C}(A^T)$, a subspace of $\mathbb{R}^n$. The spaces $\mathcal{C}(A^T A)$ and $\mathcal{C}(A)$ in general lie in different spaces ($\mathbb{R}^n$ versus $\mathbb{R}^m$). Note also that $\mathcal{C}(AA^T) = \mathcal{C}(A)$, by the theorem applied to A^T .

4.4.4 A formula for P_X **Proposition 4.4.8** (Supplementary: P_X via a generalized inverse)

For any generalized inverse $(X^T X)^-$,

$$P_X = X(X^T X)^- X^T,$$

and this matrix does not depend on the choice of generalized inverse. If $\text{rank } X = p$, then $P_X = X(X^T X)^{-1} X^T$.

Proof. Let $G = (X^T X)^-$. By the note in Section 4.2, $\hat{\beta} = GX^T Y$ solves the normal equations for every Y . By Proposition 4.4.3, $XGX^T Y = X\hat{\beta} = P_X Y$ for all Y , so $XGX^T = P_X$. The right-hand side is the unique projection matrix, so it does not depend on G . $\square$

Example 4.4.9 (P_X in the one-way model)

For Example 4.2.7 ($k = 2$, $n_1 = n_2 = 2$), $\mathcal{C}(X)$ is spanned by the orthonormal vectors $u_1 = \frac{1}{\sqrt{2}}(1, 1, 0, 0)^T$ and $u_2 = \frac{1}{\sqrt{2}}(0, 0, 1, 1)^T$. So

$$P_X = u_1 u_1^T + u_2 u_2^T = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}, \quad P_X Y = P_X \begin{pmatrix} 3 \\ 5 \\ 8 \\ 10 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 9 \\ 9 \end{pmatrix} = X\hat{\beta}(t) \quad \forall t.$$

$\text{tr } P_X = 2 = \text{rank } X$. The residual is $(I - P_X)Y = (-1, 1, -1, 1)^T$ and $R_0^2 = 4$. With $G = \text{diag}(0, \frac{1}{2}, \frac{1}{2})$ (Problem 4.2.12), XGX^T gives the same matrix.

4.4.5 Computation

`notebooks/week04/L14_matrices_projection.ipynb` computes the four subspaces of Example 4.4.5 with `scipy.linalg.null_space` and `orth`. It verifies $\mathcal{N}(A) = \mathcal{N}(A^T A)$ and the rank identities on random rank-deficient matrices, builds P_X in three ways (orthonormal basis, $X(X^T X)^+ X^T$, $X(X^T X)^- X^T$ with a hand-made G), and checks symmetry, idempotence and trace.

4.4.6 Summary

- $X\beta$ is estimable, and its least squares estimate is $X\hat{\beta} = P_X Y$, the orthogonal projection of Y onto $\mathcal{C}(X)$.
- $P_X = \sum u_i u_i^T = X(X^T X)^- X^T$: symmetric, idempotent, trace = rank X .
- $\mathcal{N}(A) = \mathcal{N}(A^T A)$, rank $A^T A = \text{rank } A$, $\mathcal{C}(A^T A) = \mathcal{C}(A^T)$ (rank-nullity).
- $\mathcal{R}(A) \perp \mathcal{N}(A)$ in $\mathbb{R}^n$ and $\mathcal{C}(A) \perp \mathcal{N}(A^T)$ in $\mathbb{R}^m$.

4.4.7 Exercises

Problem 4.4.10

Show that $I - P_X$ is the orthogonal projection onto $\mathcal{C}(X)^\perp = \mathcal{N}(X^T)$, and that $P_X(I - P_X) = 0$.

Problem 4.4.11

Compute P_X for simple linear regression with $x = (1, 2, 3, 4)$ and verify that $P_X Y = (1.9, 3.3, 4.7, 6.1)^T$ for $Y = (2, 3, 5, 6)^T$.

Problem 4.4.12

For the one-way model with general k and n_i , show that P_X is block diagonal with blocks $\frac{1}{n_i} \mathbf{1}_{n_i} \mathbf{1}_{n_i}^T$, and that $(P_X Y)_{ij} = \bar{Y}_{i\cdot}$.

Problem 4.4.13

Find the four fundamental subspaces of $B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and check rank-nullity.

Problem 4.4.14

Prove that if P is symmetric and idempotent, then P is the orthogonal projection onto $\mathcal{C}(P)$.

4.4.8 Further reading

Christensen, Appendix A (vector spaces), Appendix B.3 (projections) and §2.2; Rao, §1a–1c (vector spaces, matrices, projection operators) and §4a.

Solutions to the Exercises of Chapter 4

Problem 4.1.9. With columns $(\mu, \alpha_1, \alpha_2, \beta_1, \beta_2, \beta_3)$ and cells ordered $(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)$:

$$X = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

Column μ equals $\alpha_1 + \alpha_2$ and also $\beta_1 + \beta_2 + \beta_3$, and by Proposition 4.1.6 there is no other relation. So $\text{rank } X = 6 - 2 = 4 = I + J - 1$.

Problem 4.1.10. The γ_{22} column is zero. The non-zero cell indicators g_{11}, g_{12}, g_{21} are independent, and all other columns are sums of them ($\alpha_1 = g_{11} + g_{12}$, $\alpha_2 = g_{21}$, $\beta_1 = g_{11} + g_{21}$, $\beta_2 = g_{12}$). So $\text{rank } X = 3$, the number of non-empty cells.

Problem 4.1.11. The row of X for any observation in subgroup (i, j) is $\lambda = e_\mu + e_{\mu_i} + e_{\delta_{ij}}$, so $\mu + \mu_i + \delta_{ij} = \mathbb{E}[Y_{ijl}]$ is estimable. Linear unbiased estimates include Y_{ij1} and, better, $\bar{Y}_{ij\cdot} = \frac{1}{n_{ij}} \sum_l Y_{ijl}$.

Problem 4.1.12. The rows for cells $(1, j)$ and $(2, j)$ are $e_\mu + e_{\alpha_1} + e_{\beta_j}$ and $e_\mu + e_{\alpha_2} + e_{\beta_j}$. Their difference is $e_{\alpha_1} - e_{\alpha_2} \in \mathcal{C}(X^T)$, so $\alpha_1 - \alpha_2$ is estimable, for instance by $Y_{1j1} - Y_{2j1}$.

Problem 4.2.9. $X^T X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$ and $X^T Y = \begin{pmatrix} \sum Y_i \\ \sum x_i Y_i \end{pmatrix}$. The first equation, $n\beta_1 + \beta_2 \sum x_i = \sum Y_i$, gives $\beta_1 = \bar{Y} - \beta_2 \bar{x}$. Substituting into the second: $\beta_2(\sum x_i^2 - n\bar{x}^2) = \sum x_i Y_i - n\bar{x}\bar{Y}$, i.e. $\beta_2 S_{xx} = S_{xy}$, which is Theorem 2.2.4.

Problem 4.2.10. $(X^T Y)_\mu = \sum_{i,j} Y_{ij}$ and $(X^T Y)_{\mu_i} = \sum_j Y_{ij} = n_i \bar{Y}_{i.}$. $(X^T X \beta)_{\mu_i} = n_i(\mu + \mu_i)$, and $(X^T X \beta)_\mu = \sum_i n_i(\mu + \mu_i)$. The μ_i -equations say $\mu + \mu_i = \bar{Y}_{i.}$. The μ -equation is their weighted sum, so it adds nothing. Then $X\hat{\beta}$ has entry $\bar{Y}_{i.}$ in every row of group i , and $R_0^2 = \sum_{i,j} (Y_{ij} - \bar{Y}_{i.})^2 = \text{SSW}$.

Problem 4.2.11. If $\mathbf{1} = Xc$, then $\mathbf{1}^T \hat{e} = c^T X^T \hat{e} = 0$ by the normal equations.

Problem 4.2.12. $X^T X G = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, and multiplying by $X^T X$ gives back $\begin{pmatrix} 4 & 2 & 2 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{pmatrix}$. So G is a generalized inverse. $G X^T Y = (0, 8/2, 18/2)^T = (0, 4, 9)^T$, which is the solution with $t = 0$.

Problem 4.3.6. By Problem 4.2.10 every solution has $\hat{\mu} + \hat{\mu}_i = \bar{Y}_{i.}$. Hence $\sum c_i \hat{\mu}_i = \sum c_i (\bar{Y}_{i.} - \hat{\mu}) = \sum c_i \bar{Y}_{i.} - \hat{\mu} \sum c_i = \sum c_i \bar{Y}_{i.}$.

Problem 4.3.7. Write $\lambda = X^T X b$. Then $\lambda^T \hat{\beta} = b^T X^T Y$ and $\text{Var}(\lambda^T \hat{\beta}) = \sigma^2 b^T X^T X b$ (Proposition 3.2.10). For any generalized inverse G , $\lambda^T G \lambda = b^T X^T X G X^T X b = b^T X^T X b$. So $\text{Var}(\lambda^T \hat{\beta}) = \sigma^2 \lambda^T G \lambda$, whatever G is.

Problem 4.3.8. $X = [\mathbf{1}, x, 2x]$ has $\mathcal{C}(X) = \mathcal{C}([\mathbf{1}, x])$, so the fitted values are those of Example 4.2.6. The normal equations reduce to $\beta_1 = 0.5$ and $\beta_2 + 2\beta_3 = 1.4$. All solutions are $(0.5, 1.4 - 2s, s)^T$, $s \in \mathbb{R}$. The estimate of the estimable function $\beta_2 + 2\beta_3$ is 1.4 for every s .

Problem 4.3.9. Take $X = [x, x, x]$ with $x = (1, 2)^T$. Then $\mathcal{N}(X) = \{v : v_1 + v_2 + v_3 = 0\}$ is two-dimensional. For $\lambda = (1, 0, 0)^T$ (not estimable, since $\lambda^T (1, -1, 0)^T \neq 0$), the two solutions $\hat{\beta}$ and $\hat{\beta} + (0, 1, -1)^T$ give the same $\lambda^T \hat{\beta}$. This does not contradict the theorem, which requires equality for *all* solutions, i.e. $\lambda \perp$ every $v \in \mathcal{N}(X)$.

Problem 4.4.10. $I - P_X$ is symmetric, and $(I - P_X)^2 = I - 2P_X + P_X^2 = I - P_X$. For y , $(I - P_X)y = y - P_X y \perp \mathcal{C}(X)$. Also $y - (I - P_X)y = P_X y \in \mathcal{C}(X)$, which is orthogonal to $\mathcal{C}(X)^\perp$. By uniqueness in Lemma 4.4.2, $I - P_X$ projects onto $\mathcal{C}(X)^\perp$. This space equals $\mathcal{N}(X^T)$, because $X^T z = 0$ iff z is orthogonal to every column. Finally $P_X(I - P_X) = P_X - P_X^2 = 0$.

Problem 4.4.11. $(X^T X)^{-1} = \frac{1}{20} \begin{pmatrix} 30 & -10 \\ -10 & 4 \end{pmatrix}$, so $(P_X)_{ij} = \frac{1}{20}(30 - 10(x_i + x_j) + 4x_i x_j)$:

$$P_X = \begin{pmatrix} 0.7 & 0.4 & 0.1 & -0.2 \\ 0.4 & 0.3 & 0.2 & 0.1 \\ 0.1 & 0.2 & 0.3 & 0.4 \\ -0.2 & 0.1 & 0.4 & 0.7 \end{pmatrix}, \quad P_X(2, 3, 5, 6)^T = (1.9, 3.3, 4.7, 6.1)^T.$$

The trace is $0.7 + 0.3 + 0.3 + 0.7 = 2 = \text{rank } X$.

Problem 4.4.12. The normalised group indicators $u_i = \mathbf{1}_{(i)}/\sqrt{n_i}$ form an orthonormal basis of $\mathcal{C}(X)$, where $\mathbf{1}_{(i)}$ is the indicator of group i . They are orthogonal because their supports are disjoint. Hence $P_X = \sum_i \frac{1}{n_i} \mathbf{1}_{(i)} \mathbf{1}_{(i)}^T$, which is block diagonal with blocks $\frac{1}{n_i} \mathbf{1} \mathbf{1}^T$. Then $(P_X Y)_{ij} = \frac{1}{n_i} \sum_l Y_{il} = \bar{Y}_{i.}$.

Problem 4.4.13. $\text{rank } B = 2$, so $\mathcal{C}(B) = \mathbb{R}^2$ and $\mathcal{N}(B^T) = \{0\}$. $\mathcal{R}(B) = \text{span}\{(1, 0, 1)^T, (0, 1, 1)^T\}$ and $\mathcal{N}(B) = \text{span}\{(1, 1, -1)^T\}$, which is orthogonal to both rows. Rank-nullity: $2 + 1 = 3$.

Problem 4.4.14. For $y \in \mathbb{R}^n$, $Py \in \mathcal{C}(P)$. For any $v = Pz \in \mathcal{C}(P)$, $\langle y - Py, Pz \rangle = z^T P^T(y - Py) = z^T(Py - P^2y) = 0$. So $y - Py \perp \mathcal{C}(P)$, and Py is the orthogonal projection (Definition 4.4.1).

Chapter 5

Inference for Linear Models and Computation in R

This chapter covers Week 5 (Lectures 15–18). **Source caveat.** The instructor's Week-5 lecture notes (Feb21.pdf) were *not* accessed. The chapter is built from the Week-5 R code (Feb21.r), the Week-5 assignment (hw56.pdf), and the standard theory that the code and assignment use. The theory is written in the notation of the earlier chapters. Every numerical value shown was reproduced in the companion notebooks.

5.1 Estimating the Coefficients of Linear Models

Lecture: 15

Video: <https://youtu.be/qF9VBw3CP78>

Timestamps: not available (video transcript not accessed)

Source used: Week-5 R code Feb21.r (least squares fit of `iris`, normal equations by hand) and Week-5 assignment Q4. The Week-5 lecture PDF was *not* accessed. The derivations below follow the titles and code and are marked accordingly.

Quiding Question

By the end of this lecture you should be able to:

- compute $\hat{\beta} = (X^T X)^{-1} X^T Y$ for a full-rank model, by hand and in code;
- prove that $\hat{\beta}$ is unbiased with $\text{Cov}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$;
- derive $\text{Var}(\hat{\beta}_0)$, $\text{Var}(\hat{\beta}_1)$ and $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$ in simple linear regression;
- fit and plot a regression line with `lm/statsmodels`.

5.1.1 Recap

Week 4 established that least squares estimates are the solutions of the normal equations $X^T X \beta = X^T Y$ (Theorem 4.2.3). The solution is unique iff $\text{rank } X = p$ (Corollary 4.2.5), and $X \hat{\beta} = P_X Y$ (Proposition 4.4.3). Week 5 applies this theory to regression models, which in practice almost always have a full-rank X . Throughout this chapter we write simple linear regression as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n,$$

the notation of the R output (“(Intercept)” and the slope). In Week 2 we called these coefficients a_1, a_2 .

5.1.2 The iris example from the R code

Example 5.1.1 (Sepal length on petal width)

The R code fits `lm(Sepal.Length ~ Petal.Width, data = iris)` to Fisher’s iris data ($n = 150$ flowers), plots the points, and adds the fitted line with `abline` and with `ggplot(...)` + `stat_smooth(method = "lm")`. The estimates are

$$\hat{\beta}_0 = 4.7776, \quad \hat{\beta}_1 = 0.8886.$$

A flower whose petals are 1 cm wider has, on average, sepals 0.89 cm longer. The line passes through $(\bar{x}, \bar{y}) = (1.1993, 5.8433)$ (Corollary 2.2.5).

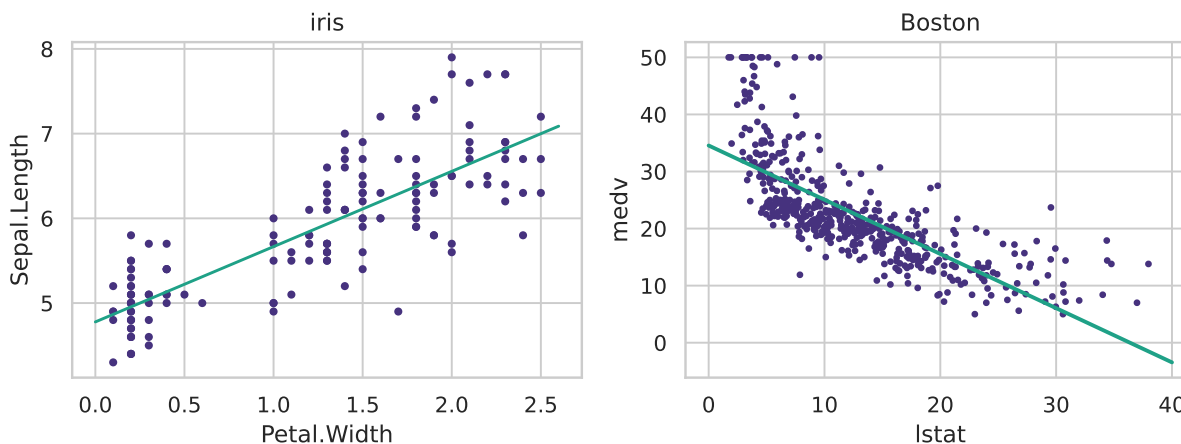


Figure 5.1: Least squares lines for `iris` (Example 5.1.1) and for `Boston` (Section 5.2).

5.1.3 Normal equations by hand

The second part of the R code solves the normal equations directly for a model with two explanatory variables:

```

x1 <- c(10,0,45,56,79); x2 <- c(2,34,15,5,4)
y <- x1 + x2 + rnorm(5, mean=0, sd=0.3)
X <- as.matrix(cbind(1,x1,x2))
betahat = solve(t(X) %*% X) %*% (t(X) %*% Y) # solution of X^TX beta = X^Ty
lm.fit <- lm(y ~ 1+x1+x2); lm.fit$coefficients

```

The data are simulated from $Y_i = 0 + 1 \cdot x_{1i} + 1 \cdot x_{2i} + \varepsilon_i$ with $\sigma = 0.3$, so $\hat{\beta}$ should be close to $(0, 1, 1)$. Here $n = 5$, $p = 3$, and

$$X^T X = \begin{pmatrix} 5 & 190 & 60 \\ 190 & 11502 & 1291 \\ 60 & 1291 & 1426 \end{pmatrix}.$$

Its determinant is non-zero ($\text{rank } X = 3$), so `solve` works. The two computations agree. The exact numbers depend on the random draw. With the notebook's seed they are $\hat{\beta} = (0.3665, 0.9905, 1.0068)$. Different runs of the R code give different numbers, because the original code sets no seed.

5.1.4 Unbiasedness and covariance of $\hat{\beta}$

Theorem 5.1.2

Let $(Y, X\beta, \sigma^2 I_n)$ be a linear model with $\text{rank } X = p$, and let $\hat{\beta} = (X^T X)^{-1} X^T Y$. Then

$$\mathbb{E}[\hat{\beta}] = \beta, \quad \text{Cov}(\hat{\beta}) = \sigma^2 (X^T X)^{-1}.$$

Proof. $\hat{\beta} = AY$ with the fixed matrix $A = (X^T X)^{-1} X^T$. By Proposition 3.2.10, $\mathbb{E}\hat{\beta} = AX\beta = (X^T X)^{-1} X^T X\beta = \beta$. Also

$$\text{Cov}(\hat{\beta}) = A(\sigma^2 I)A^T = \sigma^2 (X^T X)^{-1} X^T X (X^T X)^{-1} = \sigma^2 (X^T X)^{-1},$$

using the symmetry of $(X^T X)^{-1}$. □

Corollary 5.1.3 (Simple linear regression; Week-5 assignment Q4)

For $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with the x_i not all equal,

$$X^T X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \quad X^T Y = \begin{pmatrix} \sum Y_i \\ \sum x_i Y_i \end{pmatrix},$$

$\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$, and

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \quad \text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}}.$$

Proof. The formulas for $X^T X$, $X^T Y$ and $\hat{\beta}$ are Problem 4.2.9. Now $\det(X^T X) = n \sum x_i^2 - (\sum x_i)^2 = nS_{xx}$, so

$$(X^T X)^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix}.$$

By Theorem 5.1.2: $\text{Var}(\hat{\beta}_1) = \sigma^2 n / (n S_{xx}) = \sigma^2 / S_{xx}$ and $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2 n \bar{x} / (n S_{xx}) = -\sigma^2 \bar{x} / S_{xx}$. For the intercept, $\text{Var}(\hat{\beta}_0) = \sigma^2 \sum x_i^2 / (n S_{xx})$, and $\sum x_i^2 = S_{xx} + n \bar{x}^2$ gives $\sigma^2(1/n + \bar{x}^2 / S_{xx})$.

A direct check without matrices: $\hat{\beta}_1 = \sum_i c_i Y_i$ with $c_i = (x_i - \bar{x}) / S_{xx}$, so $\text{Var}(\hat{\beta}_1) = \sigma^2 \sum c_i^2 = \sigma^2 S_{xx} / S_{xx}^2$. Also $\text{Cov}(\bar{Y}, \hat{\beta}_1) = \frac{\sigma^2}{n} \sum c_i = 0$. Then $\text{Var}(\hat{\beta}_0) = \text{Var}(\bar{Y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) - 2\bar{x} \text{Cov}(\bar{Y}, \hat{\beta}_1)$, which gives the same result. $\square$

Example 5.1.4 (The hand example again)

For $x = (1, 2, 3, 4)$ (Example 4.2.6): $n = 4$, $\bar{x} = 2.5$, $S_{xx} = 5$, and

$$(X^T X)^{-1} = \frac{1}{20} \begin{pmatrix} 30 & -10 \\ -10 & 4 \end{pmatrix} = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{pmatrix}.$$

So $\text{Var}(\hat{\beta}_1) = 0.2\sigma^2 = \sigma^2/5$, $\text{Var}(\hat{\beta}_0) = 1.5\sigma^2 = \sigma^2(1/4 + 6.25/5)$ and $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -0.5\sigma^2 = -2.5\sigma^2/5$, in agreement with Corollary 5.1.3. The correlation of the two estimates is $-0.5/\sqrt{1.5 \times 0.2} = -0.913$. Because all $x_i > 0$, overestimating the slope goes with underestimating the intercept.

Note 5.1.5 (Supplementary: design matters)

$\text{Var}(\hat{\beta}_1) = \sigma^2 / S_{xx}$ is small when the x_i are spread out. To estimate a slope well, take observations far apart. $\text{Var}(\hat{\beta}_0)$ is smallest when $\bar{x} = 0$. Centring the covariate $(x_i - \bar{x})$ makes $\hat{\beta}_0 = \bar{Y}$ and the two estimates uncorrelated.

5.1.5 Computation

R	Python
<code>lm(Sepal.Length ~ Petal.Width, iris)</code>	<code>smf.ols("Sepal_Length ~ Petal_Width", ir).fit()</code>
<code>abline(fit1)</code>	<code>ax.plot(xs, b0 + b1*xs)</code>
<code>stat_smooth(method="lm")</code>	<code>sns.regplot(...)</code>
<code>solve(t(X)%*%X) %*% t(X)%*%Y</code>	<code>np.linalg.solve(X.T @ X, X.T @ Y)</code>
<code>vcov(fit)</code>	<code>fit.cov_params()</code>

`notebooks/week05/L15_estimating_coefficients.ipynb` also checks $\text{Cov}(\hat{\beta}) = \sigma^2(X^T X)^{-1}$ by simulation.

5.1.6 Summary

- Full rank: $\hat{\beta} = (X^T X)^{-1} X^T Y$, $\mathbb{E}\hat{\beta} = \beta$, $\text{Cov}\hat{\beta} = \sigma^2(X^T X)^{-1}$.
- SLR: $\text{Var}\hat{\beta}_1 = \sigma^2 / S_{xx}$, $\text{Var}\hat{\beta}_0 = \sigma^2(1/n + \bar{x}^2 / S_{xx})$, $\text{Cov} = -\sigma^2 \bar{x} / S_{xx}$.
- iris: $\widehat{\text{Sepal.Length}} = 4.7776 + 0.8886 \text{Petal.Width}$.

5.1.7 Exercises

Problem 5.1.6

Show that $\hat{\beta}_1 = \sum_i c_i Y_i$ with $c_i = (x_i - \bar{x})/S_{xx}$, and that $\sum c_i = 0$, $\sum c_i x_i = 1$.

Problem 5.1.7

For $x = (-1, 0, 1)$, compute $\text{Cov}(\hat{\beta})$ in simple linear regression. Why are $\hat{\beta}_0$ and $\hat{\beta}_1$ uncorrelated here?

Problem 5.1.8

Show that $\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$.

Problem 5.1.9

You may choose $n = 10$ design points $x_i \in [0, 1]$. Which choice minimises $\text{Var}(\hat{\beta}_1)$?

5.1.8 Further reading

Christensen, §2.2 and Chapter 6 (simple linear regression); Rao, §4a.2; James, Witten, Hastie & Tibshirani, *An Introduction to Statistical Learning*, §3.1 (the source of the lab in the R code).

5.2 Interval Estimates and Hypothesis Tests for the Coefficients

Lecture: 16

Video: <https://youtu.be/dXYKDr07vE8>

Timestamps: not available (video transcript not accessed)

Source used: Week-5 R code Feb21.r (Boston lab: `summary`, `confint`; t -distribution functions `pt`, `qt`). The Week-5 lecture PDF was *not* accessed. The theory is filled in from the course's own framework.

Quiding Question

By the end of this lecture you should be able to:

- show that $\text{RSS}/(n - r)$ is an unbiased estimate of σ^2 ;
- under normal errors, derive the distribution of $(\hat{\beta}_j - \beta_j)/\widehat{SE}(\hat{\beta}_j)$;
- construct confidence intervals for regression coefficients and test $H_0 : \beta_j = 0$;
- read the coefficient table of `summary(lm(...))`.

5.2.1 Recap

$\hat{\beta}$ is unbiased with $\text{Cov} \hat{\beta} = \sigma^2(X^T X)^{-1}$ (Theorem 5.1.2). To turn this into confidence intervals and tests we need two more things: an estimate of the unknown σ^2 , and the exact distribution of the estimates. The latter requires the normality assumption, which also underlies Theorem 3.1.7.

5.2.2 Estimating σ^2

Theorem 5.2.1

In the linear model $(Y, X\beta, \sigma^2 I_n)$ with $\text{rank } X = r < n$, the residual sum of squares $\text{RSS} = \|Y - X\hat{\beta}\|^2 = \|(I - P_X)Y\|^2$ satisfies $\mathbb{E}[\text{RSS}] = (n - r)\sigma^2$. Hence

$$s^2 = \frac{\text{RSS}}{n - r}$$

is an unbiased estimate of σ^2 .

Proof. Let $M = I - P_X$. It is symmetric and idempotent (Problem 4.4.10), so $\text{RSS} = Y^T M^T M Y = Y^T M Y$. Since $MX = X - P_X X = 0$ (every column of X lies in $\mathcal{C}(X)$), we get $MY = M(X\beta + \varepsilon) = M\varepsilon$ and $\text{RSS} = \varepsilon^T M \varepsilon$. For any fixed matrix M ,

$$\mathbb{E}[\varepsilon^T M \varepsilon] = \mathbb{E}\left[\sum_{i,l} M_{il} \varepsilon_i \varepsilon_l\right] = \sum_{i,l} M_{il} \sigma^2 \mathbb{1}\{i = l\} = \sigma^2 \text{tr } M,$$

using $\mathbb{E}[\varepsilon_i \varepsilon_l] = \sigma^2$ if $i = l$ and 0 otherwise. Finally $\text{tr } M = n - \text{tr } P_X = n - r$ (Lemma 4.4.2). $\square$

In simple linear regression $r = 2$ and $s^2 = \text{RSS}/(n - 2)$. Its square root is the *residual standard error* of Section 5.3. For the one-way model $r = k$ and $s^2 = \text{SSW}/(n - k)$ (Problem 3.1.16).

5.2.3 Distributions under normality

Theorem 5.2.2 (Sampling distribution of $\hat{\beta}$)

Assume $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ and $\text{rank } X = p < n$. Then

1. $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X^T X)^{-1})$;
2. $\text{RSS}/\sigma^2 \sim \chi_{n-p}^2$;
3. $\hat{\beta}$ and RSS are independent;
4. for each j , with $v_{jj} = [(X^T X)^{-1}]_{jj}$ and $\widehat{\text{SE}}(\hat{\beta}_j) = s\sqrt{v_{jj}}$,

$$T_j = \frac{\hat{\beta}_j - \beta_j}{\widehat{\text{SE}}(\hat{\beta}_j)} \sim t_{n-p}.$$

Proof. (1) $\hat{\beta} = (X^T X)^{-1} X^T Y$ is a linear function of the normal vector Y , so it is normal. Its mean and covariance are given by Theorem 5.1.2. (2) As in the proof of Theorem 5.2.1, $\text{RSS}/\sigma^2 = Z^T M Z$ with

$Z = \varepsilon/\sigma \sim \mathcal{N}(0, I_n)$ and $M = I - P_X$ symmetric idempotent of rank $n - p$. Lemma 3.1.5 gives χ^2_{n-p} . (3) $\hat{\beta}$ is a function of $U = X^T Y$, and $\text{RSS} = \|MY\|^2$ is a function of $V = MY$. The pair (U, V) is jointly normal, and $\text{Cov}(X^T Y, MY) = X^T(\sigma^2 I)M^T = \sigma^2(MX)^T = 0$. By Lemma 3.1.6, $U \perp V$, so $\hat{\beta} \perp \text{RSS}$. (4) $Z_j = (\hat{\beta}_j - \beta_j)/(\sigma\sqrt{v_{jj}}) \sim \mathcal{N}(0, 1)$ by (1). Also $W = \text{RSS}/\sigma^2 \sim \chi^2_{n-p}$ by (2), and $Z_j \perp W$ by (3). Then $T_j = Z_j/\sqrt{W/(n-p)}$, which is the definition of the t_{n-p} distribution. The unknown σ cancels. $\square$

Corollary 5.2.3 (Confidence interval and t -test)

Let $t_{n-p, 1-\alpha/2}$ be the $(1 - \alpha/2)$ -quantile of t_{n-p} (R: `qt(1-alpha/2, df)`).

1. $\hat{\beta}_j \pm t_{n-p, 1-\alpha/2} \widehat{\text{SE}}(\hat{\beta}_j)$ is a $100(1 - \alpha)\%$ confidence interval for β_j .
2. To test $H_0 : \beta_j = b$ against $H_1 : \beta_j \neq b$, use $t = (\hat{\beta}_j - b)/\widehat{\text{SE}}(\hat{\beta}_j)$. Reject at level α if $|t| > t_{n-p, 1-\alpha/2}$. The p-value is $2\mathbb{P}(T > |t|)$ for $T \sim t_{n-p}$ (R: `2*(1-pt(abs(t), df))`).

Proof. (1) $\mathbb{P}(|T_j| \leq t_{n-p, 1-\alpha/2}) = 1 - \alpha$ by symmetry of the t distribution. Rearranging the inequality $|\hat{\beta}_j - \beta_j| \leq t_{n-p, 1-\alpha/2} \widehat{\text{SE}}$ gives the interval. (2) Under H_0 , $t \sim t_{n-p}$, so $\mathbb{P}_{H_0}(|t| > t_{n-p, 1-\alpha/2}) = \alpha$. $\square$

Example 5.2.4 (By hand: the four-point data)

$x = (1, 2, 3, 4)$, $Y = (2, 3, 5, 6)$, with $\hat{\beta} = (0.5, 1.4)$ and $\text{RSS} = 0.2$ (Example 2.2.6). Here $n - p = 2$.

- $s^2 = 0.2/2 = 0.1$ and $s = 0.3162$.
- $\widehat{\text{SE}}(\hat{\beta}_1) = \sqrt{0.1 \times 0.2} = \sqrt{0.02} = 0.14142$ and $\widehat{\text{SE}}(\hat{\beta}_0) = \sqrt{0.1 \times 1.5} = 0.38730$ (Example 5.1.4).
- t for $H_0 : \beta_1 = 0$: $1.4/0.14142 = 9.8995$. With 2 degrees of freedom the p-value is $2\mathbb{P}(T > 9.8995) = 0.01005$, so we reject at 5% but not at 1%.
- $t_{2, 0.975} = 4.3027$. The 95% interval for β_1 is $1.4 \pm 4.3027 \times 0.14142 = 1.4 \pm 0.6085 = (0.7915, 2.0085)$.
- Intercept: $t = 0.5/0.3873 = 1.291$ with p-value 0.326. Its 95% interval is $0.5 \pm 1.6664 = (-1.1664, 2.1664)$, which contains 0.

With only 2 degrees of freedom, the quantile 4.30 is far larger than the normal 1.96.

5.2.4 The Boston example from the R code

Example 5.2.5 (medv on lstat)

The code follows the lab of Chapter 3 of *An Introduction to Statistical Learning*. Boston has $n = 506$ census tracts. `medv` is the median house value (in \$1000s), and `lstat` is the percentage of lower-status population. The first call, `lm(medv ~ lstat)`, gives an error because Boston was not specified as the data. The fix is `data = Boston`, or `attach(Boston)`. `summary(lm.fit)` prints

	Estimate	Std. Error	<i>t</i> value	$\mathbb{P}(> t)$
(Intercept)	34.5538	0.5626	61.415	$< 2 \times 10^{-16}$
lstat	-0.9500	0.0387	-24.528	$< 2 \times 10^{-16}$

on 504 degrees of freedom. `confint(lm.fit)` gives the 95% intervals

$$\beta_0 \in (33.4485, 35.6592), \quad \beta_1 \in (-1.0261, -0.8740),$$

i.e. $\hat{\beta}_j \pm 1.9647 \widehat{SE}$, where $t_{504,0.975} = 1.9647$. Each additional percentage point of lower-status population is associated with a decrease of between \$874 and \$1026 in median value. The relationship is overwhelmingly significant. (The residual plot in Section 5.4 shows that it is also not linear.)

5.2.5 The t distribution in R

The last part of the R code explores the t distribution.

- `1 - pt(1:5, df = 1)` gives the upper-tail probabilities 0.25000, 0.14758, 0.10242, 0.07798, 0.06283. The Cauchy (t_1) tails are very heavy.
- `qt(.975, df = c(1:10, 20, 50, 100, 1000))` gives 12.706, 4.303, 3.182, 2.776, 2.571, $\dots$, 2.228 ($df = 10$), 2.086 (20), 2.009 (50), 1.984 (100) and 1.962 (1000). As the degrees of freedom grow, the quantiles decrease to the normal value 1.960.
- The non-central t distribution (`pt(t, df=3, ncp=d)`, drawn with `image/persp`) is the distribution of T_j when H_0 is false. It determines the power of the t -test.

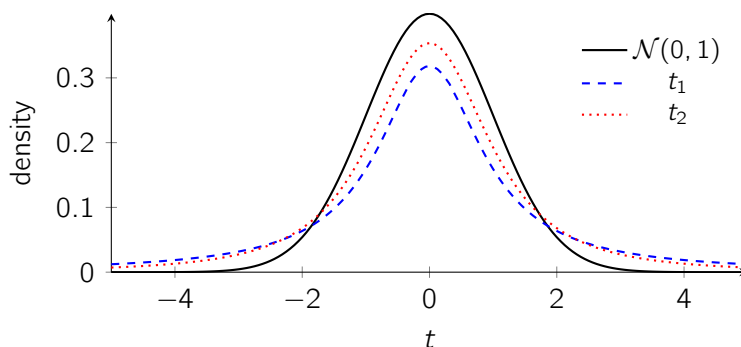


Figure 5.2: The t densities have heavier tails than the normal. This is why the critical values exceed 1.96 when the degrees of freedom are small.

5.2.6 Computation

R	Python
<code>summary(lm.fit)</code>	<code>fit.summary()</code> (or <code>fit.params</code> , <code>fit.bse</code> , <code>fit.tvalues</code> , <code>fit.pvalues</code>)
<code>confint(lm.fit)</code>	<code>fit.conf_int()</code>
<code>pt(q, df), qt(p, df)</code>	<code>scipy.stats.t.cdf(q, df), stats.t.ppf(p, df)</code>
<code>pt(t, df, ncp)</code>	<code>stats.nct.cdf(t, df, ncp)</code>

`notebooks/week05/L16_intervals_tests.ipynb` computes the coefficient table from the formulas, checks it against `statsmodels`, and verifies the 95% coverage and the t_{n-p} distribution by simulation.

5.2.7 Summary

- $s^2 = \text{RSS}/(n - r)$ is unbiased for σ^2 , because $\mathbb{E}[\varepsilon^\top M \varepsilon] = \sigma^2 \text{tr } M$.
- Under normality: $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(X^\top X)^{-1})$, $\text{RSS}/\sigma^2 \sim \chi_{n-p}^2$, and they are independent. So $(\hat{\beta}_j - \beta_j)/\widehat{\text{SE}} \sim t_{n-p}$.
- CI: $\hat{\beta}_j \pm t_{n-p, 1-\alpha/2} \widehat{\text{SE}}$. Test of $\beta_j = 0$: $t = \hat{\beta}_j/\widehat{\text{SE}}$.
- Boston: $\hat{\beta}_1 = -0.9500$ (SE 0.0387), CI $(-1.0261, -0.8740)$.

5.2.8 Exercises

Problem 5.2.6

For the snowfall data of Example 2.4.3, test $H_0 : \beta_1 = 0$ at the 5% level and give a 95% CI for the slope.

Problem 5.2.7

Show that in simple linear regression the t statistic for $\beta_1 = 0$ can be written $t = r\sqrt{n-2}/\sqrt{1-r^2}$, where r is the sample correlation.

Problem 5.2.8

Why can we not use RSS/n in place of s^2 in Theorem 5.2.2(4)? What is $\mathbb{E}[\text{RSS}/n]$?

Problem 5.2.9

In Example 5.2.4, test $H_0 : \beta_1 = 1$ against $\beta_1 \neq 1$ at the 5% level.

Problem 5.2.10

Simulate 10 000 data sets from $Y_i = 0.5 + 1.4x_i + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, 0.1)$, $x = (1, 2, 3, 4)$, and estimate the coverage of the 95% interval for β_1 .

5.2.9 Further reading

Christensen, §2.6 (sampling distributions), Chapter 3 (testing) and Chapter 6; Rao, §4b (tests of linear hypotheses); ISLR §3.1.2.

5.3 Assessing the Linear Model: RSE and R^2

Lecture: 17

Video: <https://youtu.be/g1UzPzHITGs>

Timestamps: not available (video transcript not accessed)

Source used: Week-5 R code Feb21.r (“discuss what each summary is stating”). The Week-5 lecture PDF was *not* accessed. Definitions follow ISLR §3.1.3, which the code follows.

Quiding Question

By the end of this lecture you should be able to:

- define the residual standard error (RSE) and R^2 and interpret them;
- prove the decomposition $TSS = ESS + RSS$ when the model contains an intercept;
- prove that $R^2 = r^2$ in simple linear regression;
- read the bottom part of `summary(lm(...))`.

5.3.1 Recap

Section 5.2 judged individual coefficients: is β_j zero or not? Here we judge the fit of the model as a whole: how far are the data from the fitted line, and how much of the variation in Y does the model explain?

5.3.2 Residual standard error

Definition 5.3.1 (Residual standard error)

For a model with rank $X = p$ fitted to n observations,

$$RSE = \sqrt{\frac{RSS}{n-p}} = s, \quad RSS = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

In simple linear regression, $RSE = \sqrt{RSS/(n-2)}$.

The RSE estimates σ , the standard deviation of the errors (Theorem 5.2.1). It is roughly the typical size of the amount by which a response deviates from the true regression line, measured in the units of Y . It is an absolute measure of *lack of fit*. Whether a given RSE is “small” depends on the scale of Y .

5.3.3 The coefficient of determination

Definition 5.3.2 (R^2)

For a model containing an intercept ($\mathbf{1}_n \in \mathcal{C}(X)$), let $TSS = \sum_i (Y_i - \bar{Y})^2$ and $ESS = \sum_i (\hat{Y}_i - \bar{Y})^2$ (the explained sum of squares). Then

$$R^2 = 1 - \frac{RSS}{TSS} = \frac{ESS}{TSS}.$$

The second equality is the following theorem.

Theorem 5.3.3 (Decomposition of TSS)

If $\mathbf{1}_n \in \mathcal{C}(X)$, then $TSS = ESS + RSS$, and $0 \leq R^2 \leq 1$.

Proof. Write $Y - \bar{Y}\mathbf{1} = (\hat{Y} - \bar{Y}\mathbf{1}) + (Y - \hat{Y})$. The second vector $\hat{e} = Y - \hat{Y} = (I - P_X)Y$ is orthogonal to $\mathcal{C}(X)$. The first vector lies in $\mathcal{C}(X)$, since $\hat{Y} \in \mathcal{C}(X)$ and $\mathbf{1} \in \mathcal{C}(X)$. Hence the cross term vanishes and, by Pythagoras, $\|Y - \bar{Y}\mathbf{1}\|^2 = \|\hat{Y} - \bar{Y}\mathbf{1}\|^2 + \|\hat{e}\|^2$, i.e. $TSS = ESS + RSS$. Both parts are non-negative, so $0 \leq RSS/TSS \leq 1$. $\square$

This is the same decomposition as $TSS = SSB + SSW$ in the one-way model (Theorem 3.1.4). There $\hat{Y}_{ij} = \bar{Y}_{i\cdot}$, so $ESS = SSB$ and $RSS = SSW$.

R^2 is the *proportion of the variability in Y explained by the model*. It is a relative measure, always between 0 and 1 and free of units. R^2 near 1 means that the model explains most of the variation. R^2 near 0 means that it explains little: either the linear model is wrong or the errors are large.

Theorem 5.3.4

In simple linear regression, $R^2 = r^2$, where $r = S_{XY}/\sqrt{S_{XX}S_{YY}}$ is the sample correlation.

Proof. By Theorem 2.2.4, $RSS = S_{YY} - S_{XY}^2/S_{XX}$ and $TSS = S_{YY}$. So $R^2 = 1 - RSS/TSS = \frac{S_{XY}^2}{S_{XX}S_{YY}} = r^2$. $\square$

Example 5.3.5 (The four-point data)

$Y = (2, 3, 5, 6)$, $\bar{Y} = 4$, $TSS = 4 + 1 + 1 + 4 = 10$, and $RSS = 0.2$. So $R^2 = 1 - 0.2/10 = 0.98$. Also $r = 7/\sqrt{5 \times 10} = 0.98995$ and $r^2 = 0.98$. $\checkmark$ Fitted values $(1.9, 3.3, 4.7, 6.1)$ give $ESS = 2.1^2 + 0.7^2 + 0.7^2 + 2.1^2 = 9.8 = TSS - RSS$. $RSE = \sqrt{0.2/2} = 0.3162$.

Example 5.3.6 (Boston and iris)

For `lm(medv ~ lstat, data = Boston)`: $RSS = 19472.38$, $TSS = 42716.30$, so $R^2 = 1 - 19472.38/42716.30 = 0.5441$, and $RSE = \sqrt{19472.38/504} = 6.216$. `summary` prints: "Residual standard error: 6.216 on 504 degrees of freedom, Multiple R-squared: 0.5441, Adjusted R-squared: 0.5432, F-statistic: 601.6 on 1 and 504 DF". So `lstat` explains about 54% of the variation in median house values, and a typical prediction is off by about \$6200. Since $\bar{medv} = 22.53$, that is a relative error of about 28%. For `iris` (Example 5.1.1): $R^2 = 0.6690$ and $RSE = 0.4780$ on 148 degrees of freedom.

Note 5.3.7 (Supplementary: adjusted R^2 and the F statistic)

Adding columns to X can never increase RSS: minimising over a larger column space gives a smaller or equal minimum. So R^2 always increases when variables are added, even useless ones. The *adjusted* $R^2_{adj} = 1 - \frac{RSS/(n-p)}{TSS/(n-1)}$ penalises this. For Boston it is $1 - \frac{19472.38/504}{42716.30/505} = 0.5432$. The F statistic $\frac{ESS/(p-1)}{RSS/(n-p)} = \frac{23243.91/1}{19472.38/504} = 601.6$ tests whether all slopes are zero. With one predictor, $F = t^2 = (-24.528)^2$. The F -test is the subject of Week 8.

5.3.4 Computation

`fit.rsquared`, `fit.rsquared_adj`, `np.sqrt(fit.scale)` (RSE), `fit.ssr` (RSS), `fit.centered_tss` (TSS), `fit.ess`, `fit.fvalue`. See notebooks/week05/L17_rse_rsquared.ipynb, which computes everything from the definitions and checks it against `statsmodels`. It also illustrates that R^2 never decreases when a pure-noise variable is added, while R^2_{adj} can.

5.3.5 Summary

- $RSE = \sqrt{RSS/(n-p)}$ estimates σ . It is an absolute measure of lack of fit, in the units of Y .
- With an intercept, $TSS = ESS + RSS$ and $R^2 = 1 - RSS/TSS \in [0, 1]$ is the proportion of variance explained.
- Simple linear regression: $R^2 = r^2$. Boston: $R^2 = 0.544$, $RSE = 6.216$.

5.3.6 Exercises

Problem 5.3.8

Show by example that without an intercept, $1 - RSS/TSS$ can be negative.

Problem 5.3.9

Show that R^2 equals the squared sample correlation between Y_i and $\hat{Y}_i$ in any model with an intercept.

Problem 5.3.10

If Y is measured in different units ($Y' = cY$, $c \neq 0$), how do RSE and R^2 change?

Problem 5.3.11

Compute R^2 for the one-way example of Example 3.1.9 (groups $\{5, 7\}$, $\{8, 10, 12\}$, $\{4, 6\}$).

5.3.7 Further reading

ISLR §3.1.3 (assessing the accuracy of the model); Christensen, §6.4 and Chapter 14 (coefficient of determination); Weisberg, *ALR*, §2.7 and §3.4.

5.4 Computation of Least Squares Estimates in R

Lecture: 18

Video: https://youtu.be/4gwbWg_Gq-4

Timestamps: not available (video transcript not accessed)

Source used: Week-5 R code Feb21.r (Boston lab: `names`, `coef`, `confint`, `predict`, diagnostic plots, `hatvalues`; Monte Carlo simulation; non-central t). The Week-5 lecture PDF was *not* accessed.

Quiding Question

By the end of this lecture you should be able to:

- extract every component of a fitted linear model in R and in Python;
- distinguish a confidence interval for the mean response from a prediction interval for a new response, and derive both;
- produce and interpret the four standard diagnostic plots;
- compute leverages (hat values) and identify high-leverage points;
- run a simple Monte Carlo simulation.

5.4.1 Recap

The previous three lectures gave the estimates (Section 5.1), their standard errors, intervals and tests (Section 5.2), and overall measures of fit (Section 5.3). This lecture walks through the rest of the R code, which uses these results in practice on the `Boston` data.

5.4.2 Inside a fitted model

`names(lm.fit)` lists the stored components: `coefficients`, `residuals`, `fitted.values`, `rank`, `df.residual`, the QR decomposition, and others. `coef(lm.fit)` returns $(34.5538, -0.9500)$ and `confint(lm.fit)` the intervals of Example 5.2.5.

Note 5.4.1 (How R actually solves the normal equations)

`lm` does not form $X^T X$. It computes a QR decomposition $X = QR$, where Q ($n \times p$) has orthonormal columns and R ($p \times p$) is upper triangular. Then $X^T X \beta = X^T Y$ becomes $R^T R \beta = R^T Q^T Y$, i.e. $R \beta = Q^T Y$, which is solved by back-substitution. This is numerically more stable, because the condition number of $X^T X$ is the square of that of X . It also gives $P_X = QQ^T$ directly: Q is an orthonormal basis of $\mathcal{C}(X)$, as in Lemma 4.4.2. `statsmodels` uses the Moore–Penrose pseudo-inverse by default, or QR with `method="qr"`.

5.4.3 Confidence and prediction intervals

`predict(lm.fit, data.frame(lstat = c(5,10,15)), interval = "confidence")` and `interval = "prediction"` give two different kinds of interval. At a new value x_0 , let $\hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$ and $h_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}$.

Proposition 5.4.2

Assume normal errors in simple linear regression, and let $Y_0 = \beta_0 + \beta_1 x_0 + \varepsilon_0$ be a new observation with $\varepsilon_0 \sim \mathcal{N}(0, \sigma^2)$ independent of the data.

1. $\hat{Y}_0 \pm t_{n-2, 1-\alpha/2} s \sqrt{h_0}$ is a $100(1 - \alpha)\%$ **confidence interval** for the mean response $\beta_0 + \beta_1 x_0$.
2. $\hat{Y}_0 \pm t_{n-2, 1-\alpha/2} s \sqrt{1 + h_0}$ is a $100(1 - \alpha)\%$ **prediction interval** for Y_0 .

Proof. $\hat{Y}_0 = \lambda^T \hat{\beta}$ with $\lambda = (1, x_0)^T$. It is normal with mean $\beta_0 + \beta_1 x_0$ and variance $\sigma^2 \lambda^T (X^T X)^{-1} \lambda = \sigma^2 h_0$ (Problem 5.1.8). It is independent of s^2 (Theorem 5.2.2). (1) Then $(\hat{Y}_0 - \beta_0 - \beta_1 x_0) / (s \sqrt{h_0}) \sim t_{n-2}$, exactly as in the proof of Theorem 5.2.2(4). (2) $Y_0 - \hat{Y}_0$ has mean 0 and, since ε_0 is independent of $\hat{Y}_0$, variance $\sigma^2 + \sigma^2 h_0$. It is normal, and it is independent of s^2 , because ε_0 is independent of all the data. Hence $(Y_0 - \hat{Y}_0) / (s \sqrt{1 + h_0}) \sim t_{n-2}$, and the interval follows as in Corollary 5.2.3. $\square$

The confidence interval reflects only the uncertainty in the fitted line. It shrinks to zero width as $n \rightarrow \infty$. The prediction interval also contains the irreducible noise ε_0 of a new observation, so it never becomes narrower than about $\pm 1.96\sigma$.

Example 5.4.3 (Boston)

With $n = 506$, $s = 6.2158$ and $t_{504,0.975} = 1.9647$:

lstat	fit	95% confidence interval	95% prediction interval
5	29.8036	(29.0074, 30.5998)	(17.5657, 42.0415)
10	25.0533	(24.4741, 25.6326)	(12.8276, 37.2791)
15	20.3031	(19.7316, 20.8746)	(8.0777, 32.5285)

Both intervals are centred at the same fitted value. The prediction intervals are about 20 times wider.

Example 5.4.4 (By hand)

For the four-point data at $x_0 = 5$: $\hat{Y}_0 = 0.5 + 1.4 \times 5 = 7.5$ and $h_0 = \frac{1}{4} + \frac{(5-2.5)^2}{5} = 1.5$ (an extrapolation, so h_0 is large). With $s^2 = 0.1$ and $t_{2,0.975} = 4.3027$: the confidence interval is $7.5 \pm 4.3027\sqrt{0.15} = 7.5 \pm 1.6664 = (5.8336, 9.1664)$, and the prediction interval is $7.5 \pm 4.3027\sqrt{0.25} = 7.5 \pm 2.1514 = (5.3487, 9.6514)$.

5.4.4 Diagnostic plots and leverage

`par(mfrow=c(2,2)); plot(lm.fit)` draws four diagnostic plots: residuals against fitted values, a normal Q–Q plot, a scale–location plot, and residuals against leverage. The code then draws `plot(predict(lm.fit), residuals(lm.fit))` and `plot(predict(lm.fit), rstudent(lm.fit))`. For Boston, the residuals against fitted values form a clear U-shape (Fig. 5.3). This is evidence of *non-linearity*, exactly as in Forbes' data (Section 2.3). ISLR's lab later adds a term `lstat2`.

Definition 5.4.5 (Leverage)

The **leverage** (hat value) of observation i is $h_{ii} = (P_X)_{ii}$, the i th diagonal entry of the hat matrix. The name comes from $\hat{Y} = P_X Y$: the projection matrix “puts the hat on Y ”. The **studentized residual** (`Rstudent`) is $\hat{e}_i / (s_{(i)} \sqrt{1 - h_{ii}})$, where $s_{(i)}$ is the RSE of the fit without observation i .

Proposition 5.4.6

(1) $0 \leq h_{ii} \leq 1$ and $\sum_i h_{ii} = \text{rank } X$. (2) $\text{Var}(\hat{e}_i) = \sigma^2(1 - h_{ii})$. (3) In simple linear regression $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$.

Proof. (1) Since P_X is symmetric and idempotent, $h_{ii} = (P_X^2)_{ii} = \sum_l (P_X)_{il}^2 = h_{ii}^2 + \sum_{l \neq i} (P_X)_{il}^2 \geq h_{ii}^2$. So $h_{ii} \geq h_{ii}^2$, which gives $0 \leq h_{ii} \leq 1$. Also $\sum_i h_{ii} = \text{tr } P_X = \text{rank } X$ (Lemma 4.4.2). (2) $\hat{e} = (I - P_X)Y$, so $\text{Cov}(\hat{e}) = \sigma^2(I - P_X)(I - P_X)^T = \sigma^2(I - P_X)$. Its i th diagonal entry is $\sigma^2(1 - h_{ii})$. (3) $h_{ii} = x_i^T (X^T X)^{-1} x_i$ with $x_i = (1, x_i)^T$. Using the inverse in the proof of Corollary 5.1.3, $h_{ii} = \frac{\sum_l x_l^2 - 2x_i \sum_l x_l + nx_i^2}{nS_{xx}} = \frac{S_{xx} + n(x_i - \bar{x})^2}{nS_{xx}}$. This is the stated formula, since $\sum_l x_l^2 - 2x_i \sum_l x_l + nx_i^2 = S_{xx} + n\bar{x}^2 - 2nx_i\bar{x} + nx_i^2$. $\square$

A point with large leverage has an x -value far from $\bar{x}$. It can pull the fitted line towards itself. For Boston, `which.max(hatvalues(lm.fit))` returns observation 375, which has the largest $1st$ at (37.97) and $h_{375,375} = 0.02687$. The average leverage is $p/n = 2/506 = 0.00395$, and the hat values sum to exactly 2.

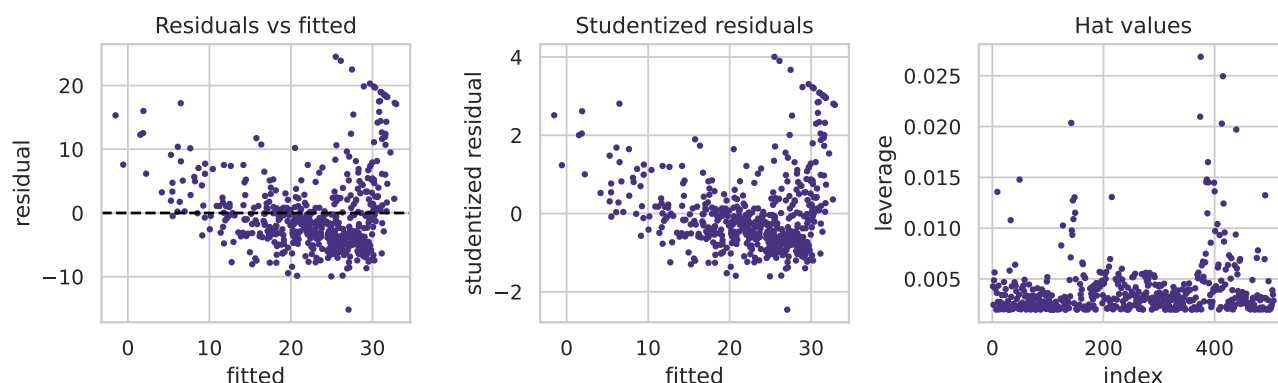


Figure 5.3: Boston: residuals and studentized residuals against fitted values (note the curvature), and hat values by observation index (the maximum is at observation 375).

5.4.5 A Monte Carlo simulation

The code ends with a simulation:

```
OneFourthPiSamples = replicate(100, { u <- runif(10000); v <- runif(10000);
  z <- sqrt(u^2 + v^2); sum(z < 1)/10000 })
```

A point (U, V) uniform on the unit square falls inside the quarter disc $\{u^2 + v^2 < 1\}$ with probability equal to its area, $\pi/4 = 0.7854$. Each replicate estimates $\pi/4$ by the proportion of 10 000 points inside, with standard error $\sqrt{(\pi/4)(1 - \pi/4)/10000} = 0.0041$. With the notebook's seed the 100 replicates have mean 0.7858 and standard deviation 0.0047. The same idea, repeat an experiment and look at the distribution of an estimate, is how the notebooks check sampling distributions such as those of Theorem 5.2.2. The instructor uses it for least squares estimators in Week 7.

5.4.6 Computation

R	Python (statsmodels)
<code>names(lm.fit), coef, confint</code>	<code>dir(fit), fit.params, fit.conf_int()</code>
<code>predict(..., interval="confidence")</code>	<code>fit.get_prediction(new).summary_frame()</code> <code>(mean_ci_*)</code>
<code>predict(..., interval="prediction")</code>	same frame, columns <code>obs_ci_*</code>
<code>plot(lm.fit)</code>	residual, Q-Q (<code>sm.qqplot</code>), leverage plots
<code>hatvalues, rstudent</code>	<code>fit.get_influence().hat_matrix_diag,</code> <code>.resid_studentized_external</code>
<code>replicate(100, {...})</code>	list comprehension with <code>rng.uniform</code>

See `notebooks/week05/L18_computation_in_R.ipynb`.

5.4.7 Summary

- At x_0 : confidence interval $\hat{Y}_0 \pm t s \sqrt{h_0}$ for the mean, prediction interval $\hat{Y}_0 \pm t s \sqrt{1 + h_0}$ for a new Y , where $h_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}$.
- Leverage $h_{ii} = (P_X)_{ii} \in [0, 1]$, $\sum h_{ii} = p$, $\text{Var } \hat{e}_i = \sigma^2(1 - h_{ii})$.
- Boston: the residuals show curvature, and the highest leverage is observation 375.
- Monte Carlo: repeat, then summarise the estimates.

5.4.8 Exercises

Problem 5.4.7

Show that the prediction interval at $x_0 = \bar{x}$ has half-width $t_{n-2, 1-\alpha/2} s \sqrt{1 + 1/n}$.

Problem 5.4.8

For the four-point data compute all four hat values and check that they sum to 2.

Problem 5.4.9

Fit $\text{medv} \sim 1\text{stat} + I(1\text{stat}^2)$ and compare R^2 and the residual plot with the straight-line fit.

Problem 5.4.10

Estimate π by Monte Carlo with 10^6 points. How many points are needed for the standard error of the estimate of π to be below 0.001?

5.4.9 Further reading

ISLR §3.1–3.3 and Lab 3.6; Weisberg, *ALR*, Chapter 9 (regression diagnostics); Christensen, Chapter 12.

Solutions to the Exercises of Chapter 5

Problem 5.1.6. Since $\sum_i (x_i - \bar{x}) = 0$, $S_{xy} = \sum_i (x_i - \bar{x})(Y_i - \bar{Y}) = \sum_i (x_i - \bar{x})Y_i$. So $\hat{\beta}_1 = S_{xy}/S_{xx} = \sum_i c_i Y_i$. Then $\sum_i c_i = \sum_i (x_i - \bar{x})/S_{xx} = 0$, and $\sum_i c_i x_i = \sum_i (x_i - \bar{x})x_i/S_{xx} = \sum_i (x_i - \bar{x})^2/S_{xx} = 1$, again because $\sum_i (x_i - \bar{x})\bar{x} = 0$. It follows that $\mathbb{E}\hat{\beta}_1 = \sum_i c_i(\beta_0 + \beta_1 x_i) = \beta_1$ and $\text{Var } \hat{\beta}_1 = \sigma^2 \sum_i c_i^2 = \sigma^2/S_{xx}$.

Problem 5.1.7. $X^T X = \begin{pmatrix} 3 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$, so $\text{Cov}(\hat{\beta}) = \sigma^2 (X^T X)^{-1} = \sigma^2 \text{diag}(1/3, 1/2)$. By Corollary 5.1.3, $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2 \bar{x}/S_{xx}$, which is 0 because the design is centred ($\bar{x} = 0$). Then $\hat{\beta}_0 = \bar{Y}$ and $\hat{\beta}_1$ is a contrast of the Y_i , orthogonal to $\mathbf{1}$.

Problem 5.1.8. Write $\hat{\beta}_0 + \hat{\beta}_1 x_0 = \bar{Y} + \hat{\beta}_1(x_0 - \bar{x})$. Now $\text{Cov}(\bar{Y}, \hat{\beta}_1) = \sum_i \frac{1}{n} c_i \sigma^2 = 0$ by Problem 5.1.6. Hence $\text{Var} = \frac{\sigma^2}{n} + (x_0 - \bar{x})^2 \frac{\sigma^2}{S_{xx}}$. (Equivalently, $\sigma^2 x_0^T (X^T X)^{-1} x_0$ with $x_0 = (1, x_0)^T$.)

Problem 5.1.9. $\text{Var} \hat{\beta}_1 = \sigma^2 / S_{xx}$, so we maximise S_{xx} . For $x_i \in [0, 1]$, $x_i^2 \leq x_i$, so $S_{xx} = \sum x_i^2 - n\bar{x}^2 \leq n\bar{x} - n\bar{x}^2 = n\bar{x}(1 - \bar{x}) \leq n/4$. Equality needs every $x_i \in \{0, 1\}$ and $\bar{x} = 1/2$: put five points at 0 and five at 1, giving $S_{xx} = 2.5$. Ten equally spaced points give only $S_{xx} = 1.019$. (This design is optimal only if the model is known to be linear. With two support points we cannot check linearity.)

Problem 5.2.6. For the Fort Collins data ($n = 93$): $\hat{\beta}_1 = 0.2035$, $\widehat{SE} = 0.1310$, $t = 1.553$ on 91 degrees of freedom, $p = 0.124 > 0.05$. We do not reject $H_0 : \beta_1 = 0$. The 95% CI is $0.2035 \pm 1.9864 \times 0.1310 = (-0.0568, 0.4638)$, which contains 0. Early-season snowfall tells us little about late-season snowfall.

Problem 5.2.7. $\hat{\beta}_1 = S_{xy}/S_{xx} = r\sqrt{S_{yy}/S_{xx}}$ and $\text{RSS} = S_{yy} - S_{xy}^2/S_{xx} = S_{yy}(1 - r^2)$. So $s^2 = S_{yy}(1 - r^2)/(n - 2)$, and

$$t = \frac{\hat{\beta}_1}{s/\sqrt{S_{xx}}} = \frac{r\sqrt{S_{yy}/S_{xx}}\sqrt{S_{xx}}\sqrt{n-2}}{\sqrt{S_{yy}(1-r^2)}} = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}.$$

For the snowfall data this gives 1.5528, the same as the regression t .

Problem 5.2.8. The t pivot uses the fact that $(n - p)s^2/\sigma^2 = \text{RSS}/\sigma^2 \sim \chi_{n-p}^2$, independent of $\hat{\beta}$. A t_{n-p} variable is $Z/\sqrt{\chi_{n-p}^2/(n-p)}$, so the divisor must be $n - p$. With RSS/n the ratio is $\sqrt{n/(n-p)} T$, not t_{n-p} . It is also biased: $\mathbb{E}[\text{RSS}/n] = \frac{n-p}{n}\sigma^2 < \sigma^2$ (Theorem 5.2.1). Intervals built from it would be too short.

Problem 5.2.9. $t = (1.4 - 1)/0.14142 = 2.828 < t_{2,0.975} = 4.303$ (two-sided $p = 0.106$). We do not reject $\beta_1 = 1$. This agrees with $1 \in (0.7915, 2.0085)$ from Example 5.2.4.

Problem 5.2.10. With $\sigma^2 = 0.1$ and 10 000 replicates, the notebook finds coverage 0.9499, within Monte Carlo error (± 0.0044) of 0.95. The interval is exact for normal errors, whatever σ^2 and n are.

Problem 5.3.8. Take $x = (1, 2, 3)$, $y = (3, 2, 1)$ and fit $y = \beta x$ through the origin. Then $\hat{\beta} = \sum x_i y_i / \sum x_i^2 = 10/14 = 5/7$, the residuals are $(16, 4, -8)/7$, $\text{RSS} = 48/7 \approx 6.857$, and the centred $\text{TSS} = 2$. So $1 - \text{RSS}/\text{TSS} = -17/7 \approx -2.43$. Without an intercept, $\mathbf{1} \notin \mathcal{C}(X)$, the decomposition $\text{TSS} = \text{ESS} + \text{RSS}$ fails, and a line through the origin can do worse than the constant $\bar{y}$.

Problem 5.3.9. With an intercept, $\mathbf{1} \in \mathcal{C}(X)$, so $\sum_i e_i = 0$ and the mean of the $\hat{Y}_i$ is $\bar{Y}$. Since $e \perp \hat{Y} - \bar{Y}\mathbf{1}$,

$$\sum_i (Y_i - \bar{Y})(\hat{Y}_i - \bar{Y}) = \sum_i (\hat{Y}_i - \bar{Y} + e_i)(\hat{Y}_i - \bar{Y}) = \text{ESS}.$$

Hence $\text{corr}(Y, \hat{Y})^2 = \text{ESS}^2/(\text{TSS} \cdot \text{ESS}) = \text{ESS}/\text{TSS} = R^2$ by Theorem 5.3.3.

Problem 5.3.10. $\hat{\beta}' = c\hat{\beta}$ and $e' = ce$, so $\text{RSS}' = c^2\text{RSS}$ and $\text{TSS}' = c^2\text{TSS}$. Thus $\text{RSE}' = |c| \text{RSE}$ (RSE is in the units of Y), while R^2 is unchanged (it is unit-free).

Problem 5.3.11. The one-way model contains the intercept (as the sum of the group indicators), so $\text{RSS} = \text{SSW} = 12 = 84/7$ and $\text{TSS} = \text{SSW} + \text{SSB} = 84/7 + 250/7 = 334/7$. Hence $R^2 = \text{SSB}/\text{TSS} = 250/334 = 125/167 \approx 0.7485$.

Problem 5.4.7. By Proposition 5.4.2 the prediction half-width is $t_{n-2, 1-\alpha/2} s\sqrt{1+h_0}$ with $h_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}$. At $x_0 = \bar{x}$, $h_0 = 1/n$. This is the narrowest prediction interval.

Problem 5.4.8. $h_i = \frac{1}{4} + \frac{(x_i - 2.5)^2}{5}$ gives $(0.7, 0.3, 0.3, 0.7)$, the diagonal of P_X in Problem 4.4.11. The sum is $2 = \text{tr } P_X = \text{rank } X$ (Proposition 5.4.6).

Problem 5.4.9. The quadratic fit is $\widehat{\text{medv}} = 42.862 - 2.3328 \text{1stat} + 0.04355 \text{1stat}^2$, with all terms highly significant. R^2 rises from 0.5441 to 0.6407 and RSE falls from 6.216 to 5.524. The U-shape in the residual-versus-fitted plot largely disappears, which confirms the non-linearity seen in Fig. 5.3.

Problem 5.4.10. $\hat{\pi} = 4\hat{p}$, where $\hat{p}$ is the fraction of m uniform points inside the quarter disc. So $\text{SE}(\hat{\pi}) = 4\sqrt{p(1-p)/m}$ with $p = \pi/4$. For $m = 10^6$ this is 0.00164 (the notebook's estimate is 3.14013). The condition $\text{SE} < 0.001$ requires $m > 16p(1-p)/10^{-6} \approx 2.70 \times 10^6$. Each extra decimal digit costs a factor of 100 in sample size.